INFINITARY REAL-VALUED LOGIC AND A REFINEMENT OF THE KOCHEN–SPECKER THEOREM

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Abstract. For nearly sixty years, refinements of the Bell–Kochen–Specker theorem have concentrated on finite configurations, combinatorics and robustness, while the logical form of the obstruction has remained essentially propositional: once a configuration is fixed, contextuality is witnessed by the inconsistency of a finite family of Boolean constraints. We replace this truth-functional analysis by infinitary real-valued model theory and model-theoretic forcing.

We axiomatize, in real-valued $L_{\omega_1\omega}$, the lattice of closed subspaces of a separable infinite-dimensional complex Hilbert space together with a quantum probability measure. In the standard models the theory recovers Gleason's theorem; after adjoining the axiom that the measure is $\{0,1\}$-valued, the resulting theory has no models, standard or nonstandard. The inconsistency is genuinely infinitary: every finite set of axioms has a model. A model existence game nevertheless converts this global failure into finite inconsistent sets of interval commitments on the values of sentences.

The same framework yields an exact definability theorem of independent descriptive-set-theoretic interest. Over the standard models, the subsets of $[0,1]$ definable from the range of the quantum probability predicate are exactly the σ-projective sets, in the sense of the least class containing the Borel sets and closed under complements, countable unions and projections along $\mathbb{N}^{\mathbb{N}}$. This classification is proved in ZFC.

For the specified Bell–Kochen–Specker strategy, plays completing the parity commitments refine the usual configuration by recording geometric identities between contextual term occurrences and their use by the measure. Probabilities are tracked up to rational tolerances, and σ-additivity along the named basis selects the four-dimensional sector in which this ray-based argument is carried out. For the eighteen-ray parity configuration of Cabello, Estebaranz and García-Alcaine, every identification required by the parity argument becomes visible. Thus the refinement is not a smaller Kochen–Specker set but a finer logical analysis of why contextuality is forced, what infinitary structure is responsible for the contradiction, and how that structure is compressed into a finite obstruction.

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2020 *Mathematics Subject Classification.* Primary 03C75, 06C15, 81P10, 81P13, 91A44; Secondary 03C25, 03C66, 03G12.

Key words and phrases. Kochen–Specker theorem, Gleason's theorem, hidden variables, quantum logic, orthomodular lattices, Solèr's theorem, infinitary logic, real-valued logic, model existence theorem, model existence game, model-theoretic forcing.

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Introduction

The Bell–Kochen–Specker theorem is one of the basic impossibility results of quantum foundations. It asserts, in particular, that a quantum probability measure on the lattice of closed subspaces of a Hilbert space cannot be dispersion-free: there is no map $\mu\colon L(H) \to \{0,1\}$ that is additive on every finite or countable orthogonal decomposition of the identity. In the usual physical reading, this rules out noncontextual hidden-variable assignments, in which every proposition is given a definite truth value independent of the compatible propositions measured together with it [BCG+22].

The theorem was announced by Specker in dimension three [Spe60], proved independently by J. S. Bell [Bel66] and by Kochen and Specker [KS67], and has since generated a large literature of increasingly efficient finite configurations and combinatorial proofs. Bell observed that the theorem follows from Gleason's theorem [Gle57] and also gave a direct continuity argument. Kochen and Specker instead exhibited a finite configuration of rays on which no noncontextual $\{0,1\}$-valuation can exist. Later derivations of finite obstructions from analytic forms of the theorem use compactness, explicitly or implicitly; see J. L. Bell [Bel96], Pitowsky [Pit98], and Hrushovski–Pitowsky [HP04].

From the standpoint of logic, however, these finite proofs use very little machinery. Once a Kochen–Specker configuration has been chosen, the contradiction is expressed by a finite family of Boolean constraints and verified by propositional inconsistency. The geometry and combinatorics of the configurations are subtle, but the logical framework in which the obstruction is certified is essentially truth-functional propositional logic. For nearly sixty years, refinements of the theorem have changed the configurations, dimensions, robustness estimates and combinatorics, while leaving this underlying logical level largely unchanged.

The purpose of the present paper is to change that level. We replace the propositional analysis by real-valued infinitary model theory and model-theoretic forcing. More precisely, we work in the $L_{\omega_1\omega}$ extension of Keisler's model theory for real-valued structures [Kei23, ADI26]. The logic allows countable suprema and infima, several interacting sorts, and quantitative predicates taking values in real intervals. These features let us place the Hilbert space, its lattice of closed subspaces, countable families of vectors and a quantum probability measure inside one formal theory, and then analyze the failure of a $\{0,1\}$-valued measure by a model existence game rather than by a fixed truth table.

This shift in logic reveals structure that the usual propositional encoding collapses. A classical Kochen–Specker graph identifies a ray across all contexts in which it appears: there is one vertex, and hence one truth value, for that ray. In the specified parity strategy, contextual term occurrences are recorded separately, together with commitments recording the geometric identities between their denotations. The equality of their probabilities is then not built into the graph but forced by the interaction between geometry and the measure. Likewise, the contradiction is reached from approximate interval commitments rather than from exact Boolean assignments, and the role of σ-additivity becomes visible as an infinitary mechanism that selects the finite-dimensional subspace in which the contradiction occurs.

The starting point is a countable theory T whose standard models are the closed unit balls of separable infinite-dimensional complex Hilbert spaces together with their lattices of closed subspaces and quantum probability measures. Solèr's representation theorem [Sol95], together with the coordinatization theorem of projective geometry, provides the lattice-theoretic part of the axiomatization. In the standard models, T recovers the setting of Gleason's theorem; this is stated precisely in Theorem A. The measure is named by a predicate μ taking values in $[0,1]$.

We then adjoin the single axiom saying that μ takes only the values 0 and 1. The resulting theory T^{01} has no models, standard or nonstandard; this is Theorem C. Its proof uses neither Gleason's theorem nor standardness. At the same time, every finite set of axioms in the displayed presentation of T^{01} has a model. Thus the inconsistency is genuinely infinitary: unlike an ordinary Kochen–Specker contradiction, it cannot be located in a finite subset of the axioms themselves.

The finite contradictions reappear at a different level, through a model existence game. One player, Olivia, maintains that the theory has no model; the other, Josh, attempts to build one by making finitely many interval commitments about the values of sentences and by answering challenges concerning witnesses, function values and countable suprema or infima. If Olivia follows a winning strategy, every play terminates after finitely many rounds at a finite position of commitments and function records that no structure satisfies. These terminal positions are the finite obstructions of the present paper. The forcing information carried by a play therefore records how an infinitary model-theoretic inconsistency is compressed into a finite contradiction. A commitment may concern an infinitary sentence; finiteness here refers to the number of commitments and records in the position.

The general game theorem applies to every winning strategy; the connection with a particular ray configuration is supplied by the strategy of Section 10.4. For that strategy, plays in which Josh makes all commitments required by the parity argument can be pictured through the eighteen-ray configuration of Cabello, Estebaranz and García-Alcaine [CEGA96]. The vertices index contextual term occurrences, and geometric axioms establish identities between their denotations. Axiom (M3), which makes the measure respect equality, transports probabilities across those identities. In the reduced system of Boolean occurrence variables and exactly-one basis constraints, all eighteen equalities are needed for the parity contradiction. The interval commitments also retain the rational tolerances under which this contradiction persists.

The strategy of Section 10.4 also displays the role of countable additivity in infinite dimension. In the models of finite axiom fragments constructed below, the measure vanishes on every finite-dimensional subspace. In the full theory, the countable-additivity scheme and all the coordinate specifications of the named orthonormal sequence force the hypothetical dispersion-free measure to give one basis ray value 1. Along the corresponding game strategy, approximate commitments single out a ray of value close to 1 and the four-dimensional block containing it. This localization belongs to the ray-based construction: on the full infinite-dimensional projection lattice, a fixed finite family of infinite-rank projections already gives a parity contradiction under finite additivity, as we explain in Section 12.

The refinement proved here is consequently not a new record for the size of a Kochen–Specker set. It is a refinement of the logical form of the theorem. The classical finite proof presents contextuality as a Boolean inconsistency on a fixed combinatorial configuration. Our analysis separates the geometric identifications, the quantitative probability commitments, the countable-additivity step and the finite contradiction, and organizes them inside a well-founded forcing tree. In this sense, the paper replaces a propositional certificate of impossibility by a model-theoretic analysis of how the impossibility is forced.

There is a second reason that the stronger logic matters. Because the probability predicate takes every value in $[0, 1]$ in the standard models, formulas can genuinely probe the full continuum of quantum probabilities. This raises a natural expressive-power question: which subsets of $[0, 1]$ can be singled out uniformly by formulas of the theory? We prove in Theorem 8.6 that the answer is exact: they are precisely the σ-projective subsets of $[0, 1]$, where σ-projective denotes the least class containing the Borel sets and closed under complements, countable unions and projections along $\mathbb{N}^{\mathbb{N}}$. The class strictly contains the projective sets. The classification is proved in ZFC; no determinacy or large-cardinal hypothesis is used. The novelty here is the exact correspondence between syntactic expressive power and a natural descriptive-set-theoretic pointclass.

The model-theoretic formulation separates the geometric identities of projection events from the probabilistic requirements imposed on a dispersion-free valuation. These dependencies enter the model existence game through its axioms, commitments and function records. In Section 12 we give the foundational interpretation after the mathematical development, distinguishing quantum probabilities from hidden-variable responses and relating the contextual presentations to earlier approaches.

Main results of the paper. The theory T speaks of three kinds of objects, which the logic keeps apart as sorts. The elements of the *Hilbert sort* are vectors; those of the *Solèr sort* carry the operations of an ortholattice. In the intended interpretation the elements of the Solèr sort are closed subspaces, identified, as above for $L(H)$, with the orthogonal projections onto them. Those of the *sequence sort* are sequences of vectors, through which the completeness

of the lattice becomes expressible. The vocabulary has a function symbol for the orthogonal projection of a vector onto an element of the Solèr sort. A vector belongs to an element of the Solèr sort when the projection onto that element leaves the vector fixed. The predicate μ assigns a probability to every element of the Solèr sort. Like every countable theory that has a model, the theory T has models whose sorts are countable. The theory also has models whose Hilbert sort and sequence sort are the full closed unit balls of Hilbert spaces. The theorems below distinguish the conclusions that hold in all models from those that require models of this second kind, which we now name. We call a model *standard* when its Hilbert and sequence sorts are the full closed unit balls of complex Hilbert spaces with their own operations, inner products and norms (see Definition 5.3 below). In a nonstandard model, the Hilbert sort lacks some of the vectors of such a ball, or contains distinct elements that no inner product or norm distinguishes, or the sequence sort is not a ball of this kind. This use of the term "nonstandard" is unrelated to nonstandard analysis. Gleason's theorem [Gle57] concerns the quantum probability measures on the lattice of closed subspaces of a separable Hilbert space of dimension at least three. It asserts that these measures are exactly the maps that send each closed subspace to $\mathrm{Tr}(\rho P)$, where P is the orthogonal projection onto the subspace and where ρ is a density operator. A density operator, that is, a positive operator of trace 1, represents a state of the quantum system. In the standard models, the theory T captures the setting of Gleason's theorem for separable infinite-dimensional complex Hilbert spaces. The point of the first theorem is therefore semantic as well as representational: it shows that the new formal language does not merely imitate the Hilbert-space picture, but recovers exactly the intended quantum-probability structures in its standard models.

Theorem A (Gleason's theorem in real-valued $L_{\omega_1\omega}$). *There is a countable theory T of real-valued $L_{\omega_1\omega}$ satisfying the following two properties.*

(1) In every standard model of T, the Hilbert sort is the closed unit ball of a separable infinite-dimensional complex Hilbert space H, and the map $a \mapsto S_a$, where S_a is the closed subspace of H spanned by the vectors of the Hilbert sort that belong to a, induces an isomorphism of ortholattices from the quotient of the Solèr sort by its equality, the relation $[a = b] = 1$ of Section 5.3, onto the lattice $L(H)$ of closed subspaces of H; moreover, there is a unique density operator ρ on H such that $\mu(a) = \mathrm{Tr}(\rho P_a)$ for every element a of the Solèr sort, where P_a is the orthogonal projection onto S_a.

(2) Conversely, for every separable infinite-dimensional complex Hilbert space H and every density operator ρ on H, there is a standard model of T whose Hilbert sort is the closed unit ball of H, whose Solèr sort is $L(H)$, and in which $\mu(a) = \mathrm{Tr}(\rho P_a)$ for every $a \in L(H)$, where P_a is the orthogonal projection onto a.

In every separable infinite-dimensional Hilbert space, the probabilities that a quantum state assigns to the closed subspaces fill the whole interval $[0, 1]$, even when the state is pure (see Lemma 8.4 below). Thus the probability predicate is not merely a Boolean remnant inside the new formalism: in the intended models it ranges over the entire continuum. This makes the range theorem the starting point for the expressive-power question answered

by Theorem 8.6: which subsets of that continuum can formulas of the theory distinguish uniformly?

Theorem B (The range of the measure). *In every standard model of T, the range of μ is the whole interval $[0,1]$. Consequently, for every Borel set $B \subseteq \mathbb{R}$ that meets $[0,1]$, the theory T, together with an axiom saying that no value of μ lies in B (see Lemma 5.2), has no standard models.*

The conclusion of Theorem B reaches well beyond the Borel sets. A subset of $[0,1]$ is *definable over* T when some formula of the logic, with one free variable a of the Solèr sort, singles it out as follows (see Definition 8.5 below). In every standard model of T and at every element a of its Solèr sort, the formula takes the value 1 if $\mu(a)$ lies in the set and the value -1 otherwise. Because formulas may quantify over the Solèr sort, and because the measure takes every value in $[0,1]$, the definable sets form a much larger class than the Borel sets. In Theorem 8.6 below we show that the definable sets are exactly the σ-projective subsets of $[0,1]$. The σ-projective sets are the members of the least class of subsets of Polish spaces that contains the Borel sets and is closed under complements, countable unions and projections along $\mathbb{N}^{\mathbb{N}}$. This class contains every projective set, and the example following Theorem 8.6 shows that the containment is strict. Consequently, the hypothesis that the probabilities avoid a given nonempty σ-projective subset of $[0,1]$ can be written as an axiom of the logic. It follows as well that the theory T together with that axiom has no standard models. The exact σ-projective classification, including the strict extension beyond the projective sets, is proved in ZFC.

The hypothesis that every probability is 0 or 1 can be written as a single axiom of the logic, displayed in Theorem C below. In that axiom, the symbol $\mathbf{1}$ denotes the sentence whose value is 1 in every structure, and for formulas φ and ψ the expression $\varphi \wedge \psi$ denotes the minimum of their values. Our next theorem shows that the axiom fails not only in the standard models but in every model of T.

Theorem C (The Bell–Kochen–Specker theorem in real-valued $L_{\omega_1\omega}$). *The theory T of Theorem A, together with the axiom $\mu(a) \wedge (\mathbf{1} - \mu(a)) \leq 0$, which, since the axioms of T confine μ to $[0,1]$, says that μ takes only the values 0 and 1, has no models, standard or not.*

The proof of Theorem C, given in Section 9, uses neither Gleason's theorem nor the standardness of the model. The inconsistency it establishes is genuinely infinitary, for every finite set of axioms of the theory of Theorem C has a model. The finite obstructions of this inconsistency are therefore not finite sets of axioms but finite sets of commitments on the values of sentences. These obstructions stand at the leaves of the well-founded trees of the game described above. The branching of these trees shows where the countable suprema and infima of the axioms enter. The rules of the game apply to every theory whose axioms are *closed conditions*, that is, assertions $\varphi \leq r$ or $\varphi \geq r$ about sentences φ, with r rational. An axiom with free variables takes this form once its variables are bound by suprema or infima, and the axioms of T are all of this kind. The next theorem isolates the general game-theoretic mechanism used later in the quantum application: from model-theoretic inconsistency one

obtains a well-founded tree whose leaves are finite inconsistent interval commitments, and whose infinite branching records precisely the infinitary operations of the logic.

Theorem D (Finite obstructions). *Let T' be a countable theory of real-valued $L_{\omega_1\omega}$ that consists of closed conditions and has no models. In the model existence game of Section 10, the player who claims that T' has no models has a winning strategy. For every such strategy, the tree of plays in which that player follows it is well-founded; at each leaf, the commitments made along the play, each of which says that the value of a sentence lies in a closed interval with rational endpoints, form a finite set that is satisfied in no structure. The tree branches infinitely only at the moves that concern countable suprema and infima; once the other player's answers to those moves are fixed, the plays have bounded length. For the theory of Theorem C, every finite set of axioms has a model; consequently, the tree of every winning strategy for that theory branches infinitely.*

The principal dependencies among the results are displayed in Figure 1. The representation and definability results concern standard models, whereas the dispersion-free inconsistency holds in every model. The general model existence theorem supplies the passage from that inconsistency to well-founded trees of finite obstructions.

Organization of the paper. The background occupies Sections 1–4 and is drawn from lattice theory, the theory of hermitian forms and projective geometry. Its principal sources are MacNeille completions [Mac37], the coordinatization of projective geometries [Bae52, MM70, Hol95], and the theorems on orthomodular spaces due to Amemiya and Araki [AA66] and to Solèr [Sol95]. From these sources we obtain the Solèr lattices, which up to isomorphism are the lattices of closed subspaces of infinite-dimensional real, complex and quaternionic Hilbert spaces. The same sources yield the pre-Solèr lattices, in which every defining condition of a Solèr lattice holds except possibly completeness. In the standard models of T, the quotient of the Solèr sort by its equality is a Solèr lattice. By contrast, in the standard models of the part of T that mentions neither the sequence sort nor the measure, this quotient need only be a pre-Solèr lattice. The logic and the theory T are presented in Sections 5 and 6. The proofs of Theorems A, B and C occupy Sections 7–9, together with that of Theorem 8.6, which identifies the sets of values that formulas define in the standard models. The model existence game, whose analysis rests on the determinacy theorem of Gale and Stewart [GS53], is the subject of Section 10, which also contains the proof of Theorem D. The reader may omit Section 4 and the remarks of Section 11 on a first reading. In Section 12 we compare the finite obstructions with the configurations of Kochen and Specker. The comparison bears on what a vertex is, the identifications and the axioms from which they follow, the tolerances of the intervals and the location of the configuration. The comparison also bears on the role of σ-additivity for configurations of rays in infinite dimension and on the topological property of the logic behind each kind of certificate. In Section A we relate the game to Keisler's model-theoretic forcing [Kei73] and compare it with the treatment of the Henkin construction in the space of Henkin types [ADI26].

1. Lattices

Birkhoff and von Neumann [BvN36] observed that the experimentally verifiable propositions about a quantum system correspond to the closed subspaces of a Hilbert space. These

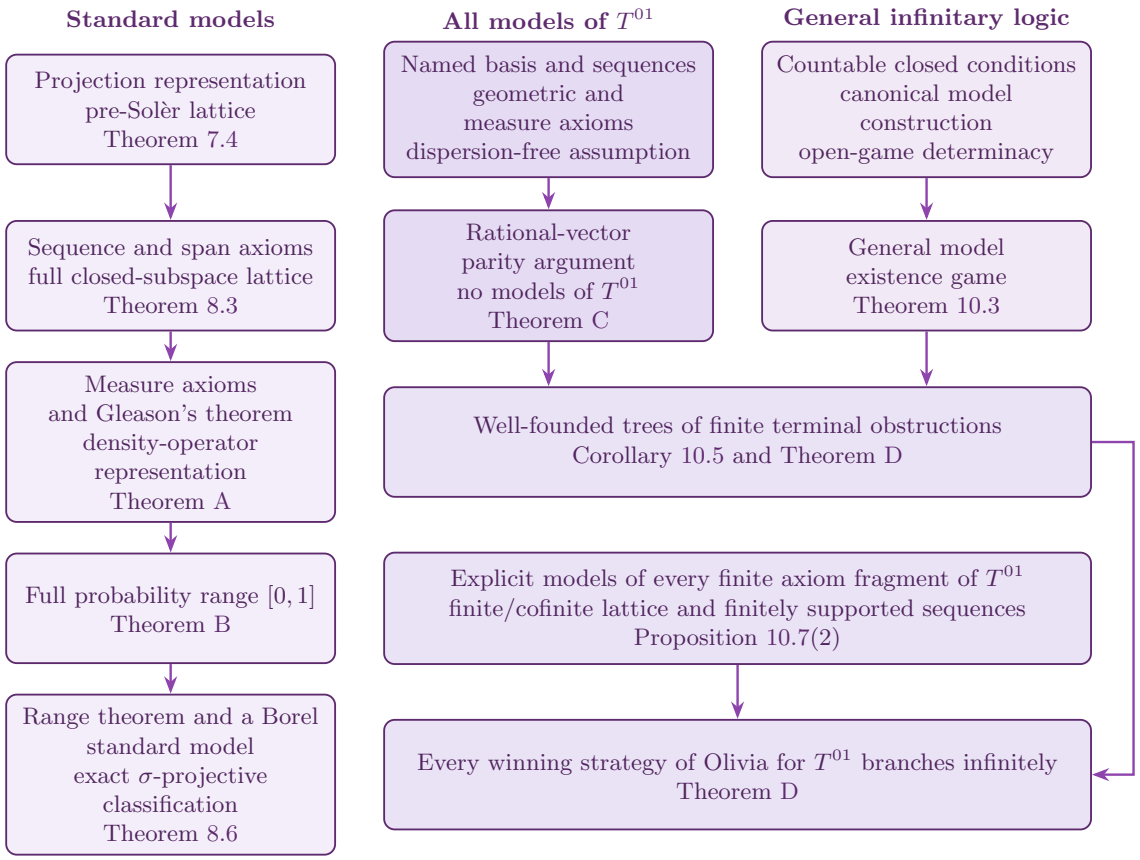


FIGURE 1. Selected proof dependencies. Arrows indicate the use of one result in another; additional inputs are named in the boxes. The left column concerns standard models. The inconsistency in the centre concerns all models and is proved by the parity argument independently of Gleason's theorem. The general game theorem applies to countable theories of closed conditions; together with the finite-fragment models, it accounts for infinite branching in every winning strategy for T^{01}.

subspaces, they further observed, form a lattice in which the distributive law fails; the logic of quantum mechanics is therefore not that of a Boolean algebra. In place of the distributive law the two authors proposed the weaker modular law, which holds in finite dimensions but fails for the closed subspaces of an infinite-dimensional Hilbert space. The law that holds in every dimension is the still weaker orthomodular law. In this section we collect the notions from lattice theory that we need: orthocomplementation and orthomodularity; atoms together with the covering property and irreducibility; and the MacNeille completion. This completion supplies the joins and meets that an incomplete lattice lacks.

Definition 1.1. A *partially ordered set* is a set with a reflexive, antisymmetric and transitive relation $\leq$. For a subset S of a partially ordered set, an *upper bound* of S is an element b with $a \leq b$ for all $a \in S$, and the *join* $\bigvee S$ of S is its least upper bound, when it exists; lower bounds and the *meet* $\bigwedge S$ are defined dually. A *lattice* is a partially ordered set in which every pair of elements a, b has a join $a \vee b$ and a meet $a \wedge b$. A lattice is *bounded* when it has a least element 0 and a greatest element 1, and *complete* when every subset has a join.

In a lattice, the order can be recovered from the meet: for all elements a and b, the relation $a \leq b$ holds if and only if $a \wedge b = a$. It is in this form that the axioms of Section 6 express the order. Completeness, likewise, can be stated through meets: a lattice is complete if and only if every subset has a meet. Indeed, when every subset has a join, the meet of a subset is the join of its lower bounds; the converse follows dually.

Definition 1.2. An *orthocomplementation* on a bounded lattice Λ is a map $a \mapsto a^\perp$ from Λ to Λ such that, for all $a, b \in \Lambda$, the equations $a^{\perp\perp} = a$, $a \wedge a^\perp = 0$ and $a \vee a^\perp = 1$ hold, and the inequality $a \leq b$ implies $b^\perp \leq a^\perp$. A bounded lattice with an orthocomplementation is an *ortholattice*. Two elements a, b of an ortholattice are *orthogonal*, written $a \perp b$, when $a \leq b^\perp$. An ortholattice is *orthomodular* when, for all elements a and b, the inequality $a \leq b$ implies $b = a \vee (b \wedge a^\perp)$.

The lattice $L(H)$ of closed subspaces of a Hilbert space H is ordered by inclusion and orthocomplemented by $M \mapsto M^\perp$. In this lattice, orthomodularity says that every closed subspace N containing a closed subspace M is the orthogonal sum of M and $N \cap M^\perp$. Orthomodular or not, every ortholattice satisfies the De Morgan law $(\bigvee S)^\perp = \bigwedge\{a^\perp : a \in S\}$ for every subset S whose join exists, since the map $a \mapsto a^\perp$ is an order-reversing bijection that is its own inverse. It is through this law that orthocomplements of joins are computed below, in the MacNeille completion and in the coordinatization theorem of Section 3.

Definition 1.3. An *atom* of a bounded lattice is a nonzero element p such that every element b with $b \leq p$ is equal to 0 or to p. A bounded lattice is *atomistic* when every element a is the join of the atoms below it (this join being required to exist). It has the *covering property* when, for every atom p and every element a with $p \not\leq a$, no element lies strictly between a and $a \vee p$. It is *irreducible* when, for any two distinct atoms p and q, some atom r distinct from p and q lies below $p \vee q$.

For complete orthomodular lattices that are atomistic and have the covering property, irreducibility in this sense has an equivalent formulation through direct products (see Maeda and Maeda [MM70] and Holland [Hol95, §4]). Such a lattice is irreducible in this sense exactly when it admits no decomposition as a direct product of two lattices, each with more than one element. We adopt the definition above because it is the form in which irreducibility enters the coordinatization theorem (see Theorem 3.3).

Definition 1.4. A *sub-ortholattice* of an ortholattice Λ is a subset that contains 0 and 1 and is closed under the operations $\vee$, $\wedge$ and $^\perp$ of Λ. An *embedding* of ortholattices is an injective map that preserves $\vee$, $\wedge$, $^\perp$, 0 and 1.

Among the pre-Solèr lattices of Section 3 are the Solèr sorts of the standard models of the part of T that mentions neither the sequence sort nor the measure. Pre-Solèr lattices, these Solèr sorts included, need not be complete (see Example 7.5). Suppose that a pre-Solèr lattice embeds in the lattice of orthogonally closed subspaces of a hermitian space, the atoms of the pre-Solèr lattice being carried onto the rays of the space. In Theorems 4.1 and 7.4 below we show that the lattice of orthogonally closed subspaces is then the MacNeille completion of the embedded lattice. This completion, which we construct presently, is characterized by a notion of density that concerns the order alone; no topology is involved. Let Γ be a complete lattice, and let Λ be a subset of Γ. We say that Λ is *join-dense* in Γ when every $g \in \Gamma$ satisfies $g = \bigvee\{a \in \Lambda : a \leq g\}$, the join being computed in Γ, and that Λ is *meet-dense* in Γ when every $g \in \Gamma$ satisfies $g = \bigwedge\{a \in \Lambda : a \geq g\}$. For instance, the set $\mathbb{Q}$ of rational numbers is join-dense and meet-dense in the extended real line $[-\infty, \infty]$.

A join-dense and meet-dense subset determines, up to isomorphism, the complete lattice that contains it (see Proposition 1.5 below), and that lattice can therefore be built from

the subset alone. The construction, due to MacNeille [Mac37], is as follows. For a partially ordered set Λ and a subset X of Λ, let X^u and X^l be the sets of upper and lower bounds of X in Λ. We call X a *cut* when $X = (X^u)^l$. Ordered by inclusion, the cuts form a complete lattice $\mathrm{DM}(\Lambda)$, the *MacNeille completion* of Λ. Into this lattice the map $a \mapsto \{b \in \Lambda : b \leq a\}$ embeds Λ, preserving all joins and meets that exist in Λ. The letters DM recall the other name of the completion, the Dedekind–MacNeille completion.

When Λ is an ortholattice, its orthocomplementation induces one on the cuts. For a cut X, let $X^\perp$ be the set $\{b \in \Lambda : b \leq a^\perp \text{ for every } a \in X\}$ of lower bounds of $\{a^\perp : a \in X\}$. The set $X^\perp$, as a set of lower bounds, is a cut, and the map $X \mapsto X^\perp$ reverses inclusion. An element b lies in $X^\perp$ exactly when $b^\perp$ is an upper bound of X, whence $X^{\perp\perp} = (X^u)^l = X$. Moreover, the intersection $X \cap X^\perp$ is $\{0\}$, because an element b with $b \leq b^\perp$ satisfies $b = b \wedge b^\perp = 0$. Since $X \mapsto X^\perp$ is an order-reversing bijection that is its own inverse, it carries the meet $X \cap X^\perp$ to the join of $X^\perp$ and X, which is therefore $\{0\}^\perp = \Lambda$. Thus $X \mapsto X^\perp$ is an orthocomplementation of $\mathrm{DM}(\Lambda)$; it extends that of Λ, since it sends the cut $\{b \in \Lambda : b \leq c\}$ to the cut $\{b \in \Lambda : b \leq c^\perp\}$, and with it we regard $\mathrm{DM}(\Lambda)$ as an ortholattice.

Proposition 1.5 (Banaschewski [Ban56], Schmidt [Sch56]). *Let Λ be a partially ordered set, and let Γ be a complete lattice that contains Λ as a join-dense and meet-dense subset, the order of Γ inducing on Λ the order of Λ. Then the map that sends each $g \in \Gamma$ to the set $\{a \in \Lambda : a \leq g\}$ is an isomorphism of Γ onto $\mathrm{DM}(\Lambda)$ that restricts on Λ to the canonical embedding $a \mapsto \{b \in \Lambda : b \leq a\}$. If Λ is an ortholattice, then the lattice Γ has at most one orthocomplementation that extends that of Λ; any such orthocomplementation is given by $g^\perp = \bigwedge\{a^\perp : a \in \Lambda,\ a \leq g\}$, and the isomorphism carries it to the orthocomplementation $X \mapsto X^\perp$ of $\mathrm{DM}(\Lambda)$.*

Proof. For $g \in \Gamma$, write $\Lambda_{\leq g} = \{a \in \Lambda : a \leq g\}$; we show that each $\Lambda_{\leq g}$ is a cut and that the map $g \mapsto \Lambda_{\leq g}$ is an order embedding whose image contains every cut. The assertions on orthocomplementations then follow from the De Morgan law. By join-density, we have $g = \bigvee \Lambda_{\leq g}$; hence the upper bounds of $\Lambda_{\leq g}$ in Λ are the elements $b \in \Lambda$ with $b \geq g$. By meet-density, the common lower bounds of these elements are the elements $a \in \Lambda$ with $a \leq g$; thus $\Lambda_{\leq g}$ is a cut. For $g, g' \in \Gamma$, join-density also shows that $g \leq g'$ if and only if $\Lambda_{\leq g} \subseteq \Lambda_{\leq g'}$; hence the map is an order embedding. Because the order of Γ induces that of Λ, the map restricts on Λ to the canonical embedding. If X is a cut and $g = \bigvee X$ in Γ, then the cut X is contained in $\Lambda_{\leq g}$. Conversely, an element $a \in \Lambda$ with $a \leq g$ lies below every $b \in X^u$ because each such b is an upper bound of X in Γ, whence $a \in X = (X^u)^l$. Thus $X = \Lambda_{\leq g}$, and the map is onto $\mathrm{DM}(\Lambda)$. The formula for $g^\perp$ follows from $g = \bigvee \Lambda_{\leq g}$ and the De Morgan law. By that formula, an element $b \in \Lambda$ lies below $g^\perp$ exactly when $b \leq a^\perp$ for every $a \in \Lambda_{\leq g}$; hence $\Lambda_{\leq g^\perp} = (\Lambda_{\leq g})^\perp$. ☻

⌛ **Historical note.** Birkhoff and von Neumann rejected the lattice of closed subspaces of an infinite-dimensional Hilbert space as the logic of quantum mechanics. Letters that von Neumann wrote to Birkhoff in 1935, while the two were preparing their paper of 1936, show why: that lattice is not modular. Von Neumann, however, wanted the probabilities of quantum propositions to behave like a dimension function (that is, like a strictly increasing valuation). Only a modular lattice carries such a function [Réd07]. In the paper itself, Birkhoff and von Neumann postulated instead an abstract projective geometry, whose lattice of subspaces is modular [BvN36]. A year later Husimi observed that the closed subspaces of a Hilbert space satisfy the orthomodular law [Hus37], which was rediscovered several times in the 1950s and

1960s. In his thesis of 1964, Piron took completeness, orthomodularity, atoms and the covering property as axioms for the propositions about a quantum system. He showed that the irreducible lattices with these properties are represented by hermitian forms over a division ring when their dimension is at least four [Pir64].

2. Hermitian spaces

The coordinatization theorem, which we state in Section 3, turns the atoms of a suitable ortholattice into the one-dimensional subspaces of a vector space over a division ring. Under the same theorem, the orthocomplementation of the ortholattice becomes orthogonality with respect to a hermitian form on that vector space. In this section we collect the facts about such spaces that we need. The most important of these facts, on which Sections 3 and 4 depend, is that orthomodularity of the lattice of orthogonally closed subspaces is equivalent to a property of the space itself. This property asks that every orthogonally closed subspace and its orthogonal complement have the whole space as their sum (see Proposition 2.8).

Definition 2.1. A *division ring* is a ring with $1 \neq 0$ in which every nonzero element is invertible; multiplication need not be commutative. An *involution* of a division ring Δ is a map $\alpha \mapsto \alpha^*$ such that $(\alpha+\beta)^* = \alpha^* + \beta^*$, $(\alpha\beta)^* = \beta^*\alpha^*$ and $\alpha^{**} = \alpha$ for all $\alpha, \beta \in \Delta$. Let E be a left vector space over Δ, that is, a left Δ-module.[1] A *hermitian form* on E is a map $\langle \cdot \mid \cdot \rangle \colon E \times E \to \Delta$ such that $\langle \alpha x + \beta y \mid z \rangle = \alpha\langle x \mid z\rangle + \beta\langle y \mid z\rangle$ and $\langle y \mid x\rangle = \langle x \mid y\rangle^*$ for all $x, y, z \in E$ and $\alpha, \beta \in \Delta$; it is *anisotropic* when $\langle x \mid x\rangle \neq 0$ for every nonzero vector $x \in E$. A *hermitian space* is a left vector space over a division ring with an involution, together with an anisotropic hermitian form.

It follows from these identities that $\langle x \mid \alpha y\rangle = \langle x \mid y\rangle\alpha^*$ for all $x, y \in E$ and $\alpha \in \Delta$. The bracket is linear in its first argument, as is customary in mathematics; in the Dirac notation used by physicists it is linear in the second. Nothing in the definition forces the scalars to be real or complex: over $\mathbb{Q}$ with the identity involution, the form $\sum_n x_n y_n$ on finitely supported rational sequences $x = (x_n)_{n\in\mathbb{N}}$ and $y = (y_n)_{n\in\mathbb{N}}$ is anisotropic. The anisotropy of this form rests on the fact that a sum of rational squares vanishes only when every square does.

Definition 2.2. In a hermitian space E, two vectors x and y are *orthogonal*, written $x \perp y$, when $\langle x \mid y\rangle = 0$. For $S \subseteq E$, let $S^\perp$ be the set $\{x \in E : \langle x \mid y\rangle = 0 \text{ for all } y \in S\}$. A subspace M is *orthogonally closed* when $M^{\perp\perp} = M$.

Orthogonality is symmetric, since $\langle y \mid x\rangle = \langle x \mid y\rangle^*$ and the involution fixes 0. In particular, every subset S of E satisfies $S \subseteq S^{\perp\perp}$.

Definition 2.3 (The lattice $\mathcal{C}(E)$). For a hermitian space E, we write $\mathcal{C}(E)$ for the set of orthogonally closed subspaces of E, ordered by inclusion.

Every set of the form $S^\perp$ is an orthogonally closed subspace: applying the inclusion-reversing operation $^\perp$ to $S \subseteq S^{\perp\perp}$ gives $S^{\perp\perp\perp} \subseteq S^\perp$, and the reverse inclusion is the same fact for $S^\perp$. If $(M_j)_{j\in J}$ is a family of orthogonally closed subspaces indexed by an arbitrary set J, then their intersection, which equals $\bigcap_j M_j^{\perp\perp} = \big(\bigcup_j M_j^\perp\big)^\perp$, is orthogonally closed. Hence $\mathcal{C}(E)$ is a complete lattice with least element $\{0\}$ and greatest element E. In this lattice the family has meet $\bigcap_j M_j$ and join $\big(\sum_j M_j\big)^{\perp\perp}$, which may be strictly larger than $\sum_j M_j$. For $M \in \mathcal{C}(E)$, anisotropy gives $M \cap M^\perp = \{0\}$, whence $M \vee M^\perp = (M^\perp \cap M)^\perp = E$. Since

[1] Physicists usually let quaternionic scalars act on the right; their quaternionic Hilbert spaces are therefore right modules. The choice of side is immaterial: a left vector space over Δ is a right vector space over the opposite division ring, and an involution is an isomorphism of Δ onto its opposite.

the map $M \mapsto M^\perp$ also reverses inclusion and is its own inverse on $\mathcal{C}(E)$, the lattice $\mathcal{C}(E)$ is, with this map, a complete ortholattice.

Definition 2.4. A hermitian space E is *orthomodular* when $M + M^\perp = E$ for every $M \in \mathcal{C}(E)$, where $M + M^\perp$ is the set of sums $v + y$ with $v \in M$ and $y \in M^\perp$.

The lattice join $M \vee M^\perp$ is always E; orthomodularity asks that the algebraic sum be E as well.

Definition 2.5. A *Hilbert space* is a hermitian space H over the real numbers $\mathbb{R}$, the complex numbers $\mathbb{C}$ or the quaternions $\mathbb{H}$, with the usual involution (the identity, complex conjugation or quaternionic conjugation), such that $\langle x \mid x\rangle > 0$ for every nonzero vector x and the space is complete in the norm $\|x\| = \langle x \mid x\rangle^{1/2}$.

Let H be a Hilbert space. By the projection theorem, every closed subspace M of H satisfies $M + M^\perp = H$, and every subspace M satisfies $M^{\perp\perp} = \overline{M}$. Hence the orthogonally closed subspaces of H are exactly its closed subspaces, which form the lattice $L(H) = \mathcal{C}(H)$. The space H is orthomodular, and so, by Proposition 2.8 below, is the lattice $L(H)$.

Lemma 2.6. *In a hermitian space E, every finite-dimensional subspace F is orthogonally closed and satisfies $F + F^\perp = E$.*

Proof. The idea is to write down the orthogonal projection onto F. Let n be the dimension of F. By anisotropy, the Gram–Schmidt process gives F an orthogonal basis $w_1, \dots, w_n$ with $\langle w_i \mid w_i\rangle \neq 0$ for $i = 1, \dots, n$. For $z \in E$, the vector $Pz = \sum_i \langle z \mid w_i\rangle \langle w_i \mid w_i\rangle^{-1} w_i$ lies in F, and the difference $z - Pz$ lies in $F^\perp$; hence $F + F^\perp = E$. If $y \in F^{\perp\perp}$, the vector $y - Py$ lies in $F^\perp \cap F^{\perp\perp}$ and is therefore orthogonal to itself; by anisotropy it is 0, whence $y \in F$. ☺

Lemma 2.7. *In a hermitian space E over a division ring Δ, the subspace $M + \Delta x$ belongs to $\mathcal{C}(E)$ for every $M \in \mathcal{C}(E)$ and every $x \in E$.*

Proof. If $x \in M$, there is nothing to prove; we therefore assume that $x \notin M$. Since $M + \Delta x$ is contained in $(M + \Delta x)^{\perp\perp}$, it suffices to show that every vector of $(M + \Delta x)^{\perp\perp}$ differs from a multiple of x by a vector of M. The idea is that such a vector induces on $M^\perp$, through the form, a functional obtained from the one induced by x by multiplication on the right by a scalar. Put $W = M^\perp$, and let ξ_x be the functional $z \mapsto \langle z \mid x\rangle$ on W. This functional is nonzero, since $x \notin W^\perp = M$, and we may choose $z_0 \in W$ with $\xi_x(z_0) = 1$. Let y be an element of $(M + \Delta x)^{\perp\perp}$, and let ξ_y be the functional $z \mapsto \langle z \mid y\rangle$ on W. Since $W \cap x^\perp = (M + \Delta x)^\perp$, the functional ξ_y vanishes on the kernel of ξ_x; as $z - \xi_x(z) z_0$ lies in that kernel for every $z \in W$, we obtain $\xi_y(z) = \xi_x(z)\alpha$ with $\alpha = \xi_y(z_0)$. Because $\langle z \mid \alpha^* x\rangle = \langle z \mid x\rangle \alpha$, the vector $y - \alpha^* x$ is orthogonal to W and therefore lies in $W^\perp = M$, whence $y \in M + \Delta x$. ☺

Proposition 2.8. *For a hermitian space E, the lattice $\mathcal{C}(E)$ is orthomodular if and only if E is an orthomodular space.*

Proof. Suppose first that E is orthomodular, and let $M \subseteq M'$ be elements of $\mathcal{C}(E)$. Every vector $z \in M'$ can be written as $v + y$ with $v \in M$ and $y \in M^\perp$; since $y = z - v$ lies in M', we have $M' = M + (M' \cap M^\perp) \subseteq M \vee (M' \wedge M^\perp) \subseteq M'$, which is the orthomodular law. Conversely, suppose that $\mathcal{C}(E)$ is orthomodular, and let Δ be the division ring of scalars of E. Let M be an element of $\mathcal{C}(E)$, and let x be a vector of $E \setminus M$. By Lemma 2.7, the subspace $M' = M + \Delta x$ belongs to $\mathcal{C}(E)$, and orthomodularity gives $M' = M \vee (M' \cap M^\perp)$. Since $M' \neq M$, the subspace $M' \cap M^\perp$ is nonzero. As this subspace meets M only in 0, and as M has codimension one in M', the subspace $M' \cap M^\perp$ is one-dimensional, spanned by a vector u. Then $M \subsetneq M + \Delta u \subseteq M'$, and the codimension count gives $M' = M + \Delta u$; hence $x \in M + M^\perp$. ☺

Historical note. In 1964 Piron published a theorem asserting that every orthomodular complex inner product space is complete, but his proof contained an error [Pir64]. Amemiya and Araki gave a correct proof in 1966, and their method also covers real and quaternionic scalars [AA66]. It was then speculated that, among infinite-dimensional hermitian spaces over arbitrary division rings, orthomodularity alone characterizes the Hilbert spaces over $\mathbb{R}$, $\mathbb{C}$ and $\mathbb{H}$. Keller ended the speculation in 1980 with a "non-classical Hilbert space", an infinite-dimensional orthomodular hermitian space over a division ring other than these three [Kel80]. Keller was a student of Herbert Gross, whose papers and book on forms in infinite-dimensional vector spaces shaped the theory; the story is told in Holland's survey of 1995 [Hol95].

3. Projective geometry, Solèr lattices and pre-Solèr lattices

In this section we explain how an ortholattice with a few combinatorial properties can be represented by orthogonally closed subspaces of a hermitian space. A hermitian space, we recall, is a vector space over a division ring with an anisotropic hermitian form. We also explain how Solèr's condition, together with completeness, forces the scalars to be real, complex or quaternionic. The ortholattices that have the combinatorial properties in question, Solèr's condition and completeness are the Solèr lattices; those that have them all except possibly completeness are the pre-Solèr lattices. In the standard models of our theory, completeness comes from the sequences of vectors placed beside the lattice rather than from axioms about the lattice alone (see Section 5 and Theorem 8.3). Pre-Solèr lattices therefore recur throughout the paper. We assume familiarity with Hilbert spaces, but none with projective geometry.

3.1. **Points and lines.** In a complex Hilbert space H, the atoms of the lattice $L(H)$ of closed subspaces are the *rays*, the one-dimensional subspaces $\mathbb{C}x$ spanned by nonzero vectors $x \in H$. The join of two distinct rays is the two-dimensional subspace, or *plane*, that they span. More generally, we call the one-dimensional subspaces of any vector space its *rays*. Which rays lie in which planes is information about the lattice alone (see Figure 2). For a vector space of dimension at least four, this information determines the space and its division ring of scalars, in the sense made precise by Theorem 3.2 below.

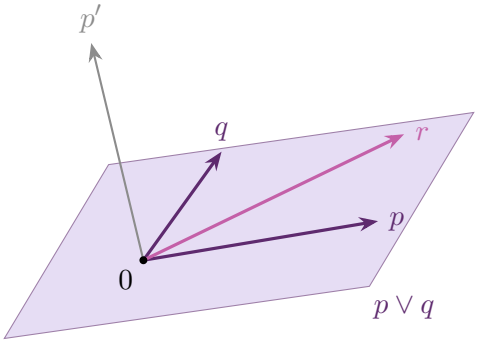


Figure 2. In the lattice $L(H)$ of closed subspaces of a complex Hilbert space H, the atoms are the rays. The join $p \vee q$ of two rays p and q is the plane they span; the ray r in that plane is an atom below $p \vee q$, while the ray p' is not.

Definition 3.1. A *projective geometry* is a set of *points* together with a family of subsets, called *lines*, such that:

(PG1) any two distinct points lie on exactly one line;
(PG2) every line contains at least three points;
(PG3) (*the Veblen–Young axiom*) if a, b, a', b' are distinct points and the line through a and b meets the line through a' and b', then the line through a and a' meets the line through b and b'.

A set of points is a *projective subspace* when it contains the whole line through any two of its points. The projective subspace *generated* by a set of points is the intersection of all projective subspaces containing it. A *projective plane* is the projective subspace generated by three points not on one line. Axiom (PG3) says that two lines lying in a common projective plane always meet (see Figure 3); a projective geometry has no parallel lines. The word *projective* comes from perspective drawing, in which parallel rails, projected onto the canvas, meet at the horizon.

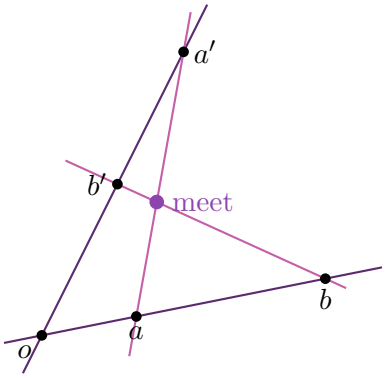


FIGURE 3. The Veblen–Young axiom. The lines ab and $a'b'$ meet at o; then the lines aa' and bb' meet as well.

The basic example comes from linear algebra. Let V be a left vector space of dimension at least two over a division ring Δ. Take as points the rays of V, and as lines the sets of rays contained in a two-dimensional subspace. Axiom (PG1) holds because two distinct rays span a unique two-dimensional subspace. Axiom (PG2) holds because the two-dimensional subspace spanned by linearly independent vectors x and y contains the three distinct rays Δx, Δy and $\Delta(x + y)$. For (PG3), suppose that the two-dimensional subspaces spanned by a, b and by a', b' have a ray in common. In that case the rays a, b, a', b' lie in a subspace of dimension at most three. By counting dimensions, any two two-dimensional subspaces of that subspace have a ray in common, in particular those spanned by a, a' and by b, b'.

Points are *independent* when none of them lies in the projective subspace generated by the others. In the basic example, rays are independent exactly when nonzero vectors spanning them, one in each ray, are linearly independent. Two projective geometries are *isomorphic* when there is a bijection between their sets of points under which the lines of the one correspond exactly to the lines of the other. Let E and E' be vector spaces over division rings Δ and Δ', respectively. A *semilinear isomorphism* from E onto E' is an additive bijection $f\colon E \to E'$ for which there is an isomorphism $\sigma\colon \Delta \to \Delta'$ with $f(\alpha x) = \sigma(\alpha) f(x)$ for all $\alpha \in \Delta$ and $x \in E$. The following theorem says that the basic example is, up to isomorphism, the only one as soon as there are four independent points.

Theorem 3.2 (Coordinatization of projective geometries; see Baer [Bae52]). *Every projective geometry that contains four independent points is isomorphic to the geometry of*

rays of a left vector space E over a division ring Δ. The pair (Δ, E) is unique up to a semilinear isomorphism.

The division ring Δ is thus recovered from the geometry alone, whatever it may be: every division ring occurs. Among the division rings so recovered are the rational numbers, the non-archimedean ordered fields (fields with infinitesimals, such as fields of formal Laurent series) and the quaternions. The hypothesis of four independent points, on the other hand, is indispensable: a projective plane regarded as a projective geometry in its own right need not come from a division ring. The planes that do are exactly those in which Desargues's theorem holds (if two triangles are in perspective from a point, the three intersection points of corresponding sides lie on one line). Planes in which Desargues's theorem fails, the *non-Desarguesian* planes, exist. In a geometry with four independent points, by contrast, that theorem follows from the axioms.

3.2. **From lattices to geometries.** To apply Theorem 3.2 to an orthomodular lattice that is atomistic and has the covering property, we take the atoms of the lattice as the points of a geometry. The lines of this geometry are the sets $\{r : r \text{ an atom with } r \leq p \vee q\}$ for distinct atoms p and q; in $L(H)$, these sets consist of the rays in the plane spanned by p and q. The covering property gives this geometry (PG1) and, with orthomodularity, (PG3); irreducibility in the sense of Definition 1.3 is exactly (PG2) (see Maeda and Maeda [MM70, §§33–34] and Holland [Hol95, §4]).

Orthocomplementation adds a relation between points: by the definition in Section 1, atoms p and q are orthogonal when $p \leq q^\perp$. In $L(H)$ this is orthogonality of rays, and it determines the inner product up to the ambiguity described at the end of Theorem 3.3 below. The following properties of an ortholattice Λ enter the theorems of this section:

(C) the lattice is complete;
(OM) it is orthomodular;
(A) it is atomistic;
(CP) it has the covering property;
(I) it is irreducible;
(D) it contains four mutually orthogonal atoms.

Condition (D) supplies the four independent points of Theorem 3.2, since mutually orthogonal atoms are independent. The coordinatization theorem below goes back, in finite dimensions, to Birkhoff and von Neumann [BvN36]. The theorem shows that every ortholattice with (OM), (A), (CP), (I) and (D) is represented in the manner of $L(H)$. Under this representation, the atoms of such an ortholattice correspond to the rays of a vector space, and orthogonality of atoms to orthogonality of rays with respect to an anisotropic hermitian form.

Theorem 3.3 (Coordinatization; Baer [Bae52], Maeda and Maeda [MM70, §34], Holland [Hol95, §3]). *Let Λ be an ortholattice that satisfies (OM), (A), (CP), (I) and (D), but not necessarily (C). Then there are a division ring Δ with an involution, a hermitian space E over Δ, and a bijection $p \mapsto \hat{p}$ from the atoms of Λ onto the rays of E, such that two atoms p and q are orthogonal exactly when $\hat{p} \perp \hat{q}$. The map*

$$\gamma\colon \Lambda \to \mathcal{C}(E), \qquad \gamma(a) = \{0\} \cup \bigcup \{\hat{p} : p \text{ an atom with } p \leq a\},$$

where $\mathcal{C}(E)$ is the lattice of orthogonally closed subspaces of E, is an embedding of ortholattices whose image contains every finite-dimensional subspace of E and the orthogonal complement of each. The triple formed by Δ, E and the form is unique up to a semilinear isomorphism followed by a rescaling $\langle x \mid y\rangle' = \langle x \mid y\rangle\zeta$, where $\zeta^* = \zeta \neq 0$, with the new involution $\alpha \mapsto \zeta^{-1}\alpha^*\zeta$.

Proof of the statements about γ. The orthocomplementation of Λ induces a *polarity* on the geometry of its atoms, in the sense of Holland [Hol95, Definition 3.2]. This polarity is the map that sends each projective subspace to the set of points orthogonal to all its points. The polarity reverses inclusion and sends the set of all points to the empty set. Under the polarity, each point p goes to the set of atoms below $p^\perp$, a hyperplane (that is, a maximal proper projective subspace) whose image is again $\{p\}$ (see Holland [Hol95, §4]). By Theorem 3.2, the atoms become the rays of a vector space E. Baer's theorem [Bae52] (see Holland [Hol95, Theorem 3.6]) shows that every polarity of this kind under which no point is orthogonal to itself is orthogonality for an anisotropic hermitian form on E. If two atoms lie below a, every atom on the line they determine lies below a; hence $\gamma(a)$ is a subspace. For an atom r the following are equivalent: $r \le a^\perp$; $a \le r^\perp$; $p \le r^\perp$ for every atom $p \le a$ (because (A) makes a the join of these atoms); $\hat{r} \perp \hat{p}$ for every atom $p \le a$. Thus $\gamma(a^\perp) = \gamma(a)^\perp$, and applying this twice gives $\gamma(a) = \gamma(a)^{\perp\perp}$. By (A), the relation $a \le b$ holds if and only if $\gamma(a) \subseteq \gamma(b)$; hence γ is injective. Meets are preserved because an atom lies below $a \wedge b$ exactly when it lies below a and b, and joins are preserved by the De Morgan law. Finally, let F be a finite-dimensional subspace of E with basis $x_1, \dots, x_k$, and let p_i be the atom with $\hat{p}_i = \Delta x_i$, for $i = 1, \dots, k$. The join $p_1 \vee \cdots \vee p_k$ exists, as every finite join does in a lattice. Because γ preserves joins, the image of this join is $(\Delta x_1 + \cdots + \Delta x_k)^{\perp\perp} = F^{\perp\perp}$, which is F by Lemma 2.6. The orthogonal complement $F^\perp$ is therefore $\gamma\big((p_1 \vee \cdots \vee p_k)^\perp\big)$. ☺

Piron [Pir64] proved this representation for complete lattices; completeness is not needed for the embedding, and in Theorem 3.6 below it serves only to make γ surjective.

3.3. **Harmonic conjugates and Solèr's condition.** The hypotheses of Theorem 3.3 do not force the division ring Δ to be $\mathbb{R}$, $\mathbb{C}$ or $\mathbb{H}$. Indeed, the lattice of all subspaces of $\mathbb{Q}^4$, orthocomplemented by orthogonality for the form $\sum_{i<4} x_i y_i$, satisfies these hypotheses with $\Delta = \mathbb{Q}$. Solèr's condition, together with completeness, singles out the real numbers, the complex numbers and the quaternions. This condition is phrased through the harmonic conjugate, a classical construction of projective geometry that uses joins and meets alone. Consider, in the hermitian space of Theorem 3.3, a ray of a plane spanned by two orthogonal rays and the harmonic conjugate of that ray with respect to these two. This harmonic conjugate is the orthocomplement of the ray in the plane exactly when the ray is spanned by the sum of two vectors of equal "length", one from each orthogonal ray. We verify this equivalence below (see Figure 5). Agreement along an infinite orthogonal sequence of atoms therefore yields, after a rescaling of the form, an infinite orthonormal sequence. This orthonormal sequence, together with orthomodularity, is what Solèr's theorem requires (see Lemma 3.4 and Theorem 3.6).

Let p and q be distinct points of a projective geometry, and write $p+q$ for the line through them. In the basic example, this line consists of the rays of the plane spanned by p and q. Fix a point c of $p+q$ other than p and q. By (PG1) and (PG3), two distinct lines in a common projective plane meet in exactly one point. With a slight abuse of notation, we denote that point by the intersection of the two lines. Choose a point r not on $p+q$ and a point r' on $r+p$ other than r and p; put $c' = (c+r') \cap (q+r)$ and $c'' = (p+c') \cap (q+r')$. The point $h = (r+c'') \cap (p+q)$ is the *harmonic conjugate* of c with respect to p and q (see Figure 4). In the basic example the point h does not depend on the choices of r and r', as the computation below shows. By Theorem 3.2, the same holds in every projective geometry with four independent points. Consider now an ortholattice that satisfies (OM), (A), (CP), (I) and (D). Its atoms form a projective geometry with four independent points, as in the

preceding subsection. The same construction applies to these atoms, with $\vee$ in place of $+$ and $\wedge$ in place of $\cap$. The relation "h is the harmonic conjugate of c with respect to p and q" is then expressed by a first-order formula in the language of lattices.

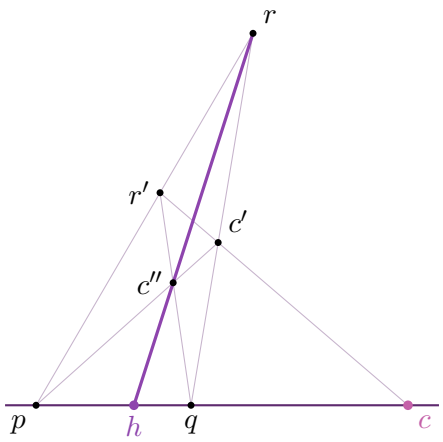


FIGURE 4. The construction of the harmonic conjugate h of a point c of the line through p and q, with respect to p and q. The construction uses the auxiliary points r, r', c' and c''.

In the basic example, the construction is simple in coordinates. Let u and v be vectors spanning p and q, respectively, scaled so that $c = \Delta(u+v)$; this is possible because c lies on $p+q$ and differs from p and q. Likewise, let w be a vector spanning r, scaled so that $r' = \Delta(u+w)$, which is possible because r' lies on $r+p$ and differs from r and p. A direct computation gives $c' = \Delta(v-w)$, $c'' = \Delta(u-v+w)$ and $h = \Delta(u-v)$. All coefficients are ± 1, so the computation is valid over every division ring:

$$\text{the harmonic conjugate of } \Delta(u+v) \text{ with respect to } \Delta u \text{ and } \Delta v \text{ is } \Delta(u-v). \tag{1}$$

Suppose now that the vector space carries an anisotropic hermitian form, as the space E of Theorem 3.3 does, and that $p \perp q$; then $u \perp v$. The orthocomplement of c within the plane $p \vee q$, the ray $c^\perp \wedge (p \vee q)$, consists of the vectors $\alpha u + \beta v$ with $\langle \alpha u + \beta v \mid u+v \rangle = \alpha\langle u \mid u\rangle + \beta\langle v \mid v\rangle = 0$. It coincides with the harmonic conjugate $\Delta(u-v)$ exactly when $\langle u \mid u\rangle = \langle v \mid v\rangle$, that is, when u and v have the same "length" (see Figure 5). In characteristic 2, that is, when $1+1=0$ in Δ, the vectors $u+v$ and $u-v$ coincide. In that case the equality $\langle u \mid u\rangle = \langle v \mid v\rangle$ would make $u+v$ isotropic, because it would give $\langle u+v \mid u+v\rangle = (1+1)\langle u \mid u\rangle = 0$. Hence, when the harmonic conjugate of c coincides with its orthocomplement in the plane $p \vee q$, the characteristic of Δ is not 2.

Solèr's condition, in the form we use, asks for an infinite orthogonal sequence of atoms along which the orthocomplement and the harmonic conjugate agree at every step:

(S) There are mutually orthogonal atoms p_n, for $n \in \mathbb{N}$, and atoms $c_n \leq p_n \vee p_{n+1}$ with $c_n \neq p_n, p_{n+1}$, such that, for every $n \in \mathbb{N}$, the harmonic conjugate of c_n with respect to p_n and p_{n+1} is $c_n^\perp \wedge (p_n \vee p_{n+1})$.

This formulation of Solèr's condition through harmonic conjugates is due to Holland [Hol95, Theorems 3.7 and 4.1]. Morash's angle-bisecting axiom [Mor73], which asks for bisectors as in the left half of Figure 5, may be used instead (see Holland [Hol95, §4]).

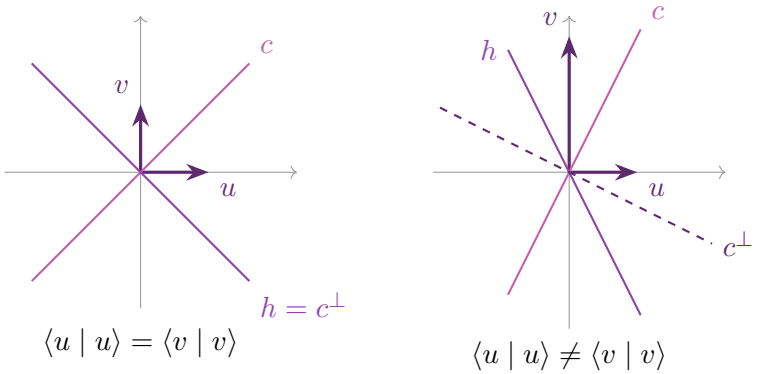


FIGURE 5. The ray $c = \Delta(u + v)$ spanned by the sum of orthogonal vectors u and v, and its harmonic conjugate $h = \Delta(u - v)$ with respect to Δu and Δv. The orthocomplement $c^\perp$ of c in the plane of u and v is drawn as well. The rays h and $c^\perp$ coincide when u and v have equal length (left), and differ otherwise (right).

Lemma 3.4. *Let Λ be an ortholattice that satisfies (OM), (A), (CP), (I), (D) and (S), and let E be a hermitian space over a division ring Δ and $p \mapsto \hat{p}$ a bijection from the atoms of Λ onto the rays of E that Theorem 3.3 provides for Λ. After a rescaling of the form, the space E contains an infinite orthonormal sequence.*

Proof. Let p_n and c_n, for $n \in \mathbb{N}$, be atoms as in (S). Choose e_0 with $\hat{p}_0 = \Delta e_0$, and inductively choose e_{n+1} with $\hat{p}_{n+1} = \Delta e_{n+1}$ and $\hat{c}_n = \Delta(e_n + e_{n+1})$. The choice of e_{n+1} is possible because $\hat{c}_n$ is a ray of the plane spanned by e_n and e_{n+1}, distinct from Δe_n and Δe_{n+1}. By (1) and the computation that follows it, condition (S) gives $\langle e_n \mid e_n \rangle = \langle e_{n+1} \mid e_{n+1} \rangle$ for every $n \in \mathbb{N}$; call this common value ζ, which satisfies $\zeta^* = \zeta \neq 0$. The form $\langle x \mid y \rangle' = \langle x \mid y \rangle \zeta^{-1}$ is hermitian for the involution $\alpha \mapsto \zeta \alpha^* \zeta^{-1}$, has the same orthogonality relation, and satisfies $\langle e_n \mid e_n \rangle' = 1$ for every n. ☺

Definition 3.5. A *Solèr lattice* is an ortholattice that satisfies (C), (OM), (A), (CP), (I), (D) and (S). A *pre-Solèr lattice* is an ortholattice that satisfies (OM), (A), (CP), (I), (D) and (S), but not necessarily (C).

Theorem 3.6 (Piron, Amemiya and Araki, Solèr; in this form Holland [Hol95, Theorem 4.1]**).** *An ortholattice is a Solèr lattice if and only if it is isomorphic to $L(H)$ for some infinite-dimensional Hilbert space H over $\mathbb{R}$, $\mathbb{C}$ or $\mathbb{H}$.*

Sketch. Let Λ be a Solèr lattice, with γ and E as in Theorem 3.3. Given $M \in \mathcal{C}(E)$, completeness provides the join a of the atoms p with $\hat{p} \subseteq M$, and this join satisfies $\gamma(a)^\perp = \gamma(a^\perp) = M^\perp$, whence $\gamma(a) = M$. Thus the embedding γ is an isomorphism onto $\mathcal{C}(E)$; the lattice $\mathcal{C}(E)$ is therefore orthomodular, and by Proposition 2.8 so is the space E. By Lemma 3.4, the space E contains, after a rescaling of the form that changes neither $\mathcal{C}(E)$ nor its orthocomplementation, an infinite orthonormal sequence. Solèr's theorem [Sol95] (see also Prestel [Pre95]) asserts that an orthomodular space containing an infinite orthonormal sequence is a Hilbert space over $\mathbb{R}$, $\mathbb{C}$ or $\mathbb{H}$. The space E, with its form so rescaled, is therefore such a Hilbert space. Conversely, the lattice $L(H)$ is complete and satisfies the other conditions. For (S), take an orthonormal sequence $(e_n)_{n \in \mathbb{N}}$ in H, and let p_n be the ray of e_n and c_n the ray of $e_n + e_{n+1}$. ☺

Condition (S) cannot be omitted from Theorem 3.6. Keller [Kel80] constructed infinite-dimensional orthomodular hermitian spaces over division rings other than $\mathbb{R}$, $\mathbb{C}$ and $\mathbb{H}$. For such a space E the lattice $\mathcal{C}(E)$ satisfies (C), (OM), (A), (CP), (I) and (D) (see Proposition 2.8 and Lemmas 2.7 and 2.6). Nevertheless, by the uniqueness clause of Theorem 3.3, the lattice $\mathcal{C}(E)$ is isomorphic to no lattice $L(H)$ of a Hilbert space H over $\mathbb{R}$, $\mathbb{C}$ or $\mathbb{H}$.

Historical note. Von Staudt's *Geometrie der Lage* of 1847 was the first treatment of projective geometry free of any metric. His later calculus of "throws" outlined the algebraization of an axiomatic geometry. Hilbert showed in his *Grundlagen der Geometrie* of 1899 that Desargues's theorem follows from the axioms of space but cannot be proved in the plane without the axioms of congruence. On that theorem he founded an algebra of segments that does not use those axioms [Hil99]. Veblen and Young gave axioms for projective geometry in 1908 [VY08]. In their book of 1910 they introduced coordinates through an algebra of points that descends from von Staudt's algebra of throws [VY10]. Solèr, a student of Herbert Gross, had just begun work on Gross's orthomodular problem when Gross died suddenly in 1989. She completed the work under the informal guidance of Alexander Prestel, a friend of Gross's at Konstanz, and her theorem appeared in 1995 [Sol95]. In a survey of the same year, Holland restated her condition through harmonic conjugates, constructed as in the book of Veblen and Young, in the form used in this section [Hol95].

4. What a pre-Solèr lattice is missing (optional)

A pre-Solèr lattice satisfies every condition in the definition of a Solèr lattice except, possibly, completeness. We show that the MacNeille completion of such a lattice supplies the joins and meets that the lattice may lack. We show further that the completion is a Solèr lattice exactly when it is orthomodular, which need not be the case. Neither completeness nor the orthomodularity of the completion is expressed, among pre-Solèr lattices, by any sentence of two-valued $L_{\omega_1\omega}$ about the lattice alone (see Proposition 4.5 below). The theory of Section 6 secures both in its standard models, through the sorts for vectors and for sequences of vectors that the vocabulary of Section 5 places beside the Solèr sort.

Throughout this section we fix a pre-Solèr lattice Λ and a hermitian space E that Theorem 3.3 provides for Λ. Under the embedding γ of that theorem, we identify Λ with its image $\gamma(\Lambda) \subseteq \mathcal{C}(E)$.

Theorem 4.1. *Let Λ be a pre-Solèr lattice, let E be a hermitian space that Theorem 3.3 provides for Λ, and identify Λ with its image $\gamma(\Lambda)$ in $\mathcal{C}(E)$ under the embedding γ of that theorem. Then Λ is join-dense and meet-dense in $\mathcal{C}(E)$; consequently $\mathcal{C}(E)$, with $M \mapsto M^\perp$, is isomorphic as an ortholattice to the MacNeille completion of Λ.*

Proof. Every $M \in \mathcal{C}(E)$ is the join in $\mathcal{C}(E)$ of the rays it contains, which belong to Λ by Theorem 3.3. If $M^\perp = \{0\}$, then the subspace M is E, which belongs to Λ. Otherwise M equals $M^{\perp\perp} = \bigcap\{y^\perp : 0 \neq y \in M^\perp\}$, a meet of hyperplanes, each of which belongs to Λ. Both assertions now follow from Proposition 1.5. ☺

The elements of the MacNeille completion that do not come from Λ are therefore orthogonally closed subspaces M such that M and $M^\perp$ are both infinite-dimensional. Indeed, the lattice Λ contains every finite-dimensional subspace and the orthogonal complement of each (see Theorem 3.3). Adjoining these elements to Λ need not produce a lattice isomorphic to some $L(H)$, because the completion may fail to be orthomodular (see Example 4.3). As the next theorem shows, orthomodularity of the completion is exactly what is needed.

Theorem 4.2. *Let Λ be a pre-Solèr lattice, and let E be a hermitian space that Theorem 3.3 provides for Λ. Then the following are equivalent:*

(1) the MacNeille completion of Λ is orthomodular;
(2) the space E is orthomodular;
(3) the lattice Λ is isomorphic to a join-dense and meet-dense sub-ortholattice of $L(H)$ for some infinite-dimensional Hilbert space H over $\mathbb{R}$, $\mathbb{C}$ or $\mathbb{H}$.

When these conditions hold, the space E, with its form suitably rescaled, is a Hilbert space as in (3). Moreover, the lattice Λ is isomorphic to $L(H)$ for such an H if and only if Λ is complete.

Proof. Conditions (1) and (2) are equivalent by Theorem 4.1 and Proposition 2.8. If (2) holds, then Lemma 3.4 and Solèr's theorem show that the space E, with its form rescaled, is a Hilbert space. Rescaling changes neither $\mathcal{C}(E)$ nor its orthocomplementation, and condition (3) follows by Theorem 4.1. If (3) holds, then by Proposition 1.5 the orthomodular lattice $L(H)$ is isomorphic as an ortholattice to the MacNeille completion of Λ; this is (1). For the last assertion, a complete pre-Solèr lattice is a Solèr lattice and by Theorem 3.6 is therefore isomorphic to $L(H)$ for an infinite-dimensional Hilbert space H over $\mathbb{R}$, $\mathbb{C}$ or $\mathbb{H}$. Conversely, every lattice $L(H)$ is complete. ☺

Example 4.3. Let E_0 be the space of finitely supported complex sequences $x = (x_n)_{n\in\mathbb{N}}$, with the inner product $\langle x \mid y\rangle = \sum_n x_n\overline{y_n}$ of the space ℓ^2 of square-summable sequences. Let Λ_0 be the set of finite-dimensional subspaces of E_0 and their orthogonal complements in E_0. Then Λ_0 is a sub-ortholattice of $\mathcal{C}(E_0)$: its elements are orthogonally closed (the finite-dimensional ones by Lemma 2.6). Moreover, for finite-dimensional F and G the joins $F \vee G = F + G$, $F^\perp \vee G^\perp = (F \cap G)^\perp$ and $F \vee G^\perp = (F^\perp \cap G)^\perp$ belong to Λ_0, whence the De Morgan law places the meets in Λ_0 as well. The lattice Λ_0 is orthomodular, because every $M \in \Lambda_0$ satisfies $M + M^\perp = E_0$ by Lemma 2.6, and the argument of Proposition 2.8 applies. As a sub-ortholattice of $\mathcal{C}(E_0)$ that contains every ray of E_0, the lattice Λ_0 inherits (A), (CP) and (I) from $\mathcal{C}(E_0)$, in which (CP) holds by Lemma 2.7. Conditions (D) and (S) hold with $p_n = \mathbb{C}e_n$ and $c_n = \mathbb{C}(e_n + e_{n+1})$, where $(e_n)_{n\in\mathbb{N}}$ is the sequence of standard unit vectors. Consequently Λ_0 is a pre-Solèr lattice, and the space E_0, with $\hat{p} = p$ for every atom p of Λ_0, is a hermitian space that Theorem 3.3 provides for Λ_0. The map γ of that theorem is then the inclusion of Λ_0 in $\mathcal{C}(E_0)$. The MacNeille completion of Λ_0, which is $\mathcal{C}(E_0)$ by Theorem 4.1, is not orthomodular. Indeed, by the Amemiya–Araki theorem [AA66], a complex inner product space whose lattice of orthogonally closed subspaces is orthomodular is complete, which E_0 is not. Hence Λ_0 is an orthomodular lattice whose MacNeille completion is not orthomodular; Harding [Har91] studies such lattices.

Orthomodularity of the completion can be stated through the atoms of Λ alone, without reference to the completion or to the space E. It is in this form that Proposition 4.5 below shows that orthomodularity of the completion cannot be expressed by a sentence of two-valued $L_{\omega_1\omega}$ about the lattice. For a set $\mathcal{A}$ of atoms, let $\mathcal{A}^\perp$ be the set of atoms orthogonal to every element of $\mathcal{A}$, and call $\mathcal{A}$ *orthoclosed* when $\mathcal{A} = \mathcal{A}^{\perp\perp}$. Consider the following condition, named (MOM) for MacNeille orthomodularity:

(MOM) for every orthoclosed set $\mathcal{A}$ of atoms and every atom $p \notin \mathcal{A}$, some atom $q \in \mathcal{A}^\perp$ lies in $(\mathcal{A} \cup \{p\})^{\perp\perp}$.

Theorem 4.4. *The MacNeille completion of a pre-Solèr lattice is orthomodular if and only if the lattice satisfies (MOM).*

Proof. Let $\hat{p}$ denote, for each atom p, the ray of E that corresponds to p under Theorem 3.3. The map $\mathcal{A} \mapsto \{0\} \cup \bigcup\{\hat{p} : p \in \mathcal{A}\}$ is a bijection from the orthoclosed sets of atoms onto $\mathcal{C}(E)$, carrying $\mathcal{A}^\perp$ to the orthogonal complement of the image of $\mathcal{A}$. If M is the image of $\mathcal{A}$ and $\hat{p} = \Delta x$, where Δ is the division ring of scalars of E, then by Lemma 2.7 the set $(\mathcal{A} \cup \{p\})^{\perp\perp}$ corresponds to $M + \Delta x$. Condition (MOM) therefore says that $(M + \Delta x) \cap M^\perp \neq \{0\}$ whenever $M \in \mathcal{C}(E)$ and $x \notin M$. If $y = v + \alpha x$ is nonzero, with $v \in M$ and $y \in M^\perp$, then the scalar α is nonzero because $y \notin M$, whence the vector $x = \alpha^{-1}(y - v)$ lies in $M + M^\perp$. Conversely, if $x = v + y$ with $v \in M$ and $y \in M^\perp$, then the vector $y = x - v$ is a nonzero element of $(M + \Delta x) \cap M^\perp$. Thus (MOM) is equivalent to the orthomodularity of E, and the theorem follows from Theorem 4.2. ☺

By Lemma 2.6, the requirement in (MOM) holds automatically when $\mathcal{A}$ or $\mathcal{A}^\perp$ corresponds to a finite-dimensional subspace. The content of (MOM) lies in the orthoclosed sets $\mathcal{A}$ for which both $\mathcal{A}$ and $\mathcal{A}^\perp$ correspond to infinite-dimensional subspaces.

Proposition 4.5. *Neither (C) nor (MOM) is equivalent, among pre-Solèr lattices, to a sentence of two-valued $L_{\omega_1\omega}$ in the vocabulary of ortholattices expanded by constants for atoms p_n and c_n, for $n \in \mathbb{N}$, that witness (S).*

Proof. Suppose that a sentence φ of $L_{\omega_1\omega}$ were equivalent to (MOM) among pre-Solèr lattices. The axioms of ortholattices and the conditions (OM), (A), (CP), (I), (D) and (S) are expressed by countably many sentences of $L_{\omega_1\omega}$ in this vocabulary; together with φ, they hold in $L(\ell^2)$. By the downward Löwenheim–Skolem theorem for a countable fragment, they hold in a countable structure Λ', which is a pre-Solèr lattice satisfying (MOM). By Theorems 4.4 and 4.2, the lattice Λ' contains every ray of an infinite-dimensional Hilbert space, and is therefore uncountable, a contradiction. The same argument applies to (C), using Theorem 3.6. ☺

⧖ **Historical note.** MacNeille introduced the completion by cuts in 1937 [Mac37], generalizing the cuts by which Dedekind had constructed the real numbers from the rationals in 1872 [Ded72]. Banaschewski and Schmidt characterized it in 1956 by join-density and meet-density [Ban56, Sch56]. In 1969 Adams solved negatively Problem 36 of Birkhoff's *Lattice Theory*, in its third edition of 1967, by giving an orthocomplemented modular lattice whose completion by cuts is not orthomodular [Ada69]. His lattice consists of the finite-dimensional subspaces of an incomplete complex inner product space and their orthogonal complements, as in Example 4.3, and his argument also rests on the Amemiya–Araki theorem. Harding gave in 1991, without recourse to the Amemiya–Araki theorem, a construction of orthomodular lattices whose completions by cuts are not orthomodular. Following a construction that Kalmbach introduced in 1977, his construction glues together the Boolean algebras generated by the chains of a lattice [Har91]. Harding noted that whether every orthomodular lattice can be embedded in a complete one was still open.

5. The logic and the vocabulary

Completeness is the one condition in the definition of a Solèr lattice that a pre-Solèr lattice need not satisfy. The downward Löwenheim–Skolem theorem puts this condition beyond the reach of the sentences of $L_{\omega_1\omega}$, two-valued or real-valued, that speak of the lattice alone. Consider any countable fragment that contains sentences expressing the axioms of ortholattices and the conditions in the definition of a Solèr lattice other than completeness. A Solèr lattice, expanded by constants for atoms that witness Solèr's condition (S), has a countable substructure that is elementary for this fragment, in the sense that every formula of the

fragment has the same value in the two structures at every tuple from the substructure of the appropriate sorts. This substructure, taken in the real-valued case modulo the formula that expresses equality, is therefore a pre-Solèr lattice, yet it is not complete. Were it complete, it would be a Solèr lattice, which by Theorem 3.6 is isomorphic to the lattice of closed subspaces of an infinite-dimensional Hilbert space and is therefore uncountable. Consequently, no sentence of the fragment is true in exactly those pre-Solèr lattices that are complete. In the two-valued case, this argument is given in detail in the optional Proposition 4.5. Because completeness thus eludes the sentences that speak of the lattice alone, it enters our theory through a Hilbert space placed beside the lattice. The joins that a pre-Solèr lattice may lack are supplied by that Hilbert space, through sequences of vectors whose closed spans the lattice is required to contain. In this section we first recall the logic that we use. We then present a vocabulary in which the lattice stands beside a Hilbert space and a space of sequences of vectors. The three are linked by function symbols for orthogonal projection, for the passage between vectors and atoms, for the passage between vectors and sequences, and for the closed span of a sequence.

5.1. **Real-valued $L_{\omega_1\omega}$.** We use the real-valued infinitary logic of Aubone, Dueñez and Iovino [ADI26], whose structures are Keisler's general structures [Kei23, Section 2.1]. We recall its definitions as far as we use them.

Vocabularies and structures. A vocabulary consists of sorts, of function symbols, each with the sorts of its arguments and of its values, and of predicate symbols. Each predicate symbol R comes with the sorts of its arguments and with a nonnegative real number r_R, its *a priori bound*. In this paper the a priori bound of every predicate symbol is a positive integer. The vocabulary contains a nullary predicate symbol $\mathbf{1}$, the *logical unity constant*, with a priori bound 1. A structure for the vocabulary interprets each sort as a nonempty set, each function symbol as a function from the product of its argument sorts to its value sort, and each predicate symbol R as a function from the product of its argument sorts to the interval $[-r_R, r_R]$; the logical unity constant is interpreted as 1. No topology is prescribed with respect to which the functions or the predicates must be continuous, and the vocabulary has no equality symbol. Eagle's metric structures, by contrast, have a master metric with respect to which their predicates, which take values in $[0,1]$, are uniformly continuous [Eag14, Definitions 1.4 and 1.5].

Formulas. The atomic formulas are the expressions $R(\bar{t})$, where R is a predicate symbol and $\bar{t}$ is a tuple of terms of the appropriate sorts. Formulas are built from the atomic formulas by the connectives $s \cdot \varphi$, for $s \in \mathbb{Q}$, $\varphi + \psi$ and $\varphi \vee \psi$, by the connectives $\sup_x$, one for each variable x, and by countable joins $\bigvee_n \varphi_n$. A countable join is formed from a family $(\varphi_n)_{n\in\mathbb{N}}$ of formulas whose free variables lie in one finite set and whose a priori bounds, defined next, have a finite supremum. The a priori bound of $R(\bar{t})$ is r_R; the a priori bounds of $s \cdot \varphi$, $\varphi + \psi$, $\varphi \vee \psi$, $\sup_x \varphi$ and $\bigvee_n \varphi_n$ are $|s| r_\varphi$, $r_\varphi + r_\psi$, $\max(r_\varphi, r_\psi)$, r_φ and $\sup_n r_{\varphi_n}$, respectively. In a structure, the connectives $s \cdot \varphi$, $\varphi + \psi$ and $\varphi \vee \psi$ are interpreted as multiplication by s, as the sum and as the maximum, the connective $\sup_x$ as the supremum over the sort of x, and a countable join as the supremum of the values of its members. By induction on formulas, every formula φ takes its values in $[-r_\varphi, r_\varphi]$ in every structure, and hence in the interval $I_\varphi = [-\lceil r_\varphi \rceil, \lceil r_\varphi \rceil]$, whose endpoints are integers. The number r_φ does not change when terms are substituted for the free variables of φ.

Abbreviations. For a rational number q we write q for the sentence $q \cdot \mathbf{1}$, whose value is q in every structure; in particular, we write 0 for $0 \cdot \mathbf{1}$. We write $-\varphi$ for $(-1) \cdot \varphi$ and $\varphi - \psi$ for $\varphi + (-\psi)$. The minimum, the infimum over the sort of x and the countable infimum are

expressed by $\varphi \wedge \psi$, which abbreviates $-(-\varphi \vee -\psi)$, by $\inf_x \varphi$, which abbreviates $-\sup_x -\varphi$, and by $\bigwedge_n \varphi_n$, which abbreviates $-\bigvee_n -\varphi_n$. Finally, the absolute value $|\varphi|$ abbreviates $\varphi \vee -\varphi$. The connectives do not include multiplication, but the square of a formula, and hence the product of two formulas, can be written as a countable join of affine expressions. For a formula φ, let φ^2 denote the countable join of the formulas $(2q)\cdot\varphi - q^2$, for the rational numbers q with $|q| \leq r_\varphi$.

Lemma 5.1. *For every formula φ, the join φ^2 is a formula with a priori bound at most $3r_\varphi^2$, and its value in every structure, at every tuple, is the square of the value of φ.*

Proof. The member of index q has a priori bound $2|q|r_\varphi + q^2 \leq 3r_\varphi^2$, and its free variables are those of φ; the join is therefore a formula. The values of φ lie in $[-r_\varphi, r_\varphi]$, and for a real number u in this interval we have $2qu - q^2 = u^2 - (u-q)^2 \leq u^2$; since the rational numbers of the interval are dense in it, the term $(u-q)^2$ can be made arbitrarily small, and the supremum of the values of the members at u is therefore u^2. ☺

For formulas φ and ψ, the product $\varphi\psi$ abbreviates $\frac{1}{4}(\varphi+\psi)^2 - \frac{1}{4}(\varphi-\psi)^2$, whose value is the product of the values of φ and ψ. Polynomial expressions with rational coefficients in formulas are therefore formulas. For the pair of predicates $\langle x \mid y\rangle$ introduced below, we write $|\langle x \mid y\rangle|^2$ for $(\mathrm{Re}\langle x \mid y\rangle)^2 + (\mathrm{Im}\langle x \mid y\rangle)^2$.

Conditions and theories. The statements of the logic are not formulas but conditions, which compare the value of a sentence with a rational number. For a sentence φ and a rational number r, a *condition* is an expression $\varphi > r$ or $\varphi < r$. A structure $\mathfrak{M}$ satisfies $\varphi > r$ when the value $\varphi^{\mathfrak{M}}$ of φ in $\mathfrak{M}$ is greater than r, and it satisfies $\varphi < r$ when that value is less than r. A *theory* is a set of conditions, and a *model* of a theory is a structure that satisfies every condition in it. Non-strict inequalities are recovered as countable theories: the *closed condition* $\varphi \geq r$ denotes the theory $\{\varphi > r - \frac{1}{n} : n \geq 1\}$, the closed condition $\varphi \leq r$ denotes $\{\varphi < r + \frac{1}{n} : n \geq 1\}$, and the closed condition $\varphi = r$ denotes the union of the two. An *axiom* is a closed condition on a formula $\varphi(\bar{x})$, possibly without free variables, required of all values of $\bar{x}$. The axiom $\varphi(\bar{x}) \leq r$ is the closed condition $\sup_{\bar{x}} \varphi(\bar{x}) \leq r$, and the axiom $\varphi(\bar{x}) \geq r$ is $\inf_{\bar{x}} \varphi(\bar{x}) \geq r$; the axiom $\varphi(\bar{x}) = r$ is the union of the two. For formulas φ and ψ, we write $\varphi \leq \psi$ and $\varphi = \psi$ for the axioms $\varphi - \psi \leq 0$ and $\varphi - \psi = 0$, which are closed conditions when φ and ψ are sentences.

Truth values. As Aubone, Dueñez and Iovino do [ADI26], we think of 1 as true and of -1 as false. A formula is *two-valued* in a structure if it takes only the values 1 and -1 there. For formulas θ and θ' that are two-valued in a structure, negation, disjunction and conjunction are expressed there by $-\theta$, $\theta \vee \theta'$ and $\theta \wedge \theta'$. The existential and the universal quantifier are expressed by $\sup_x$ and $\inf_x$, and countable disjunction and conjunction by $\bigvee_n$ and $\bigwedge_n$, the latter only for families whose a priori bounds have a finite supremum. A supremum of values in $\{-1, 1\}$ over a nonempty set is attained, and the connectives $\sup_x$ and $\inf_x$, applied to a two-valued formula, are therefore the traditional quantifiers over the sort of x; this is how we use them over the Solèr sort, on which equality and order are given by two-valued formulas. A two-valued formula θ is asserted by the axiom $\theta \geq 1$, and the implication from θ to θ' is asserted by the axiom $-\theta \vee \theta' \geq 1$. If θ is two-valued and ψ is any formula, the axiom $\theta \wedge \psi \leq 0$ says that θ implies $\psi \leq 0$: where θ is false, the minimum is at most -1. Real quantities such as inner products, norms and the measure keep their numerical values; every statement about these quantities is a closed condition or an axiom with a rational threshold. A measure or an assignment that takes only the values 0 and 1 is always described as such, or as $\{0, 1\}$-valued, never as two-valued.

Truncations. Since the a priori bound of a countable join is the supremum of the a priori bounds of its members, the steep functions $u \mapsto \min(1, \max(0, nu))$, for $n \in \mathbb{N}$, applied to a formula φ with $r_\varphi > 0$, cannot be gathered into a single countable join: although their values lie in $[0,1]$, the formulas $((n \cdot \varphi) \wedge 1) \vee 0$ have the a priori bounds $\max(nr_\varphi, 1)$, which are unbounded. As usual in multisorted languages, steepness is supplied by the vocabulary instead.[2] In Section 5.3, each real quantity named by a predicate symbol comes with its truncations at every scale n, as predicate symbols of a priori bound 1, and the axioms of group (Tr) of Section 6 fix their values. Countable joins of these truncations record whether a quantity exceeds a rational number.

5.2. **Henkin types and model existence.** The model existence argument proceeds by assigning values to sentences and requiring those assignments to respect the semantics of the logic. Henkin types express these requirements for structures whose elements are named by closed terms. We recall their definition here so that the logical framework remains self-contained; in Section 10 we express the model existence argument through challenges and answers, and in Section A we compare that game with the topological treatment of Aubone, Dueñez and Iovino.

Following Aubone, Dueñez and Iovino [ADI26], we call a *fragment* a set of formulas that contains the first-order formulas and is closed under subformulas, under substitution of terms and under the connectives $s \cdot \varphi$, $\varphi + \psi$, $\varphi \vee \psi$ and $\sup_x$. For a countable fragment $\mathcal{F}$ of real-valued $L_{\omega_1\omega}$ for a countable vocabulary, such as $\mathcal{L}$, let C be a set of new constants, countably many of each sort. Let $\mathcal{F}_C$ be the fragment that $\mathcal{F}$ generates in the vocabulary enlarged by C, and let $S_C^{\mathcal{F}}$ be the set of sentences of $\mathcal{F}_C$. An *$S_C^{\mathcal{F}}$-Henkin type* is an element $\mathfrak{m}$ of the product of the intervals $[-r_\varphi, r_\varphi]$, over the sentences φ of $S_C^{\mathcal{F}}$, that satisfies the following *Henkin conditions*:

- (H1′) $\mathbf{1}^{\mathfrak{m}} = 1$;
- (H2′) $(s \cdot \varphi)^{\mathfrak{m}} = s\,\varphi^{\mathfrak{m}}$ for every rational number s;
- (H3′) $(\bigvee_n \varphi_n)^{\mathfrak{m}} = \sup_n \varphi_n^{\mathfrak{m}}$;
- (H4′) $(\sup_x \varphi)^{\mathfrak{m}} = \sup_t \varphi(t)^{\mathfrak{m}}$, the supremum being taken over the closed terms t of the sort of x;
- (H5′) $(\varphi \vee \psi)^{\mathfrak{m}} = \max(\varphi^{\mathfrak{m}}, \psi^{\mathfrak{m}})$;
- (H6′) $(\varphi + \psi)^{\mathfrak{m}} = \varphi^{\mathfrak{m}} + \psi^{\mathfrak{m}}$.

Here $\varphi^{\mathfrak{m}}$ denotes the coordinate of $\mathfrak{m}$ at φ, and the sentences involved range over $S_C^{\mathcal{F}}$. With the topology induced by the product topology, the $S_C^{\mathcal{F}}$-Henkin types form the *space of Henkin types* $\mathcal{H}_C^{\mathcal{F}}$, a G_δ subset of the product and therefore a Čech-complete space [ADI26, Proposition 5 and Corollary 6]. Every Henkin type is the $S_C^{\mathcal{F}}$-type of a structure for the enlarged vocabulary in which every element is the value of a closed term [ADI26, Lemma 2]. By Čech's form of the Baire category theorem, the space of Henkin types is therefore a Baire space, and in Section A we return to it.

[2] In the model theory of Banach spaces, a function or predicate defined on a normed space E stands for a family of symbols, one on each ball of radius n of E, linked by the inclusion maps between these balls; the map $x \mapsto nx$ from the unit ball to the ball of radius n, followed by a predicate of a priori bound 1 on the latter ball, then yields a predicate of a priori bound 1 that varies n times as fast as one on the unit ball. The truncated predicates of Section 5.3 are symbols of this kind, which we list directly to avoid the notation of these families of sorts; for the measure, which lives on the Solèr sort, they are simply new symbols, whose values the axioms (Tr) fix.

5.3. **The vocabulary.** The vocabulary $\mathcal{L}$ has three sorts. Two of them, the Hilbert sort and the sequence sort, are *vector sorts*, intended to be the closed unit balls of Hilbert spaces. Working in unit balls keeps every predicate bounded.

- ♠ The *Hilbert sort* $\mathbf{H}$ has a binary function $(x, y) \mapsto \lambda x + \lambda' y$ for each pair λ, λ' of elements of the field $\mathbb{Q}(i)$ of complex numbers with rational real and imaginary parts such that $|\lambda| + |\lambda'| \leq 1$. The sort $\mathbf{H}$ also has the predicates $\mathrm{Re}\langle x \mid y\rangle$, $\mathrm{Im}\langle x \mid y\rangle$ and $\|x\|$, with a priori bound 1, and constants e_n for $n \in \mathbb{N}$. We write 0 for the closed term $0e_0 + 0e_0$, and $\langle x \mid y\rangle$ for the pair of predicates $\mathrm{Re}\langle x \mid y\rangle$ and $\mathrm{Im}\langle x \mid y\rangle$.
- ♠ The *sequence sort* $\mathbf{K}$ has the same functions and predicates as $\mathbf{H}$, one constant ε, and unary functions θ_N for $N \in \mathbb{N}$; in this sort, we write 0 for the closed term $0\varepsilon + 0\varepsilon$.
- ♠ The *Solèr sort* $\mathbf{L}$ has the constants 0 and 1, the binary functions $\wedge$ and $\vee$ and the unary function $^\perp$.
- ♠ Function symbols connecting the sorts:

$$\pi\colon \mathbf{L} \times \mathbf{H} \to \mathbf{H}, \quad \mathrm{at}\colon \mathbf{H} \to \mathbf{L}, \quad \mathrm{rep}\colon \mathbf{L} \to \mathbf{H},$$
$$\iota_n\colon \mathbf{H} \to \mathbf{K}, \quad \kappa_n\colon \mathbf{K} \to \mathbf{H}\ (n \in \mathbb{N}), \quad \mathrm{span}\colon \mathbf{K} \to \mathbf{L}.$$

- ♠ A predicate μ on $\mathbf{L}$, the *measure*, with a priori bound 2. The bound is wider than 1 so that the inequality $\mu(a) \leq 1$, like $\mu(a) \geq 0$, is imposed by the axiom (M1) of Section 6 below, and not by the vocabulary.
- ♠ *Truncations.* For every predicate symbol R listed above, every positive integer n and every rational number q, two predicate symbols $R^+_{n,q}$ and $R^-_{n,q}$ with the arguments of R and with a priori bound 1.

The intended values of the truncations are

$$R^+_{n,q}(\bar{x}) = \min\bigl(1, \max(0, n(R(\bar{x}) - q))\bigr), \qquad R^-_{n,q}(\bar{x}) = \min\bigl(1, \max(0, n(q - R(\bar{x})))\bigr),$$

which the axioms (Tr) of Section 6 impose. For an atomic formula $R(\bar{t})$, where R is one of the inner-product, norm or measure predicates listed above, and a rational number q, let

$$\lceil R(\bar{t}) > q\rceil = \bigvee_{n\geq 1} R^+_{n,q}(\bar{t}), \qquad \lceil R(\bar{t}) < q\rceil = \bigvee_{n\geq 1} R^-_{n,q}(\bar{t}),$$

countable joins of members with a priori bound 1. In every structure that satisfies (Tr), the formula $\lceil R(\bar{t}) > q\rceil$ takes the value 1 where $R(\bar{t}) > q$ and the value 0 elsewhere, and likewise for $\lceil R(\bar{t}) < q\rceil$. Several symbols now have more than one meaning: the symbol 0 names a sentence, a closed term of each vector sort and a constant of the Solèr sort, and the symbol 1 names a sentence and a constant of the Solèr sort. The symbols $\wedge$ and $\vee$, for their part, name both connectives and functions of the Solèr sort, and the arguments to which they are applied always show which is meant. For vector terms t_1, t_2 of the same sort, the equation $t_1 \doteq t_2$ abbreviates the axiom $\|\frac{1}{2}t_1 - \frac{1}{2}t_2\| \leq 0$, a closed condition when t_1 and t_2 are closed. A vector x *belongs* to an element a of the Solèr sort, or is a *member* of a, when $\pi(a, x) \doteq x$.

The intended meanings are as follows. Each element a of the Solèr sort names a closed subspace of the Hilbert space whose closed unit ball is the Hilbert sort. The function $\pi(a, \cdot)$ is the orthogonal projection onto that subspace, and the members of a are therefore the vectors of the unit ball that lie in the subspace. Vectors and elements of the Solèr sort are linked by at and rep. The function at sends a nonzero vector x to the element whose members are the multiples of x in the unit ball, and it sends 0 to 0. In the opposite direction, each nonzero element a is sent by rep to a unit vector that belongs to a, and 0 is sent to the zero vector. The constants e_n, for $n \in \mathbb{N}$, form an orthonormal basis of the Hilbert space. The elements of the sequence sort are the square-summable sequences of vectors of that space whose ℓ^2 norm

is at most 1. These sequences are linked to the Hilbert sort by ι_n and κ_n. The first sends a vector x to the sequence that has x in position n and 0 elsewhere, and the second reads the entry in position n. The function θ_N keeps the entries of a sequence in the positions $n < N$ and replaces the others by 0. The constant ε is the sequence $(2^{-n-1}e_n)_{n\in\mathbb{N}}$. The function span sends a sequence s to the element whose members are the vectors of the unit ball in the closed span of the entries of s.

Equality and order on the Solèr sort are not symbols of the vocabulary. They are expressed by formulas built from the norms of the vectors $\mathrm{rep}(a)$. For elements a, b of the Solèr sort, let

$$[a \neq 0] = 2\|\mathrm{rep}(a)\| - 1, \qquad [a \leq b] = 1 - 2\|\mathrm{rep}(a \wedge (a \wedge b)^{\perp})\|, \qquad [a = b] = [a \leq b] \wedge [b \leq a].$$

In the intended interpretation, $\|\mathrm{rep}(a)\|$ is 1 when a is nonzero and 0 when a is zero, whence $[a \neq 0]$ is two-valued, with the value 1 exactly when $a \neq 0$. The element $a \wedge (a \wedge b)^{\perp}$ is the orthogonal complement of $a \wedge b$ in a, which is zero exactly when $a \leq b$. Hence $[a \leq b]$ and $[a = b]$ are two-valued, with the value 1 exactly when $a \leq b$ and when $a = b$, respectively. In Lemma 6.4 below we show that the same holds in every model of the theory T_0 of Section 6. For terms t_1, t_2 of the Solèr sort, the equation $t_1 = t_2$ abbreviates the axiom $[t_1 = t_2] \geq 1$. An equation that stands as an axiom in Section 6 is an axiom in the sense of Section 5, required of all values of its free variables.

The axiom of Theorem B excludes the points of a Borel set from the range of the measure. This axiom needs a two-valued formula that says whether a value of the measure lies in that set. For a Borel set $B \subseteq \mathbb{R}$, we write $\beta_B(a)$ for any formula of real-valued $L_{\omega_1\omega}$, with the single free variable a of the Solèr sort, whose values are prescribed as follows. At every element a of every structure for $\mathcal{L}$ that satisfies the axioms (Tr), the formula takes the value 1 if $\mu(a) \in B$ and -1 otherwise.

Lemma 5.2. *For every Borel set $B \subseteq \mathbb{R}$, there is a formula $\beta_B(a)$ as just described.*

Proof. Let $\mathcal{S}$ be the class of sets $B \subseteq \mathbb{R}$ for which such a formula exists with a priori bound at most 3. For rational numbers $q \leq q'$, the formula

$$1 - 2\big(\lceil \mu(a) > q' \rceil \vee \lceil \mu(a) < q \rceil\big)$$

takes the value 1 when $\mu(a) \in [q, q']$ and -1 otherwise, in every structure that satisfies (Tr). Its a priori bound is 3, and the class $\mathcal{S}$ therefore contains the closed intervals with rational endpoints. If $B \in \mathcal{S}$, then the negation $-\beta_B(a)$, which has the same a priori bound, is a formula for $\mathbb{R} \setminus B$. If $B_n \in \mathcal{S}$ for every $n \in \mathbb{N}$, then the countable join of the formulas $\beta_{B_n}(a)$, whose a priori bounds are at most 3, is a formula for $\bigcup_n B_n$ with a priori bound at most 3. The class $\mathcal{S}$ is therefore a σ-algebra, and since every open subset of $\mathbb{R}$ is a countable union of closed intervals with rational endpoints, it contains every Borel set. ☺

When a formula $\beta_B(a)$ is obtained by following, as in the proof of Lemma 5.2, a construction of B from closed intervals with rational endpoints by complements and countable unions, its negations and countable joins are nested as in that construction. The depth of this nesting is a countable ordinal, unbounded below ω_1 as B varies. The axioms of Theorem B therefore involve formulas whose joins are nested to arbitrarily large countable depth, although none of these formulas contains a quantifier.

Definition 5.3. Let $\mathfrak{M}$ be a structure for $\mathcal{L}$, or for a vocabulary obtained from $\mathcal{L}$ by removing the predicate μ, the sequence sort with the symbols that involve it, or both, the truncations of the removed predicates being removed with them. The structure $\mathfrak{M}$ is *standard* when its Hilbert sort is the closed unit ball of a complex Hilbert space H, when its sequence sort, if it

has one, is the closed unit ball of a complex Hilbert space K, and when the functions $\lambda x+\lambda' y$ and the predicates $\mathrm{Re}\langle x \mid y\rangle$, $\mathrm{Im}\langle x \mid y\rangle$ and $\|x\|$ are the restrictions of the operations, the inner products and the norms of these spaces.

Besides T, we define in Section 6 below two theories T_0 and T_1 contained in it. The theory T_0 is formulated in the vocabulary without the measure and the sequence sort, and the theory T_1 in the vocabulary without the measure alone. A model of one of the theories T_0, T_1 and T is called *nonstandard* when it is not standard. The word has nothing to do with the nonstandard models of arithmetic or of nonstandard analysis: every predicate takes real values, and no element is infinitely large or infinitesimal. In Lemma 6.3 below, we show that the Hilbert sort of every model of T_0 maps onto a subset of the closed unit ball of a complex Hilbert space. The $\mathbb{Q}(i)$-linear span of that subset is dense in the space, and the map carries the functions $\lambda x + \lambda' y$ and the predicates of the Hilbert sort to the operations, the inner product and the norm of the space. Hence the same holds for the Hilbert sort of every model of T_1 or of T. In a standard model of T_0, the map is the identity of the whole ball. A nonstandard model of T_0 may lack some of the vectors of the ball, and it may contain distinct elements that the map identifies. The sequence sort, which lies outside the scope of Lemma 6.3, may fail to be the closed unit ball of a complex Hilbert space in a nonstandard model of T_1 or of T. Such a failure can occur whether or not the Hilbert sort is the whole ball.

Standardness enters some of our theorems as a hypothesis on the model rather than as a consequence of axioms. Indeed, no countable theory in one of the vocabularies of Definition 5.3 that contains T_0 and has a model has only standard models. Every standard model of T_0 has an uncountable Hilbert sort, the closed unit ball of an infinite-dimensional Hilbert space (see Proposition 7.1 below). The downward Löwenheim–Skolem theorem, by contrast, gives every such theory a model whose sorts are countable [ADI26, Lemma 3]. In particular, the theories T_0, T_1 and T (which have standard models by Theorem A(2) and by taking reducts) have nonstandard models whose sorts are countable. The conclusion of Theorem C applies to these countable models as well: in each of them, as in every model of T, the measure takes some value other than 0 and 1. The conclusions of Theorem A(1) and Theorem B, which concern the whole lattice of closed subspaces and the whole range of the measure, are by contrast proper to the standard models (see Remark 11.2).

⧖ **Historical note.** Carol Karp, a student of Henkin's, wrote her dissertation of 1959 on languages with expressions of infinite length and published a book on them in 1964 [Kar64]. In 1969 Makkai applied Smullyan's method of consistency properties to $L_{\omega_1\omega}$, building models in the manner of Henkin's completeness proof, and extended it to the infinitary logics on admissible sets [Mak69b, Mak69a]. In 1971 Keisler made the model existence theorem for consistency properties a central method of his book on $L_{\omega_1\omega}$ [Kei71, Chapter 3], which also contains the omitting types theorem for countable fragments [Kei71, Chapter 11]. In 1973 Keisler built countable models by forcing with finite sets of sentences satisfiable in a given class of models [Kei73, p. 100, Example 1.2]. In the same work he replaced principal types by $\forall\exists$ formulas over a set of formulas of a countable fragment [Kei73, p. 107]. His general omitting types theorem gives conditions under which a $\forall\exists$ class contains a countable model in which countably many given $\forall\exists$ sentences hold [Kei73, p. 108]. Keisler's finite conditions and $\forall\exists$ formulas have counterparts in the model existence game of Section 10 and in the contextual presentations of Section 12 (see Section A). In 1974 Morley derived the omitting types theorem for countable fragments from the Baire category theorem, applied to a space of countable structures, which he showed to be a Baire space [Mor74]. In 2001 Keisler and Miller carried the theorem into first-order logic without equality, where its usual formulation

does not hold. There the model is built from closed terms, and principal types give way to locally realized types [KM01, Section 4].

Model theory with truth values in the unit interval goes back to Chang's work of 1958 and 1961 on the infinite-valued logic of Łukasiewicz [Cha58, Cha61]. The continuous model theory of Chang and Keisler of 1966 took its truth values in a compact Hausdorff space [CK66]. Henson introduced the logic of positive bounded formulas in 1976 to study a Banach space through its nonstandard hull [Hen76]. In metric approaches to real-valued model theory, a distance predicate takes the place of equality, and continuity requirements ensure that the other symbols respect the resulting uniform structure.[3] Uniform continuity with moduli prescribed by the signature was the setting of Iovino's thesis of 1994 and of his papers of 1999 on stable Banach spaces [Iov94, Iov99a, Iov99b]; in 2002 Henson and Iovino presented the model theory of normed space structures in it [HI02, Sections 2 and 3]. In 2008 Ben Yaacov, Berenstein, Henson and Usvyatsov set out the model theory of metric structures in continuous logic [BYBHU08]. In 2009 Ben Yaacov and Iovino introduced an infinitary continuous logic for model-theoretic forcing [BYI09]. In 2010 Ben Yaacov and Usvyatsov built moduli of uniform continuity into the signatures of continuous logic [BYU10, Section 2]. In 2014 Eagle proved the omitting types theorem for infinitary $[0, 1]$-valued logic, whose metric structures have a master metric with respect to which the predicates are uniformly continuous [Eag14, Definitions 1.4 and 1.5]. In the model theory for real-valued structures that Keisler set out in 2023, a general structure requires no metric signature: its function and predicate symbols may be interpreted by arbitrary mappings, subject only to the boundedness of the predicates [Kei23, Sections 1 and 2.1]. General structures correspond to first-order structures without equality, as metric structures with a distance predicate correspond to first-order structures with equality [Kei23, Section 1]. For vocabularies with countably many function and predicate symbols, every general structure expands to one with a distance predicate that is a uniform limit of formulas [Kei23, Theorem 3.3.4]. The logic of this paper, whose conditions are strict inequalities, is the infinitary extension of the logic of Keisler's theory due to Aubone, Dueñez and Iovino, who build its models from closed terms and prove its omitting types theorem by Morley's method in the Čech-complete space of Henkin types [ADI26].

6. The axioms

In this section we state the axioms of the theory T of Theorem A and of two smaller theories contained in it. The theory T_0 governs the Hilbert sort and its relation to the Solèr sort. The theory T_1 adds the sequence sort and the spans of its sequences, and T adds the measure. This division separates three concerns: the representation of the lattice by projections, the completeness that the lattice acquires in standard models, and the quantum probability measure. Interpreted in a standard model, the axioms first make the constants e_n an orthonormal basis, and they then attach to each element of the Solèr sort an orthogonal projection acting on the Hilbert sort. These projections are combined, by the axioms, as the closed subspaces onto which they project are combined. In the same reading, the function at sends each nonzero vector to the element whose members are its multiples, while rep chooses a unit vector in each nonzero element. The function span sends each sequence of the sequence sort to the element of the Solèr sort whose members are the vectors of the unit ball lying in the closed span of the entries of that sequence. The last axioms concern the measure μ,

[3] In Łukasiewicz predicate logic, the congruence axioms for a similarity, or graded equality, make every predicate 1-Lipschitz for the distance given by one minus the similarity (see [Háj98, Section 5.6] and the earlier work collected there).

which must respect equality and take its values in $[0,1]$, with the value 1 at the greatest element of the Solèr sort. The measure must, moreover, be additive on orthogonal elements and continuous along the finite joins that exhaust a span. The axioms fall into nine groups: one for each of the three sorts, four for the remaining symbols that connect the sorts, the group (Tr) for the truncations, and the group (M) for the measure. The groups for the sorts are (H), (OL) and (K), the last of which also governs the functions ι_n, κ_n and θ_N. The groups (P) and (Sub) concern the projection π and its relation to the lattice operations, the group (At) governs at and rep, and the group (Sp) governs span.

Free variables are universally quantified: the variables x, y, z range over $\mathbf{H}$, the variables s, t over $\mathbf{K}$, and the variables a, b, c over $\mathbf{L}$. The letters λ, λ' range over the elements of $\mathbb{Q}(i)$ with $|\lambda| + |\lambda'| \leq 1$, and the letters n, m, k, N range over $\mathbb{N}$. These letters are not variables of the logic: an axiom in which they occur free stands for countably many axioms, one for each choice of their values. A bound occurrence of one of these letters, by contrast, indexes a finite sum, a countable join or a countable infimum. Each axiom is understood in the sense of Section 5, as a closed condition required of all values of its free variables. The equations written with $\doteq$ between vector terms and with $=$ between terms of the Solèr sort are the abbreviations introduced in Section 5.3. As in Section 5, an implication between two-valued formulas θ and θ' is asserted by the axiom $-\theta \vee \theta' \geq 1$, and an implication from a two-valued formula θ to a closed condition $\psi \leq 0$ by the axiom $\theta \wedge \psi \leq 0$. Some axioms are followed by a reading in words, in italics, which describes what the axiom says in standard models. For $N \in \mathbb{N}$, let b_N be the closed term $\mathrm{at}(e_0) \vee \cdots \vee \mathrm{at}(e_{N-1})$ of the Solèr sort, with $b_0 = 0$.

(H) The Hilbert sort.

(H1) *Inner product space.* For any two terms built from variables by the functions $\lambda x + \lambda' y$ that denote the same $\mathbb{Q}(i)$-linear combination of the variables, the equation between them; the closed conditions $\langle \lambda x + \lambda' y \mid z \rangle = \lambda \langle x \mid z \rangle + \lambda' \langle y \mid z \rangle$ and $\langle y \mid x \rangle = \overline{\langle x \mid y \rangle}$, each written as two real closed conditions; and $\|x\| \geq 0$ and $\|x\|^2 = \mathrm{Re}\langle x \mid x \rangle$.

(H2) *Separability.* $\langle e_n \mid e_m \rangle = \delta_{nm}$, where δ_{nm} is the Kronecker delta, and

$$\bigwedge_N \left\| \tfrac{1}{2} x - \tfrac{1}{2} \pi(b_N, x) \right\| \leq 0.$$

The constants e_n are orthonormal, and every vector is the limit of its projections onto the spans of $e_0, \ldots, e_{N-1}$.

(OL) The Solèr sort.

(OL) The equations of ortholattices:

$$a \wedge b = b \wedge a, \quad a \wedge (b \wedge c) = (a \wedge b) \wedge c, \quad a \wedge (a \vee b) = a,$$
$$a \vee b = b \vee a, \quad a \vee (b \vee c) = (a \vee b) \vee c, \quad a \vee (a \wedge b) = a,$$
$$a \wedge 0 = 0, \quad a \vee 1 = 1, \quad a^{\perp\perp} = a, \quad (a \wedge b)^{\perp} = a^{\perp} \vee b^{\perp}, \quad a \wedge a^{\perp} = 0.$$

(P) Each element of the Solèr sort determines a projection.

(P1) $\pi(a, \lambda x + \lambda' y) \doteq \lambda \pi(a, x) + \lambda' \pi(a, y)$.

(P2) $\pi(a, \pi(a, x)) \doteq \pi(a, x)$.

(P3) $\langle \pi(a, x) \mid y \rangle = \langle x \mid \pi(a, y) \rangle$.

Axioms (Sub1) and (Sub2) express norm convergence by a limit superior: for a sequence $(d_n)_{n\in\mathbb{N}}$ of nonnegative real numbers, the infimum over N of the suprema of the tails $(d_n)_{n\geq N}$ vanishes exactly when the sequence converges to zero. Here the sequences are the errors in the alternating-projection approximations to the meet and the join; in Lemma 6.4 we identify their limits by von Neumann's alternating projection theorem.

(Sub) The operations of the Solèr sort are those of subspaces.

(Sub0) $\pi(0,x) \doteq 0$ and $\pi(1,x) \doteq x$.
Only the zero vector belongs to 0*; every vector belongs to* 1.

(Sub1) $\bigwedge_N \bigvee_{n\geq N} \left\|\frac{1}{2}\pi(a\wedge b,x) - \frac{1}{2}(P_aP_b)^n x\right\| \leq 0$, where $(P_aP_b)^n x$ is the term obtained by applying $\pi(b,\cdot)$ and then $\pi(a,\cdot)$, n times.
The members of $a\wedge b$ are the common members of a and b.

(Sub2) $\bigwedge_N \bigvee_{n\geq N} \left\|\frac{1}{4}\pi(a\vee b,x) - \frac{1}{4}x + \frac{1}{4}(P_{a^\perp}P_{b^\perp})^n x\right\| \leq 0$, where $(P_{a^\perp}P_{b^\perp})^n x$ is the term of (Sub1) for $a^\perp$ and $b^\perp$, and the vector inside the norm is denoted by a term built from the functions $\lambda x + \lambda' y$.
The closed subspace spanned by the members of $a\vee b$ is the closure of the sum of the closed subspaces spanned by the members of a and by those of b.

(Sub3) $\frac{1}{2}\pi(a^\perp,x) \doteq \frac{1}{2}x - \frac{1}{2}\pi(a,x)$.
The members of $a^\perp$ are the vectors orthogonal to all members of a.

(At) Atoms and unit vectors.

(At1) $\pi(\mathrm{at}(x),x) \doteq x$, and $\|\pi(\mathrm{at}(x),y)\|^2\,\|x\|^2 = |\langle y \mid x\rangle|^2$.
Every vector x belongs to $\mathrm{at}(x)$, and for $x\neq 0$ the projection onto $\mathrm{at}(x)$ is the projection onto the line spanned by x.

(At2) $\pi(a,\mathrm{rep}(a)) \doteq \mathrm{rep}(a)$; $\|\mathrm{rep}(a)\| \wedge \big(1 - \|\mathrm{rep}(a)\|\big) \leq 0$; and $\|\pi(a,x)\| \leq \|\mathrm{rep}(a)\|$.
The vector $\mathrm{rep}(a)$ belongs to a; its norm is 0 or 1; and it is 1 unless a has no nonzero member.

(At3) $\|\mathrm{rep}(\mathrm{at}(x))\| \leq \lceil \|x\| > 0 \rceil$.
If $\|x\| = 0$, then $\mathrm{at}(x)$ is 0.

(K) The sequence sort.

(K1) The axioms (H1), written for $\mathbf{K}$.

(K2) $\iota_n(\lambda x + \lambda' y) \doteq \lambda\iota_n x + \lambda'\iota_n y$ and $\kappa_n(\lambda s + \lambda' t) \doteq \lambda\kappa_n s + \lambda'\kappa_n t$; $\langle \iota_n x \mid \iota_m y\rangle = \delta_{nm}\langle x \mid y\rangle$; and $\langle \kappa_n s \mid x\rangle = \langle s \mid \iota_n x\rangle$.

(K3) $\langle \theta_N s \mid t\rangle = \sum_{n<N} \langle \kappa_n s \mid \kappa_n t\rangle$, written as two real closed conditions, and $\bigwedge_N \left\|\frac{1}{2}s - \frac{1}{2}\theta_N s\right\| \leq 0$.
The sequence $\theta_N s$ keeps the first N entries of s, and these truncations converge to s.

(K4) $\kappa_n\varepsilon \doteq 2^{-n-1}e_n$.

(Sp) Spans.

(Sp1) $\pi(\mathrm{span}(s), \kappa_n s) \doteq \kappa_n s$.
Every entry of s belongs to $\mathrm{span}(s)$.

(Sp2) $\Big(-\bigwedge_n [\mathrm{at}(\kappa_n s) \le b]\Big) \vee [\mathrm{span}(s) \le b] \ge 1$.
If every entry of s belongs to b, then $\mathrm{span}(s) \le b$.

(Tr) Truncations.

(Tr) For every predicate symbol R of $\mathcal{L}$ other than $\mathbf{1}$ and the truncation symbols, every positive integer n and every rational number q:
$$R^+_{n,q}(\bar{x}) = \big((n \cdot (R(\bar{x}) - q)) \wedge 1\big) \vee 0, \qquad R^-_{n,q}(\bar{x}) = \big((n \cdot (q - R(\bar{x}))) \wedge 1\big) \vee 0.$$

For each n and q, the axiom (Tr) consists of two equations between formulas whose a priori bounds grow with n; only the countable joins $\lceil R(\bar{t}) > q \rceil$ and $\lceil R(\bar{t}) < q \rceil$ of Section 5.3 gather the truncations for all n, and their members have a priori bound 1. The axioms (Tr) for the predicates of the Hilbert sort belong to T_0, those for the predicates of the sequence sort to T_1, and those for μ to T.

Axiom (M5) below compares the measure of the span of a sequence with the measures of the finite joins of the atoms of its entries. For $N \in \mathbb{N}$, let $c_N(s)$ be the term $\mathrm{at}(\kappa_0 s) \vee \cdots \vee \mathrm{at}(\kappa_{N-1} s)$ of the Solèr sort, with $c_0(s) = 0$.

(M) The measure.

(M1) $\mu(a) \ge 0$ and $\mu(a) \le 1$.

(M2) $\mu(1) = 1$.

(M3) $[a = b] \wedge |\mu(a) - \mu(b)| \le 0$.
Equal elements have equal measures.

(M4) $\mu\big(a \vee (b \wedge a^\perp)\big) = \mu(a) + \mu(b \wedge a^\perp)$.
The measure is additive on orthogonal elements.

(M5) $\mu(\mathrm{span}(s)) = \bigvee_N \mu(c_N(s))$.
The measure of the span of a sequence is the supremum of the measures of the joins of the atoms of its first N entries.

Axiom (M4) needs no hypothesis of orthogonality, because the element $b \wedge a^\perp$ is orthogonal to a for every b, and every element c orthogonal to a is equal to $c \wedge a^\perp$, whence (M3) carries (M4) over to c. Axiom (M5) states continuity along an increasing sequence of elements whose join is a span; together with (M4), it yields σ-additivity (see the proof of Theorem A).

Definition 6.1. The theory T_0 consists of the axioms (H), (OL), (P), (Sub), (At) and the axioms (Tr) for the predicates of the Hilbert sort, in the vocabulary obtained from $\mathcal{L}$ by removing the sequence sort with its constant ε, the functions ι_n, κ_n, θ_N and span, and the predicate μ, together with the truncations of the removed predicates. The theory T_1 consists of T_0 together with (K), (Sp) and the axioms (Tr) for the predicates of the sequence sort, in the vocabulary $\mathcal{L}$ without μ and its truncations; the theory T consists of T_1 together with (M) and the axioms (Tr) for μ.

The analysis of the models of T_0 rests on a Hilbert space built from the Hilbert sort, and on von Neumann's alternating projection theorem, which reaches the projection onto the intersection of two closed subspaces by projecting alternately onto each.

Theorem 6.2 (von Neumann's alternating projection theorem [vN49, Lemma 22]**).** *Let U and V be closed subspaces of a Hilbert space, and let P_U, P_V and $P_{U\cap V}$ be the orthogonal projections onto U, V and $U \cap V$. Then, for every vector x of the space, the sequence $\big((P_U P_V)^n x\big)_{n\in\mathbb{N}}$ converges in norm to $P_{U\cap V}x$.*

Lemma 6.3. *Let $\mathfrak{M}$ be a model of T_0. There are a complex Hilbert space $\bar{E}$, a subset Y of the closed unit ball of $\bar{E}$ whose $\mathbb{Q}(i)$-linear span E is dense in $\bar{E}$, and a map $x \mapsto \hat{x}$ from the Hilbert sort of $\mathfrak{M}$ onto Y with the following properties. The map sends $\lambda x + \lambda' y$ to $\lambda\hat{x} + \lambda'\hat{y}$, and the predicates $\langle x \mid y\rangle$ and $\|x\|$ to $\langle \hat{x} \mid \hat{y}\rangle$ and $\|\hat{x}\|$. For every element a of the Solèr sort, the map $\hat{x} \mapsto \widehat{\pi(a,x)}$ is well defined and extends uniquely to an orthogonal projection $\bar{P}_a$ of $\bar{E}$. If $\mathfrak{M}$ is standard, then $\bar{E}$ is the Hilbert space whose closed unit ball is the Hilbert sort, and the map $x \mapsto \hat{x}$ is the identity.*

Proof. The idea is to recover a vector space from the Hilbert sort by formal multiples. By (H1), the form given by the pair of predicates $\langle x \mid y\rangle$ is hermitian and satisfies $\mathrm{Re}\langle x \mid x\rangle = \|x\|^2 \geq 0$; the Cauchy–Schwarz inequality therefore holds for it. Each pair (k, x), with $k \geq 1$ an integer and x an element of the Hilbert sort, stands for kx. We define the sum of two pairs by $(k, x) + (l, y) = (2kl, \frac{1}{2l}x + \frac{1}{2k}y)$ and, for an element λ of $\mathbb{Q}(i)$, the multiple $\lambda(k, x) = (km, \frac{\lambda}{m}x)$, where m is a positive integer with $m \geq |\lambda|$; the form is given by $\langle (k, x) \mid (l, y)\rangle = kl\langle x \mid y\rangle$. We identify two pairs when the form vanishes on their difference. Axiom (H1) makes the quotient a $\mathbb{Q}(i)$-vector space E with a positive definite hermitian form, and we let $\bar{E}$ be its completion and $\hat{x}$ the class of $(1, x)$. For an element a of the Solèr sort, consider the assignment Q that sends a pair (k, x) to $(k, \pi(a, x))$; by (P1)–(P3), it respects sums and multiples, it is idempotent and it is symmetric with respect to the form, in each case up to pairs on which the form vanishes. For every pair v, symmetry and idempotence therefore give $\langle Qv \mid Qv\rangle = \langle v \mid Qv\rangle$, which is at most $\langle v \mid v\rangle^{1/2}\langle Qv \mid Qv\rangle^{1/2}$ by the Cauchy–Schwarz inequality, whence $\langle Qv \mid Qv\rangle \leq \langle v \mid v\rangle$. In particular, the assignment Q sends pairs on which the form vanishes to such pairs, and it therefore induces on E a $\mathbb{Q}(i)$-linear, idempotent, symmetric and contractive map Q_a, which extends by continuity to a self-adjoint idempotent $\bar{P}_a$ of $\bar{E}$. In a standard model, the Hilbert sort is the closed unit ball of a Hilbert space H, and the form is positive definite on it. In that case, the map sending the class of (k, x) to kx identifies E with H, since every vector of H is a positive integer multiple of a vector of that ball. As H is complete, it follows that $\bar{E} = H$ and that $x \mapsto \hat{x}$ is the identity of the ball. ☺

For every element a of the Solèr sort of a model of T_0, we write S_a for the range of $\bar{P}_a$. In the following lemma we show that the formulas $[a \neq 0]$, $[a \leq b]$ and $[a = b]$ of Section 5.3 are two-valued in every model of T_0, and that they express, respectively, that S_a is nonzero, that S_a is contained in S_b, and that the two subspaces are equal.

Lemma 6.4. *Let $\mathfrak{M}$ be a model of T_0, and let $\bar{E}$ and $\bar{P}_a$ be as in Lemma 6.3.*

1. *We have $\bar{P}_0 = 0$, $\bar{P}_1 = \mathrm{Id}$ and $\bar{P}_{a^\perp} = \mathrm{Id} - \bar{P}_a$; moreover, $S_{a\wedge b} = S_a \cap S_b$ and $S_{a\vee b} = \overline{S_a + S_b}$.*
2. *The norm $\|\mathrm{rep}(a)\|$ is 1 if $S_a \neq \{0\}$ and 0 otherwise; hence $[a \neq 0]$ is two-valued, with the value 1 exactly when $S_a \neq \{0\}$.*
3. *The formula $[a \leq b]$ is two-valued, with the value 1 exactly when $S_a \subseteq S_b$, and the formula $[a = b]$ is two-valued, with the value 1 exactly when $S_a = S_b$.*

(4) *For every x with $\|x\| > 0$, the subspace $S_{\mathrm{at}(x)}$ is $\mathbb{C}\hat{x}$, and for every x with $\|x\| = 0$ it is $\{0\}$.*

Consequently, the relation $[a = b] = 1$ is an equivalence relation on the Solèr sort that is respected by $\wedge$, $\vee$ and $^\perp$. The quotient $\mathbf{L}^{\mathfrak{M}}/{=}$ by this relation is an ortholattice, and the map that sends the class of a to S_a is an embedding of ortholattices into the lattice $L(\bar{E})$ of closed subspaces of $\bar{E}$, which carries the order of the quotient to inclusion.

Proof. (1) The first three equalities follow from (Sub0) and (Sub3) by continuity, since E is dense in $\bar{E}$. By (Sub1), the projection $\bar{P}_{a\wedge b}$ agrees on the vectors $\hat{x}$, and hence on $\bar{E}$, with the strong limit of the operators $(\bar{P}_a\bar{P}_b)^n$, which is the projection onto $S_a \cap S_b$ by Theorem 6.2. Likewise, axiom (Sub2) and the equality $\bar{P}_{c^\perp} = \mathrm{Id} - \bar{P}_c$ show that $\bar{P}_{a\vee b} = \mathrm{Id} - P_{S_a^\perp \cap S_b^\perp}$, whose range is $(S_a^\perp \cap S_b^\perp)^\perp = \overline{S_a + S_b}$.

(2) By (At2), together with the axiom $\|x\| \geq 0$ of (H1), the norm $\|\mathrm{rep}(a)\|$ is 0 or 1; by (At2) again, the vector $\widehat{\mathrm{rep}(a)}$ lies in S_a, and $\|\bar{P}_a\hat{x}\| \leq \|\mathrm{rep}(a)\|$ for every x. If $S_a \neq \{0\}$, then $\bar{P}_a$ does not vanish on the dense subspace E, and hence not on some $\hat{x}$, whence $\|\mathrm{rep}(a)\| > 0$ and so $\|\mathrm{rep}(a)\| = 1$. If $S_a = \{0\}$, then $\widehat{\mathrm{rep}(a)} \in S_a$ gives $\|\mathrm{rep}(a)\| = 0$.

(3) By (1), the subspace $S_{a\wedge(a\wedge b)^\perp}$ is $S_a \cap (S_a \cap S_b)^\perp$, the orthogonal complement of $S_a \cap S_b$ in S_a. It is $\{0\}$ exactly when $S_a \cap S_b = S_a$, that is, when $S_a \subseteq S_b$. By (2), the formula $[a \leq b]$ is therefore two-valued, with the value 1 exactly when $S_a \subseteq S_b$, and the assertion about $[a = b]$ follows.

(4) Let $P = \bar{P}_{\mathrm{at}(x)}$ and suppose that $\|x\| > 0$, so that $\hat{x} \neq 0$. By (At1), the vector $\hat{x}$ lies in $S_{\mathrm{at}(x)}$, and $\|P\hat{y}\|^2\|\hat{x}\|^2 = |\langle \hat{y} \mid \hat{x}\rangle|^2$ for every y. Let P_x be the projection onto $\mathbb{C}\hat{x}$, which satisfies $\|P_x\hat{y}\|^2\|\hat{x}\|^2 = |\langle \hat{y} \mid \hat{x}\rangle|^2$ as well; since the vector $\hat{x}$ lies in the range of P, we have $P_x \leq P$. Then $\|P\hat{y}\|^2 = \|P_x\hat{y}\|^2 + \|(P - P_x)\hat{y}\|^2$ gives $(P - P_x)\hat{y} = 0$ for every y, whence $P = P_x$ by density. Finally, if $\|x\| = 0$, then (At3) and the axioms (Tr) for the norm give $\|\mathrm{rep}(\mathrm{at}(x))\| = 0$, whence $S_{\mathrm{at}(x)} = \{0\}$ by (2).

Part (3) shows that the relation $[a = b] = 1$ holds exactly when $S_a = S_b$, and the relation is therefore an equivalence relation. Since, by (1), the subspaces $S_{a\wedge c}$, $S_{a\vee c}$ and $S_{a^\perp}$ are determined by S_a and S_c, the relation is respected by the operations. The same part (1) shows that the map sending the class of a to S_a preserves the operations; this map is injective by (3), and its image is therefore a sub-ortholattice of $L(\bar{E})$ isomorphic to the quotient. Finally, the order of the quotient, in which $a \leq b$ means $a \wedge b = a$, corresponds to inclusion, again by (3). ☺

In Lemma 6.4, the axioms (OL) are not used; they hold in every structure that satisfies the other axioms of T_0, by the embedding into $L(\bar{E})$. We keep them, so that the ortholattice structure of the Solèr sort has axioms of its own (see Remark 11.6).

⧖ **Historical note.** In the continuous logic set out by Ben Yaacov, Berenstein, Henson and Usvyatsov in 2008, the unit ball of a Banach space carries the maps $(x, y) \mapsto \alpha x + \beta y$ with $|\alpha| + |\beta| \leq 1$ as functions. In that logic, a Hilbert space carries in addition its inner product as a predicate [BYBHU08], and the Hilbert sort follows both conventions. Von Neumann's alternating projection theorem, on which axioms (Sub1) and (Sub2) rest, comes from lecture notes that he distributed in mimeographed form in 1933. The notes were printed in 1950 as the second volume of his *Functional Operators* [vN50].

7. Standard models of T_0: pre-Solèr lattices

In this section we study the standard models of T_0, the theory whose vocabulary has neither the sequence sort nor the measure. Since the vocabulary has no equality symbol, two elements of the Solèr sort may have the same members, and equality on the Solèr sort is the relation $[a = b] = 1$ of Section 5.3. The object of study is therefore the quotient of the Solèr sort by this relation, which is an ortholattice by Lemma 6.4. We show that this quotient is a pre-Solèr lattice, that is, an ortholattice that satisfies every condition in the definition of a Solèr lattice except, possibly, completeness. To each element of the Solèr sort corresponds the closed subspace spanned by its members, and this correspondence induces on the quotient an embedding of ortholattices into the lattice of closed subspaces of the Hilbert space whose closed unit ball is the Hilbert sort. We show further that the image of this embedding is join-dense and meet-dense in that lattice, in the order-theoretic sense of Section 1, whence the lattice of closed subspaces is, up to isomorphism, the MacNeille completion of the quotient. Gleason's theorem already holds at this level, in the sense that the σ-additive states on the quotient, defined below, are given by density operators. Completeness, the one condition that the quotient may lack, need not hold (see Example 7.5); it is obtained in Section 8 from the sequence sort.

Throughout this section, we fix a standard model $\mathfrak{M}$ of T_0. We write H for the Hilbert space whose closed unit ball is the Hilbert sort of $\mathfrak{M}$ (see Definition 5.3), and $\mathbf{L}^{\mathfrak{M}}$ for the Solèr sort of $\mathfrak{M}$. By Lemma 6.3, the space $\bar{E}$ of that lemma is H. For each element a of $\mathbf{L}^{\mathfrak{M}}$, we write P_a for the projection $\bar{P}_a$, which extends $\pi(a, \cdot)$, and S_a for the range of P_a. This range is the closed subspace of H spanned by the members of a, which are the vectors x of the unit ball with $\pi(a, x) = x$. We write $\tilde{a}$ for the class of a in $\mathbf{L}^{\mathfrak{M}}/{=}$; by Lemma 6.4, we have $\tilde{a} = \tilde{b}$ exactly when $S_a = S_b$.

Proposition 7.1. *The space H is a separable infinite-dimensional complex Hilbert space, and the constants e_n of the Hilbert sort, for $n \in \mathbb{N}$, form an orthonormal basis of H.*

Proof. By (H2), the vectors e_n are orthonormal. By Lemma 6.4, the projection P_{b_N} is the projection onto the span of $e_0, \dots, e_{N-1}$, and (H2) places every vector of the unit ball, and hence every vector of H, in the closed span of the vectors e_n. Thus $(e_n)_{n\in\mathbb{N}}$ is an orthonormal basis of H, which is therefore separable and infinite-dimensional; it is complex by Definition 5.3. ☺

Proposition 7.2. *The map $\tau\colon \tilde{a} \mapsto S_a$ is an embedding of ortholattices from $\mathbf{L}^{\mathfrak{M}}/{=}$ into the lattice $L(H)$ of closed subspaces of H; that is, it is injective and satisfies*

$$S_0 = \{0\}, \quad S_1 = H, \quad S_{a\wedge b} = S_a \cap S_b, \quad S_{a\vee b} = \overline{S_a + S_b}, \quad S_{a^\perp} = S_a^\perp.$$

Moreover, the relation $\tilde{a} \le \tilde{b}$ holds if and only if $S_a \subseteq S_b$, that is, if and only if $[a \le b] = 1$.

Proof. The proposition restates the last assertion of Lemma 6.4 for the standard model $\mathfrak{M}$, in which $\bar{E} = H$; part (3) of that lemma gives the description of the order. ☺

Proposition 7.3. *For every nonzero x in the unit ball, the subspace $S_{\widetilde{\mathrm{at}(x)}}$ is $\mathbb{C}x$. The atoms of $\mathbf{L}^{\mathfrak{M}}/{=}$ are exactly the classes $\widetilde{\mathrm{at}(x)}$ with $x \neq 0$, and the map $x \mapsto \widetilde{\mathrm{at}(x)}$ induces a bijection from the unit sphere of H, modulo multiplication by complex numbers of modulus one, onto the set of atoms of $\mathbf{L}^{\mathfrak{M}}/{=}$, whose inverse sends an atom $\tilde{a}$ to the class of $\mathrm{rep}(a)$.*

Proof. The first assertion is part (4) of Lemma 6.4, since in a standard model every nonzero vector x has positive norm. Since the ray $\mathbb{C}x$ is an atom of $L(H)$ and the map τ is an

order embedding, the class $\widetilde{\mathrm{at}(x)}$ is an atom of $\mathbf{L}^{\mathfrak{M}}/{=}$. Let $\tilde{a}$ be an atom of $\mathbf{L}^{\mathfrak{M}}/{=}$. By part (2) of Lemma 6.4, the vector $\mathrm{rep}(a)$ is a unit vector, and it belongs to a by (At2). Hence $S_{\mathrm{at}(\mathrm{rep}(a))} = \mathbb{C}\mathrm{rep}(a)$ is a nonzero subspace of S_a, and $\widetilde{\mathrm{at}(\mathrm{rep}(a))}$ is a nonzero element below $\tilde{a}$; since $\tilde{a}$ is an atom, we have $\tilde{a} = \widetilde{\mathrm{at}(\mathrm{rep}(a))}$. For unit vectors x and y, the classes $\widetilde{\mathrm{at}(x)}$ and $\widetilde{\mathrm{at}(y)}$ coincide exactly when $\mathbb{C}x = \mathbb{C}y$, that is, when y is a multiple of x by a complex number of modulus one. In particular, for a unit vector x, the unit vector $\mathrm{rep}(\mathrm{at}(x))$ is such a multiple of x, because it lies in $S_{\mathrm{at}(x)} = \mathbb{C}x$. The inverse of the bijection is therefore as stated. ☺

Theorem 7.4. *Let $\mathfrak{M}$ be a standard model of T_0, let H be the Hilbert space whose closed unit ball is its Hilbert sort, and, for each element a of its Solèr sort $\mathbf{L}^{\mathfrak{M}}$, let S_a be the closed subspace of H spanned by the members of a. Then $\mathbf{L}^{\mathfrak{M}}/{=}$ is a pre-Solèr lattice, and the image of the map $\tau\colon \tilde{a} \mapsto S_a$ is join-dense and meet-dense in the lattice $L(H)$ of closed subspaces of H, which is therefore isomorphic to the MacNeille completion of $\mathbf{L}^{\mathfrak{M}}/{=}$. The image of τ contains every finite-dimensional subspace of H and the orthogonal complement of each, and the map τ sends every join that exists in $\mathbf{L}^{\mathfrak{M}}/{=}$ to the corresponding join in $L(H)$.*

Proof. We begin with the finite-dimensional subspaces and then verify, condition by condition, that $\mathbf{L}^{\mathfrak{M}}/{=}$ is a pre-Solèr lattice. The zero subspace is S_0; now let F be a nonzero finite-dimensional subspace of H with a basis $x_1, \dots, x_k$ of unit vectors. By Propositions 7.2 and 7.3, the map τ sends $(\mathrm{at}(x_1) \vee \cdots \vee \mathrm{at}(x_k))^{\sim}$ to the closure of F, which is F itself, and $((\mathrm{at}(x_1) \vee \cdots \vee \mathrm{at}(x_k))^{\perp})^{\sim}$ to $F^{\perp}$. Condition (OM) holds because $\tau(\mathbf{L}^{\mathfrak{M}}/{=})$ is a sub-ortholattice of the orthomodular lattice $L(H)$. For (A), let $\tilde{a}$ be a nonzero element of $\mathbf{L}^{\mathfrak{M}}/{=}$; the atoms below $\tilde{a}$ are the classes $\widetilde{\mathrm{at}(x)}$ with $x \in S_a$ and $0 < \|x\| \leq 1$. Any upper bound $\tilde{b}$ of these atoms satisfies $S_b \supseteq S_a$, hence $\tilde{b} \geq \tilde{a}$, which makes $\tilde{a}$ their join. For (CP), let $\tilde{a}$ be an element of $\mathbf{L}^{\mathfrak{M}}/{=}$, and let $\widetilde{\mathrm{at}(x)}$ be an atom not below $\tilde{a}$; then the subspace S_a does not contain x. By Proposition 7.2 and Lemma 2.7, the subspace $S_{a \vee \mathrm{at}(x)}$ is $S_a + \mathbb{C}x$, in which S_a has codimension one. Since τ is an order embedding, no element lies strictly between $\tilde{a}$ and $(a \vee \mathrm{at}(x))^{\sim}$. Condition (I) holds because the join of two distinct atoms corresponds to a plane, which contains infinitely many rays, all in $\tau(\mathbf{L}^{\mathfrak{M}}/{=})$. The atoms $\widetilde{\mathrm{at}(e_n)}$, for $n \in \mathbb{N}$, give (D). For (S), take $p_n = \widetilde{\mathrm{at}(e_n)}$ and $c_n = \widetilde{\mathrm{at}(\tfrac{1}{2}(e_n + e_{n+1}))}$; since τ is injective and preserves the lattice operations and the orthocomplementation, it suffices to verify the condition for the images of these atoms. These images are the rays of e_n and of $e_n + e_{n+1}$, where the condition follows from (1) and the computation following it, because $\|e_n\| = \|e_{n+1}\|$.

Each closed subspace S of H is the join in $L(H)$ of its rays, and it equals $\bigcap\{y^{\perp} : y \in S^{\perp},\ 0 < \|y\| \leq 1\}$, where each $y^{\perp} = S_{\mathrm{at}(y)^{\perp}}$ lies in the image of τ. These two facts make the image of τ join-dense and meet-dense in $L(H)$. By Proposition 1.5, the lattice $L(H)$ is therefore isomorphic to the MacNeille completion of $\mathbf{L}^{\mathfrak{M}}/{=}$. Finally, let $A \subseteq \mathbf{L}^{\mathfrak{M}}/{=}$ have a join $\tilde{c}$ in $\mathbf{L}^{\mathfrak{M}}/{=}$, and let W be the closed span of the subspaces S_a, for $\tilde{a} \in A$; clearly $W \subseteq S_c$. If y is a vector of $S_c \cap W^{\perp}$ with $0 < \|y\| \leq 1$, then every element of A lies below $(\mathrm{at}(y)^{\perp})^{\sim}$, and hence so does $\tilde{c}$. The vector y, which lies in S_c, is then orthogonal to itself, which is absurd. Since $S_c = W + (S_c \cap W^{\perp})$ by the projection theorem, it follows that $S_c = W$. ☺

Example 7.5. Let L_{fc} (for "finite dimension or finite codimension") be the set of finite-dimensional subspaces of the space ℓ^2 of square-summable complex sequences and their orthogonal complements. By the argument of Example 4.3, the set L_{fc} is a sub-ortholattice of $L(\ell^2)$. Take the closed unit ball of ℓ^2 as Hilbert sort and the standard unit vectors as the

constants e_n, and take L_{fc}, with its lattice operations and orthocomplementation, as Solèr sort. Put $\pi(S,x) = P_S x$ and $\mathrm{at}(x) = \mathbb{C}x$, let $\mathrm{rep}(S)$ be a unit vector chosen in S for each $S \neq \{0\}$, and let $\mathrm{rep}(\{0\})$ be 0. Each truncation $R^{\pm}_{n,q}$ of a predicate of the Hilbert sort is interpreted by its intended value $\min\bigl(1, \max(0, \pm n(R-q))\bigr)$. Axioms (Sub1) and (Sub2) hold by Theorem 6.2, and the other axioms of T_0 by construction; the result is therefore a standard model of T_0, in which $[S = S'] = 1$ exactly when $S = S'$. Its Solèr sort L_{fc}, which is its own quotient, is not complete. Indeed, an upper bound in L_{fc} of the rays $\mathbb{C}e_{2n}$, for $n \in \mathbb{N}$, is infinite-dimensional and hence of the form $G^{\perp}$ for a finite-dimensional subspace G orthogonal to every e_{2n}. The closed span of the vectors e_{2n+1} is infinite-dimensional, whereas G is finite-dimensional; that span therefore contains a unit vector y orthogonal to G. The subspace $(G + \mathbb{C}y)^{\perp}$, which contains every e_{2n} but not y, is an upper bound of the same rays that lies strictly below $G^{\perp}$. The rays $\mathbb{C}e_{2n}$ therefore have no join in L_{fc}.

In the form we need, Gleason's theorem concerns two notions: density operators, by which quantum mechanics represents the states of a system, and σ-additive states on an ortholattice. A *density operator* on H is a positive operator ρ whose trace $\operatorname{Tr}\rho = \sum_k \langle \rho u_k \mid u_k \rangle$, computed in any orthonormal basis $(u_k)_{k\in\mathbb{N}}$ of H, equals 1. A *σ-additive state* on an ortholattice Λ is a map $\nu\colon \Lambda \to [0,1]$ with $\nu(1) = 1$ such that $\nu(\bigvee_{n\in J} a_n) = \sum_{n\in J} \nu(a_n)$ for every family $(a_n)_{n\in J}$ of mutually orthogonal elements of Λ, indexed by a set $J \subseteq \mathbb{N}$, whose join exists in Λ.

Theorem 7.6 (Gleason's theorem for T_0). *Let $\mathfrak{M}$ be a standard model of T_0, let H be the Hilbert space whose closed unit ball is its Hilbert sort, and, for each element a of its Solèr sort $\mathbf{L}^{\mathfrak{M}}$, let P_a be the orthogonal projection onto the closed subspace spanned by the members of a. The σ-additive states on $\mathbf{L}^{\mathfrak{M}}/{=}$ are exactly the maps $\tilde{a} \mapsto \operatorname{Tr}(\rho P_a)$, where ρ ranges over the density operators on H, and each such state determines its density operator.*

Proof. We first verify that density operators give σ-additive states, and then recover a density operator from a state by applying Gleason's theorem to its values on rays. The map $\tilde{a} \mapsto \operatorname{Tr}(\rho P_a)$ is well defined, since P_a depends only on $\tilde{a}$. Let ρ be a density operator on H. Suppose that the elements $\tilde{a}_n$, for n in a set $J \subseteq \mathbb{N}$, are mutually orthogonal and have a join $\tilde{c}$ in $\mathbf{L}^{\mathfrak{M}}/{=}$. By Proposition 7.2 and Theorem 7.4, the subspace S_c is then the closure of the sum of the mutually orthogonal subspaces S_{a_n}, whence $P_c x = \sum_{n\in J} P_{a_n} x$ for every $x \in H$. We compute the traces in an orthonormal basis of H assembled from orthonormal bases of the subspaces S_{a_n} and of $S_c^{\perp}$; since the terms $\langle \rho u \mid u \rangle$ are nonnegative, they may be grouped by n, and we obtain $\operatorname{Tr}(\rho P_c) = \sum_{n\in J} \operatorname{Tr}(\rho P_{a_n})$. Moreover, the map $\tilde{a} \mapsto \operatorname{Tr}(\rho P_a)$ takes the value $\operatorname{Tr}\rho = 1$ at $\tilde{1}$, and its values lie in $[0,1]$. Conversely, let ν be a σ-additive state on $\mathbf{L}^{\mathfrak{M}}/{=}$. Additivity on the orthogonal pair $\tilde{1}, \tilde{0}$, whose join is $\tilde{1}$, gives $\nu(\tilde{0}) = 0$. For a unit vector x, put $f(x) = \nu(\widetilde{\mathrm{at}(x)})$. For $S_a \neq \{0\}$, let $\mathcal{U}_a$ be an orthonormal basis of S_a, which is countable because H is separable. The atoms $\widetilde{\mathrm{at}(u)}$, for $u \in \mathcal{U}_a$, are mutually orthogonal, and their join in $\mathbf{L}^{\mathfrak{M}}/{=}$ is $\tilde{a}$. Indeed, the element $\tilde{a}$ is an upper bound of these atoms, and every upper bound $\tilde{b}$ of them satisfies $S_b \supseteq S_a$. Hence the value $\nu(\tilde{a})$ is the sum $\sum_{u\in\mathcal{U}_a} f(u)$. Taking $a = 1$ shows that f is a nonnegative *frame function* of weight 1: its sum over every orthonormal basis of H is 1. Since H is separable of dimension at least three, Gleason's theorem [Gle57] gives a unique density operator ρ with $f(x) = \langle \rho x \mid x \rangle$ for every unit vector x. It follows that $\nu(\tilde{a}) = \sum_{u\in\mathcal{U}_a} \langle \rho u \mid u \rangle = \operatorname{Tr}(\rho P_a)$ whenever $S_a \neq \{0\}$; the same equality holds when $S_a = \{0\}$, since then $\tilde{a} = \tilde{0}$ and $\nu(\tilde{0}) = 0$. ☻

⧖ **Historical note.** Gleason proved his theorem in 1957 to answer a question of Mackey's. Mackey had asked whether every countably additive assignment of probabilities to the closed

subspaces of a separable Hilbert space is given by the Born rule [Mac57, Gle57]. The Born rule gives a closed subspace, in the state described by a density operator ρ, the probability $\mathrm{Tr}(\rho P)$, where P is the orthogonal projection onto that subspace. Frame functions are the heart of Gleason's proof: he shows first, by means of spherical harmonics, that the continuous frame functions on the unit sphere of $\mathbb{R}^3$ come from quadratic forms. Gleason then shows, in the hardest step, that every nonnegative frame function on that sphere is continuous. Finally, he passes from $\mathbb{R}^3$ to real and complex separable Hilbert spaces of dimension at least three. Gleason noted that the restriction on the dimension is essential: in dimension two the theorem fails. Cooke, Keane and Moran gave in 1985 an elementary proof, longer than Gleason's but with fewer prerequisites [CKM85].

8. Standard models of T_1, proofs of Theorems A and B, and definable sets of values

The quotient of the Solèr sort in a standard model of T_0 may represent only the subspaces of finite dimension or finite codimension, as the lattice L_{fc} of Example 7.5 shows. To obtain every closed subspace, we use the fact that a closed subspace of a separable Hilbert space is the closed span of a square-summable sequence of vectors. We identify the sequence sort of a standard model of T_1 with the unit ball of the space of square-summable sequences of vectors of its Hilbert space, and show that span represents the closed span of each sequence in the Solèr sort. We prove that the quotient of the Solèr sort is therefore isomorphic to the whole lattice of closed subspaces, and use this identification to prove Theorems A and B. In Section 8.1 we then show that the sets of values defined uniformly in the standard models are exactly the σ-projective subsets of $[0,1]$.

Throughout this section we fix a standard model $\mathfrak{M}$ of T_1, whose Solèr sort we denote by $\mathbf{L}^{\mathfrak{M}}$. We write H and K for the Hilbert spaces whose closed unit balls are the Hilbert sort and the sequence sort of $\mathfrak{M}$, respectively. Since T_1 contains T_0, the reduct of $\mathfrak{M}$ to the vocabulary of T_0 is a standard model of T_0, to which the results of Section 7 apply. As there, for each element a of $\mathbf{L}^{\mathfrak{M}}$ we write P_a for the orthogonal projection of H that extends $\pi(a,\cdot)$. Its range, the closed subspace spanned by the members of a, is again denoted by S_a; we write $\tilde{a}$ for the class of a in $\mathbf{L}^{\mathfrak{M}}/{=}$, and τ for the map $\tilde{a} \mapsto S_a$.

Lemma 8.1. *For each $n \in \mathbb{N}$, the map κ_n from the sequence sort to the Hilbert sort, which in the intended interpretation reads the entry in position n of a sequence (see Section 5.3), extends to a bounded linear map from K to H, and the map $s \mapsto (\kappa_n s)_{n\in\mathbb{N}}$ is a linear isometry of K onto the space $\ell^2(\mathbb{N}, H)$ of sequences $(x_n)_{n\in\mathbb{N}}$ in H with $\sum_n \|x_n\|^2 < \infty$.*

Proof. By (K2), each map κ_n is linear on the unit ball of K, and the maps ι_n, which in the intended interpretation place a vector in position n of a sequence, extend to linear isometries from H into K with mutually orthogonal ranges. By the same axiom, we have $\langle \kappa_m s \mid x\rangle = \langle s \mid \iota_m x\rangle$, whence $\|\kappa_m s\| \le \|s\|$, and $\kappa_m \iota_n x = \delta_{mn} x$; each map κ_m therefore extends to a bounded linear map from K to H. For s and t in the unit ball of K, axiom (K3) gives $\langle \theta_N s \mid t\rangle = \sum_{n<N} \langle \kappa_n s \mid \kappa_n t\rangle = \sum_{n<N} \langle \iota_n \kappa_n s \mid t\rangle$, whence $\theta_N s = \sum_{n<N} \iota_n \kappa_n s$. Since the identity $\langle \kappa_m s \mid x\rangle = \langle s \mid \iota_m x\rangle$ makes κ_m the adjoint of the isometry ι_m, the operator $\sum_{n<N} \iota_n \kappa_n$ is the orthogonal projection onto the sum of the mutually orthogonal ranges of $\iota_0, \dots, \iota_{N-1}$. These projections increase with N, whence the norms $\|s - \theta_N s\|$ decrease, and by the second part of (K3) their infimum, and hence their limit, is 0. Consequently, the numbers $\langle \theta_N s \mid s\rangle = \sum_{n<N} \|\kappa_n s\|^2$ converge to $\|s\|^2$, which is therefore equal to $\sum_n \|\kappa_n s\|^2$, and the map $s \mapsto (\kappa_n s)_{n\in\mathbb{N}}$ is isometric on the unit ball and, being linear, on K. Given $(x_n)_{n\in\mathbb{N}} \in \ell^2(\mathbb{N}, H)$, the partial sums $\sum_{n<N} \iota_n x_n$ form a Cauchy sequence in K, since the

maps ι_n are isometries with mutually orthogonal ranges. Since K is complete, these partial sums converge to some s, which satisfies $\kappa_n s = x_n$ by the continuity of κ_n. ☺

Lemma 8.2. *For every s in the unit ball of K, the subspace $S_{\mathrm{span}(s)}$ is the closed span of the entries $\kappa_n s$.*

Proof. Let W be that closed span. By (Sp1), the subspace $S_{\mathrm{span}(s)}$ contains every entry of s, and hence it contains W. Conversely, let y be a vector of $W^\perp$ with $0 < \|y\| \leq 1$, and let $b = \mathrm{at}(y)^\perp$, so that $S_b = y^\perp$ by Lemma 6.4. Every entry $\kappa_n s$ lies in S_b, and so does the subspace $S_{\mathrm{at}(\kappa_n s)}$, which is $\mathbb{C}\kappa_n s$ or $\{0\}$; by Lemma 6.4, we therefore have $[\mathrm{at}(\kappa_n s) \leq b] = 1$ for every n. Axiom (Sp2) gives $[\mathrm{span}(s) \leq b] = 1$, that is, $S_{\mathrm{span}(s)} \subseteq y^\perp$. Since y was arbitrary, the subspace $S_{\mathrm{span}(s)}$ is orthogonal to $W^\perp$, whence $S_{\mathrm{span}(s)} \subseteq W^{\perp\perp} = W$. ☺

Theorem 8.3. *Let $\mathfrak{M}$ be a standard model of T_1, let H be the Hilbert space whose closed unit ball is its Hilbert sort, and, for each element a of its Solèr sort, let S_a be the closed subspace of H spanned by the members of a. Then H is a separable infinite-dimensional complex Hilbert space, and the map $\tau\colon \tilde{a} \mapsto S_a$ is an isomorphism of ortholattices from the quotient $\mathbf{L}^{\mathfrak{M}}/{=}$ of the Solèr sort of $\mathfrak{M}$ onto the lattice $L(H)$ of closed subspaces of H; in particular, that quotient is a Solèr lattice. Conversely, for every separable infinite-dimensional complex Hilbert space H', there is a standard model of T_1 whose Hilbert sort is the closed unit ball of H' and whose Solèr sort is $L(H')$.*

Proof. By Proposition 7.1, the space H is separable and infinite-dimensional. By Proposition 7.2, the map τ is an embedding; it remains to be shown that it is onto, for then the quotient $\mathbf{L}^{\mathfrak{M}}/{=}$, being isomorphic to $L(H)$, is a Solèr lattice by Theorem 3.6. Let S be a closed subspace of H, and let $(u_n)_{n\in\mathbb{N}}$ enumerate an orthonormal basis of S, which is countable by Proposition 7.1, followed by zeros if the basis is finite. By Lemma 8.1 there is s in the unit ball of K with $\kappa_n s = 2^{-n-1}u_n$ for all n, since $\sum_n 4^{-n-1} < 1$, and the subspace $S_{\mathrm{span}(s)}$ is S by Lemma 8.2.

For the converse, we build the model sort by sort. The Hilbert sort is the closed unit ball of H', and the constants e_n are interpreted as an orthonormal basis $(e_n)_{n\in\mathbb{N}}$ of H'. The Solèr sort is $L(H')$, with $\pi(S, x) = P_S x$ for the orthogonal projection P_S onto S and with $\mathrm{at}(x) = \mathbb{C}x$. For each $S \neq \{0\}$ we let $\mathrm{rep}(S)$ be a unit vector chosen in S, and we let $\mathrm{rep}(\{0\})$ be 0. The sequence sort is the closed unit ball of $\ell^2(\mathbb{N}, H')$, on which the function ι_n places a vector x in position n of a sequence whose other entries are 0, and the function κ_n reads the entry of a sequence s in position n. The function θ_N keeps the entries of s in the positions $n < N$ and replaces the others by 0, while the constant ε is the sequence $(2^{-n-1}e_n)_{n\in\mathbb{N}}$, whose norm is less than 1. The function span sends each element s of the sequence sort to the closed span of its entries. Finally, each truncation $R^{\pm}_{n,q}$ is interpreted by its intended value $\min\bigl(1, \max(0, \pm n(R - q))\bigr)$, so that the axioms (Tr) hold. Axioms (Sub1) and (Sub2) hold by Theorem 6.2, and the others by construction; in this model, $[S = S'] = 1$ exactly when $S = S'$. ☺

Proof of Theorem A. The idea is to derive finite additivity of the measure from (M4) and countable additivity from (M5), and then to apply Gleason's theorem in the form of Theorem 7.6. Let $\mathfrak{M}$ be a standard model of T. Its reduct to the vocabulary of T_1 is a standard model of T_1, to which the notation fixed at the beginning of this section applies. By Theorem 8.3, the space H is a separable infinite-dimensional complex Hilbert space, and the map

τ is an isomorphism of $\mathbf{L}^{\mathfrak{M}}/{=}$ onto $L(H)$. By (M3), the value $\mu(a)$ depends only on the class $\tilde{a}$, and we write $\mu(\tilde{a})$ for it.

Step 1: finite additivity. Let a and c be elements of $\mathbf{L}^{\mathfrak{M}}$ with $S_c \subseteq S_a^{\perp}$. By Lemma 6.4, the elements $c \wedge a^{\perp}$ and c have the same class, and so have $a \vee (c \wedge a^{\perp})$ and $a \vee c$. Axioms (M3) and (M4) therefore give $\mu(a \vee c) = \mu(a) + \mu(c)$. Taking $a = c = 0$ gives $\mu(0 \vee 0) = 2\mu(0)$, and since $0 \vee 0$ and 0 have the same class, (M3) turns this equation into $\mu(0) = 2\mu(0)$, whence $\mu(0) = 0$.

Step 2: sums along orthonormal bases. Let a be an element of $\mathbf{L}^{\mathfrak{M}}$, and let $(u_k)_{k\in\mathbb{N}}$ enumerate an orthonormal basis of S_a, followed by zeros if it is finite. By Lemma 8.1 there is s in the unit ball of K with $\kappa_k s = 2^{-k-1}u_k$; by Lemma 8.2, the element $\mathrm{span}(s)$ has the class $\tilde{a}$. For $u_k \neq 0$, the subspaces $S_{\mathrm{at}(\kappa_k s)}$ and $S_{\mathrm{at}(u_k)}$ are both $\mathbb{C}u_k$ by Proposition 7.3, and for $u_k = 0$ both are $\{0\}$ by part (4) of Lemma 6.4. By Step 1 and (M3), the measure of $c_N(s)$ is therefore $\sum_{k<N} \mu(\mathrm{at}(u_k))$, and axiom (M5) gives

$$\mu(a) = \sum_k \mu(\mathrm{at}(u_k)). \tag{$*$}$$

Step 3: application of Gleason's theorem. We show that the map $\tilde{a} \mapsto \mu(\tilde{a})$ is a σ-additive state on $\mathbf{L}^{\mathfrak{M}}/{=}$, in order to apply Theorem 7.6. Its values lie in $[0,1]$ by (M1), and by (M2) it takes the value 1 at $\tilde{1}$. Let $(\tilde{a}_n)_{n\in J}$, for a set $J \subseteq \mathbb{N}$, be a family of mutually orthogonal elements of $\mathbf{L}^{\mathfrak{M}}/{=}$. This family has a join $\tilde{c}$, because the map τ is onto $L(H)$. By Proposition 7.2 and Theorem 7.4, the subspaces S_{a_n} are mutually orthogonal, and the closure of their sum is S_c; orthonormal bases of the subspaces S_{a_n} therefore form together an orthonormal basis of S_c. Applied to c, the identity $(*)$ expresses $\mu(c)$ as the sum of the values $\mu(\mathrm{at}(u))$ over the vectors u of that basis. Applied to each a_n, the identity expresses $\mu(a_n)$ as the sum of the same values over the vectors that come from S_{a_n}. Since these values are nonnegative by (M1), the first sum may be grouped by n, whence $\mu(\tilde{c}) = \sum_{n\in J} \mu(\tilde{a}_n)$. The reduct of $\mathfrak{M}$ to the vocabulary of T_0 is a standard model of T_0 with the same Hilbert sort and the same Solèr sort, to which Theorem 7.6 applies. By that theorem, there is a unique density operator ρ on H with $\mu(a) = \mathrm{Tr}(\rho P_a)$ for every $a \in \mathbf{L}^{\mathfrak{M}}$.

The converse. Let a separable infinite-dimensional complex Hilbert space H' and a density operator ρ on H' be given. We expand a standard model of T_1 built for H' as in the proof of Theorem 8.3 by $\mu(a) = \mathrm{Tr}(\rho P_a)$, where P_a is the orthogonal projection onto the closed subspace a of H', and we interpret the truncations of μ by their intended values, so that the axioms (Tr) for μ hold. Axiom (M3) holds because the formula $[a = b]$ takes the value 1 only when $a = b$. Axioms (M1) and (M2) hold because every orthogonal projection P satisfies $0 \leq \mathrm{Tr}(\rho P) \leq 1$ and because the trace of ρ is 1. Axiom (M4) holds because the subspaces a and $b \wedge a^{\perp}$ are orthogonal, so that the projection onto their join is the sum of their projections. For (M5), let s be an element of the sequence sort. The subspaces $c_N(s)$ increase with N, and the closure of their union is the closed span $\mathrm{span}(s)$ of the entries of s. The projections onto them therefore converge strongly to $P_{\mathrm{span}(s)}$, and since ρ is of trace class, the numbers $\mathrm{Tr}(\rho P_{c_N(s)})$ increase to $\mathrm{Tr}(\rho P_{\mathrm{span}(s)})$. ☺

The proof of Theorem B rests on a property of the density operators on a separable infinite-dimensional complex Hilbert space. For each such operator ρ, the map $S \mapsto \mathrm{Tr}(\rho P_S)$ on the closed subspaces of that space takes every value in $[0,1]$, unlike a measure with only the values 0 and 1.

Lemma 8.4. *For every separable infinite-dimensional complex Hilbert space H and every density operator ρ on H, the set $\{\mathrm{Tr}(\rho P_S) : S \in L(H)\}$, where P_S is the orthogonal projection onto S, is the interval $[0,1]$.*

Proof. The values 0 and 1 are attained at $\{0\}$ and H, respectively. Fix α with $0 < \alpha < 1$; to show that α is attained, we rotate one vector of a subspace spanned by eigenvectors of ρ continuously and apply the intermediate value theorem. The rotation leads from a subspace whose value under $S \mapsto \mathrm{Tr}(\rho P_S)$ is at least α to one whose value is smaller. Since ρ is a positive compact operator of trace 1, there are orthonormal eigenvectors $v_1, v_2, \dots$ (finitely or infinitely many) whose eigenvalues $\eta_1 \geq \eta_2 \geq \cdots > 0$ are the nonzero eigenvalues of ρ, repeated according to multiplicity, and these eigenvalues satisfy $\sum_i \eta_i = 1$. Write $\eta_{\leq k} = \eta_1 + \cdots + \eta_k$, with $\eta_{\leq 0} = 0$, and choose an integer $k \geq 0$ with $\eta_{\leq k} < \alpha \leq \eta_{\leq k+1}$. On the unit vectors of $W = \{v_1, \dots, v_{k+1}\}^\perp$, the quantity $\langle \rho x \mid x \rangle$ has infimum 0. Indeed, if ρ has only finitely many nonzero eigenvalues, then its kernel has finite codimension in the infinite-dimensional space H, and it therefore meets W, which also has finite codimension, in a nonzero vector; otherwise the eigenvectors v_i with $i > k+1$ lie in W, and their eigenvalues η_i tend to 0. Choose a unit vector $u \in W$ with $\langle \rho u \mid u \rangle < \alpha - \eta_{\leq k}$. For $0 \leq \vartheta \leq 1$, let w_ϑ be the unit vector $\sqrt{1-\vartheta}\, v_{k+1} + \sqrt{\vartheta}\, u$, and let S_ϑ be the subspace spanned by $v_1, \dots, v_k$ and w_ϑ. Since $\rho v_{k+1} = \eta_{k+1} v_{k+1}$ and the vector u is orthogonal to $v_1, \dots, v_{k+1}$, the cross terms vanish. Consequently, we have

$$\mathrm{Tr}(\rho P_{S_\vartheta}) = \eta_{\leq k} + (1-\vartheta)\eta_{k+1} + \vartheta \langle \rho u \mid u \rangle,$$

a continuous function of ϑ that equals $\eta_{\leq k+1} \geq \alpha$ at $\vartheta = 0$ and is smaller than α at $\vartheta = 1$. By the intermediate value theorem, it takes the value α. ☺

Proof of Theorem B. Let $\mathfrak{M}$ be a standard model of T, and let H be the Hilbert space whose closed unit ball is its Hilbert sort. By Theorem A, there is a density operator ρ on H with $\mu(a) = \mathrm{Tr}(\rho P_a)$ for every element a of the Solèr sort, and the map $\tilde{a} \mapsto S_a$ is onto $L(H)$. By Lemma 8.4, the range of μ is $[0,1]$. Since B meets $[0,1]$, some element a of the Solèr sort satisfies $\mu(a) \in B$, and the formula $\beta_B(a)$ takes the value 1 at that element. Hence the axiom $\beta_B(a) \leq -1$ of Theorem B fails in $\mathfrak{M}$. ☺

⌛ **Historical note.** Sierpiński proved in 1922 that a measure without atoms takes, on the measurable subsets of a set, every value between 0 and the measure of that set [Sie22]. A $\{0,1\}$-valued measure, which is what an assignment of truth values amounts to, is the opposite extreme: the whole space is an atom. By contrast, the measures $S \mapsto \mathrm{Tr}(\rho P_S)$ that density operators ρ define on the closed subspaces of a separable infinite-dimensional Hilbert space take every value in $[0,1]$ (see Lemma 8.4). Such a measure takes all these values even when ρ is a pure state, the orthogonal projection onto a ray, whose classical counterpart would be a point mass.

8.1. **Definable sets of values.** The formulas $\beta_B(a)$ constructed in the proof of Lemma 5.2 use the connectives of the logic but none of its quantifiers. With quantifiers over the Solèr sort, formulas define more sets of values in the standard models, because the measure takes every value in $[0,1]$ there. In Theorem 8.6 below we show that the sets so defined are exactly the σ-projective subsets of $[0,1]$, and that the conclusion of Theorem B extends to all of them.

Aguilera, Müller and Schlicht [AMS21] study the σ-projective sets, under this name, as subsets of the finite powers of $\mathbb{N}^{\mathbb{N}}$. Let Γ assign to each Polish space X a class $\Gamma(X)$ of subsets of X that contains the Borel subsets of X and is closed under complements in X

and countable unions. Suppose moreover that, whenever $A \in \Gamma(X \times \mathbb{N}^{\mathbb{N}})$, the set $\{x \in X : (x, z) \in A \text{ for some } z \in \mathbb{N}^{\mathbb{N}}\}$ belongs to $\Gamma(X)$, and take for Γ the least assignment with these properties. A subset of a Polish space X is *σ-projective* when it belongs to $\Gamma(X)$. On the finite powers of $\mathbb{N}^{\mathbb{N}}$, the σ-projective sets are the sets of Aguilera, Müller and Schlicht. Their class is the least that contains the open sets and is closed under complements, countable unions and projections along $\mathbb{N}^{\mathbb{N}}$. This class is therefore contained in Γ, which has these properties on the finite powers of $\mathbb{N}^{\mathbb{N}}$. Conversely, for a Polish space X, consider the sets $A \subseteq X$ such that $g^{-1}[A]$ belongs to the class of Aguilera, Müller and Schlicht for every k and every Borel map $g\colon (\mathbb{N}^{\mathbb{N}})^k \to X$. The class of Aguilera, Müller and Schlicht contains the Borel sets and is closed under the operations that define Γ. Hence these sets A include the Borel sets and are closed under complements, countable unions and projections along $\mathbb{N}^{\mathbb{N}}$. These sets therefore include $\Gamma(X)$, as the identity maps show for $X = (\mathbb{N}^{\mathbb{N}})^k$. Every projective set [Kec95, Section 37] is σ-projective, and so is every countable union of projective sets. The σ-projective sets are closed under preimages by Borel maps between Polish spaces. Indeed, the sets $A \subseteq X$ such that $g^{-1}[A]$ is σ-projective for every Borel map g from a Polish space into X include the Borel sets and are closed under complements and countable unions. They are also closed under projections along $\mathbb{N}^{\mathbb{N}}$, since, for a set $A \subseteq X \times \mathbb{N}^{\mathbb{N}}$ and a Borel map g from a Polish space Z into X, the preimage under g of the projection of A is the projection of the preimage of A under the Borel map $g \times \mathrm{id}$ from $Z \times \mathbb{N}^{\mathbb{N}}$ into $X \times \mathbb{N}^{\mathbb{N}}$. These sets therefore include every σ-projective set. The σ-projective sets are also closed under projections along every nonempty Polish space Y. Given a continuous surjection $f\colon \mathbb{N}^{\mathbb{N}} \to Y$ [Kec95, Theorem 7.9] and a set $A \subseteq X \times Y$, the projection of A to X is the projection along $\mathbb{N}^{\mathbb{N}}$ of the preimage of A under $\mathrm{id} \times f$. A function f from a Polish space to $\mathbb{R}$ is *Γ-measurable* when the set $\{f > r\}$ is σ-projective for every real number r. The set $\{f \geq r\}$, which is the intersection of the sets $\{f > r - \frac{1}{n}\}$, is then σ-projective as well.

Definition 8.5. A set $D \subseteq [0,1]$ is *definable over T* if there is a formula $\gamma(a)$ of real-valued $L_{\omega_1\omega}$, with the single free variable a of the Solèr sort, whose value, in every standard model of T and at every element a of its Solèr sort, is 1 if $\mu(a) \in D$ and -1 otherwise; we say that *γ defines D*.

Only subsets of $[0,1]$ need be considered in Definition 8.5, since (M1) confines μ to $[0,1]$ in every model of T: a formula that, in every standard model of T and at every element a of its Solèr sort, takes the value 1 if $\mu(a)$ lies in a set $E \subseteq \mathbb{R}$ and -1 otherwise defines $E \cap [0,1]$.

Theorem 8.6. *A subset of $[0,1]$ is definable over T if and only if it is σ-projective. Consequently, if $D \subseteq [0,1]$ is a nonempty σ-projective set and $\gamma(a)$ defines D, then the theory T together with the axiom $\gamma(a) \leq -1$, which says that no value of μ lies in D, has no standard models.*

The class of σ-projective subsets of $[0,1]$ is strictly larger than that of projective ones. For each $n \geq 1$, the hierarchy theorem for projective sets [Kec95, Section 37] provides a $\mathbf{\Sigma}^1_{n+1}$ set $C_n \subseteq [0,1]$ that is not $\mathbf{\Sigma}^1_n$, and we let h_n be a homeomorphism of $[0,1]$ onto $[2^{-n}, 2^{-n+1}]$. The set $C = \bigcup_n h_n[C_n \cap (0,1)]$ is a countable union of projective sets, and it contains none of the points 2^{-m}. If C were $\mathbf{\Sigma}^1_k$, then for $n > k$ the set $h_n^{-1}[C]$, which equals $C_n \cap (0,1)$, would be $\mathbf{\Sigma}^1_k$. The set C_n, which differs from $C_n \cap (0,1)$ by at most two points, would then be $\mathbf{\Sigma}^1_k$ and hence $\mathbf{\Sigma}^1_n$, which is false. If C were $\mathbf{\Pi}^1_k$, it would be $\mathbf{\Sigma}^1_{k+1}$, which is excluded in the same way. Hence C is not projective, and by Theorem 8.6 it is definable over T.

The two directions use different features of the standard models. To define every σ-projective set, we use the surjectivity of μ: quantification over the Solèr sort then realizes projection along the interval of probabilities. To bound the expressive power of formulas,

we choose one standard model in which every symbol is interpreted by a Borel function on Polish spaces. In that model, a continuous map that assigns to each probability t a projection of measure t transfers the descriptive complexity of a formula to the subset of $[0,1]$ that it defines.

Proof of Theorem 8.6. Every σ-projective subset of $[0,1]$ is definable. For $m \geq 1$, consider a formula $\gamma(a_1, \ldots, a_m)$ with free variables $a_1, \ldots, a_m$ of the Solèr sort. The formula γ *strongly defines* a set $A \subseteq [0,1]^m$ if it has a priori bound at most 3, is two-valued in every model of T and, in every standard model of T, equals 1 at $(a_1, \ldots, a_m)$ exactly when $(\mu(a_1), \ldots, \mu(a_m)) \in A$. Let $\mathcal{A}$ be the class of sets $A \subseteq [0,1]^m$, for $m \geq 1$, that some formula strongly defines. We show that $\mathcal{A}$ contains every σ-projective subset of every cube $[0,1]^m$; a formula that strongly defines a set $D \subseteq [0,1]$ defines it in the sense of Definition 8.5.

First, $\mathcal{A}$ contains the Borel sets. We repeat the proof of Lemma 5.2 with the closed boxes $[q_1, q_1'] \times \cdots \times [q_m, q_m']$, for rational numbers $q_i \leq q_i'$, in place of the closed intervals with rational endpoints. The formula for such a box is the minimum

$$\beta_{[q_1,q_1']}(a_1) \wedge \cdots \wedge \beta_{[q_m,q_m']}(a_m)$$

of the formulas of that proof for the intervals $[q_i, q_i']$. For every Borel set $B \subseteq \mathbb{R}^m$, the modified proof gives a formula $\beta_B(a_1, \ldots, a_m)$. In every structure for $\mathcal{L}$ that satisfies (Tr), its value at every tuple of elements of the Solèr sort is 1 if $(\mu(a_1), \ldots, \mu(a_m)) \in B$ and -1 otherwise. The formula $\beta_B(a_1, \ldots, a_m)$ strongly defines $B \cap [0,1]^m$. Next, if γ strongly defines $A \subseteq [0,1]^m$, then the formula $-\gamma$ strongly defines $[0,1]^m \setminus A$. If γ_n strongly defines $A_n \subseteq [0,1]^m$ for every $n \in \mathbb{N}$, then the join $\bigvee_n \gamma_n$, whose members have a priori bounds at most 3, strongly defines $\bigcup_n A_n$. Finally, if $\gamma(a_1, \ldots, a_m, b)$ strongly defines $A \subseteq [0,1]^{m+1}$, then the formula $\sup_b \gamma(a_1, \ldots, a_m, b)$ is two-valued in every model of T. Indeed, the Solèr sort contains the constant 0, and a supremum of values in $\{-1, 1\}$ over a nonempty set is attained. In a standard model of T, the formula $\sup_b \gamma(a_1, \ldots, a_m, b)$ takes the value 1 at $(a_1, \ldots, a_m)$ exactly when $(\mu(a_1), \ldots, \mu(a_m), \mu(b)) \in A$ for some element b. By Theorem B, such an element b exists exactly when $(\mu(a_1), \ldots, \mu(a_m), t) \in A$ for some $t \in [0,1]$. Thus $\mathcal{A}$ contains the Borel subsets of the cubes $[0,1]^m$ and is closed under complements, countable unions and projections along $[0,1]$.

It remains to be shown that these operations produce every σ-projective subset of every cube $[0,1]^m$. Consider the least class Γ_0 of subsets of the cubes $[0,1]^m$, for $m \geq 1$, containing their Borel subsets and closed under complements in the cube, countable unions and projections from $[0,1]^{m+1}$ to $[0,1]^m$. We have already shown that the class Γ_0 is contained in $\mathcal{A}$. Fix a Borel isomorphism χ of $[0,1]$ onto $\mathbb{N}^\mathbb{N}$ [Kec95, Theorem 15.6]. For each Polish space X, let $\Gamma'(X)$ be the class of sets $A \subseteq X$ such that $g^{-1}[A] \in \Gamma_0$ for every $m \geq 1$ and every Borel map $g\colon [0,1]^m \to X$. Each class $\Gamma'(X)$ contains the Borel subsets of X and is closed under complements in X and countable unions. Let $A \in \Gamma'(X \times \mathbb{N}^\mathbb{N})$ be given, and let $g\colon [0,1]^m \to X$ be a Borel map. Since χ is onto $\mathbb{N}^\mathbb{N}$, we have

$$g^{-1}\big[\{x : \exists z \in \mathbb{N}^\mathbb{N}\ (x,z) \in A\}\big] = \{u \in [0,1]^m : \exists t \in [0,1]\ (u,t) \in (g \times \chi)^{-1}[A]\},$$

the projection along $[0,1]$ of a set of Γ_0, because $g \times \chi$ is a Borel map on $[0,1]^{m+1}$; this projection belongs to Γ_0. Hence the assignment Γ' is closed under projections along $\mathbb{N}^\mathbb{N}$, and we have $\Gamma(X) \subseteq \Gamma'(X)$ for every Polish space X. For a σ-projective set $A \subseteq [0,1]^m$, taking for g the identity shows that $A \in \Gamma_0$ and therefore that $A \in \mathcal{A}$.

Every definable subset of $[0,1]$ is σ-projective. Let $\gamma(a)$ define a set $D \subseteq [0,1]$. Let $H = \ell^2(\mathbb{N})$, let v and w be orthogonal unit vectors of H, and let ρ be the projection onto

$\mathbb{C}v$, a density operator. By Theorem A(2) and its proof, there is a standard model $\mathfrak{M}$ of T that can be described as follows. Its Hilbert sort is the closed unit ball B_H of H, its sequence sort is the closed unit ball B_K of $K = \ell^2(\mathbb{N}, H)$, and its Solèr sort is $L(H)$. The symbols are interpreted as in that proof, with the standard unit vectors of $\ell^2(\mathbb{N})$ as the constants e_n and with $\mu(S) = \mathrm{Tr}(\rho P_S)$. In that proof, the value $\mathrm{rep}(S)$ may be any unit vector of S when $S \neq \{0\}$, and here we let it be $P_S e_n / \|P_S e_n\|$ for the least n with $P_S e_n \neq 0$. We identify $L(H)$ with the set of orthogonal projections on H, with the strong operator topology. Fix a dense sequence $(x_k)_{k\in\mathbb{N}}$ in B_H, and give the set of operators of norm at most 1 the strong operator topology. For such operators, convergence at the vectors x_k implies convergence at every vector. The map $Q \mapsto (Qx_k)_{k\in\mathbb{N}}$ is therefore a homeomorphism of the set of operators of norm at most 1 onto its image in the Polish space $B_H^{\mathbb{N}}$. The image is closed: if $(y_k)_{k\in\mathbb{N}}$ is the limit of sequences $(Q_j x_k)_{k\in\mathbb{N}}$ in the image, then we have $\|\sum_k c_k y_k\| \leq \|\sum_k c_k x_k\|$ for every finitely supported family of complex numbers c_k. The map $x_k \mapsto y_k$ therefore extends to an operator of norm at most 1. The projections form a closed subset, since a strong limit Q of projections satisfies $\langle Qx \mid x\rangle = \|Qx\|^2 = \langle Q^*Qx \mid x\rangle$ for every x, whence $Q = Q^*Q$ by polarization; the operator Q is therefore self-adjoint, and it is idempotent, since $Q^2 = Q^*Q = Q$. Thus the space $L(H)$ is Polish, and so are B_H and B_K with their norm topologies.

Every symbol of $\mathcal{L}$ is interpreted in $\mathfrak{M}$ by a Borel function on a product of these spaces. The vector operations, the inner products and the norms are continuous, as are the maps ι_n, κ_n and θ_N, the constants and the orthocomplement $Q \mapsto I - Q$. The map $(Q, x) \mapsto Qx$, which interprets π, is continuous because $\|Qx - Q_0x_0\| \leq \|x - x_0\| + \|(Q - Q_0)x_0\|$. The measure, given by $\mu(Q) = \mathrm{Tr}(\rho Q) = \langle Qv \mid v\rangle$, is continuous as well. The meet $Q \wedge Q'$ is the strong limit of $(QQ'Q)^n$ as $n \to \infty$, by Theorem 6.2 and the identity $(QQ'Q)^n = (QQ')^nQ$. As a pointwise limit of continuous maps, the meet is Borel, and the join is given by $Q \vee Q' = I - ((I - Q) \wedge (I - Q'))$. The truncations are continuous functions of the predicates that they truncate. The map at sends $x \neq 0$ to the projection $y \mapsto \langle y \mid x\rangle x/\|x\|^2$, continuously, and 0 to 0. For rep, the set of projections Q with $Qe_k = 0$ for $k < n$ and $Qe_n \neq 0$ is Borel, and on it $\mathrm{rep}(Q) = Qe_n/\|Qe_n\|$ is continuous. For span, let $P_N(s)$ be the projection onto the span of the entries $\kappa_0 s, \ldots, \kappa_{N-1} s$ of s. The ball B_K splits into finitely many sets according to the set of indices $i < N$ such that $\kappa_i s$ is not in the span of $\kappa_0 s, \ldots, \kappa_{i-1} s$. On each of these finitely many sets, the map P_N is continuous, being given by Gram–Schmidt orthonormalization of the corresponding entries. These sets are Borel, since the rank of the Gram matrix of $(\kappa_0 s, \ldots, \kappa_i s)$ is a Borel function of s, the set where it is at least r being open. The projections $P_N(s)$ increase with N to the projection $\mathrm{span}(s)$ onto the closed span of the entries of s, strongly, and the map span is Borel. It follows by induction on terms that every term is interpreted by a Borel function.

We show by induction on formulas that, for every formula $\varphi(\bar{x})$, the function $\bar{x} \mapsto \varphi^{\mathfrak{M}}(\bar{x})$ is Γ-measurable on the product of the Polish spaces that interpret the sorts of $\bar{x}$. Atomic formulas, those built from the truncations included, are continuous functions of Borel functions of terms, hence Borel. For the connectives $s \cdot \varphi$, $\varphi + \psi$ and $\varphi \vee \psi$, the set where the value exceeds r is the preimage under (φ, ψ) of an open subset of $\mathbb{R}^2$, which is a countable union of open boxes with rational corners. The preimage of each of these boxes is a finite intersection of sets $\{\varphi > q\}$, $\{\psi > q\}$ and complements of sets $\{\varphi \geq q'\}$, $\{\psi \geq q'\}$. For a countable join, the set $\{\bigvee_n \varphi_n > r\}$ is the union $\bigcup_n \{\varphi_n > r\}$. For a supremum over a variable y, the set $\{\bar{x} : \sup_y \varphi(\bar{x}, y) > r\}$ is the projection of $\{(\bar{x}, y) : \varphi(\bar{x}, y) > r\}$ along the Polish space that interprets the sort of y. In each case the set is σ-projective.

Now the function $\gamma^{\mathfrak{M}}$ is Γ-measurable, and the set $\{Q \in L(H) : \gamma^{\mathfrak{M}}(Q) > 0\}$ is σ-projective. For $t \in [0,1]$, let $\xi(t)$ be the projection at$(\sqrt{t}\,v + \sqrt{1-t}\,w)$ onto the line spanned by the unit vector $\sqrt{t}\,v + \sqrt{1-t}\,w$. The map ξ is continuous and satisfies $\mu(\xi(t)) = |\langle v \mid \sqrt{t}\,v + \sqrt{1-t}\,w\rangle|^2 = t$. Since $\gamma^{\mathfrak{M}}$ takes the value 1 exactly at the elements Q with $\mu(Q) \in D$,

$$D = \{t \in [0,1] : \gamma^{\mathfrak{M}}(\xi(t)) > 0\} = \xi^{-1}\big[\{\gamma^{\mathfrak{M}} > 0\}\big],$$

a continuous preimage of a σ-projective set, hence σ-projective.

The consequence follows as in the proof of Theorem B, with γ in place of β_B: in a standard model, some element a has $\mu(a) \in D$, and $\gamma(a)$ takes the value 1 there. ☺

The second half of the proof yields more than σ-projectivity: it bounds the projective complexity of a defined set by the nesting of quantifiers in a defining formula, and it thereby shows that countable joins are indispensable in every formula that defines the set C above. To make this precise, we define the *quantifier depth* $\mathrm{qd}(\varphi)$ of a formula φ, a countable ordinal, by induction. Atomic formulas have depth 0, and the depth of $s \cdot \varphi$, of $\varphi + \psi$ and of $\varphi \vee \psi$ is the largest depth of the formulas to which the connective is applied. The depth of a countable join is given by $\mathrm{qd}(\bigvee_n \varphi_n) = \sup_n \mathrm{qd}(\varphi_n)$, and that of a supremum by $\mathrm{qd}(\sup_x \varphi) = \mathrm{qd}(\varphi)+1$. In the model $\mathfrak{M}$ of the proof, the induction on formulas given there shows that, for every $k < \omega$ and every formula φ with $\mathrm{qd}(\varphi) \le k$, each set $\{\varphi^{\mathfrak{M}} > r\}$ is $\mathbf{\Delta}^1_{k+1}$. For $k = 0$, the formula φ contains no supremum over a variable, and these sets are Borel. For $k \ge 1$, if $\mathrm{qd}(\sup_y \varphi) \le k$, then the formula φ has depth at most $k-1$, and the set $\{(\sup_y \varphi)^{\mathfrak{M}} > r\}$ is the projection of a $\mathbf{\Delta}^1_k$ set. This projection is therefore $\mathbf{\Sigma}^1_k$, and hence $\mathbf{\Delta}^1_{k+1}$. The other cases use only countable unions, countable intersections and complements, under which $\mathbf{\Delta}^1_{k+1}$ is closed. If a formula γ with $\mathrm{qd}(\gamma) \le k$ defines D, then the set $D = \xi^{-1}[\{\gamma^{\mathfrak{M}} > 0\}]$ is therefore $\mathbf{\Delta}^1_{k+1}$. Since the set C above is not projective, every formula that defines it has quantifier depth at least ω. As the subformula relation is well founded, any formula that defines C has a subformula of depth at least ω whose proper subformulas all have finite depth. That subformula is a countable join whose members have unbounded finite depths.

Uniformity across standard models. The definition asks for one formula that defines D in every standard model, whatever orthonormal basis interprets the constants e_n, whatever density operator gives the measure and whatever unit vectors interpret rep. The requirement cannot, in general, be weakened to definability in a single standard model. For $i \in \mathbb{N}$, we write $z_i(b)$ for the pair of predicates $\langle \mathrm{rep}(b) \mid e_i\rangle$, and $\mathrm{nonatom}(b)$ for the two-valued formula $\sup_c\big([c \ne 0] \wedge -[c = b] \wedge [c \le b]\big)$, which takes the value 1 exactly when some element lies strictly between 0 and b. For an atomic formula $R(\bar t)$ with R one of the inner-product, norm or measure predicates, let $[R(\bar t) > 0] = 2\lceil R(\bar t) > 0\rceil - 1$ and $[R(\bar t) = 0] = 1 - 2\big(\lceil R(\bar t) > 0\rceil \vee \lceil R(\bar t) < 0\rceil\big)$, two-valued formulas in every structure that satisfies (Tr). Let $\delta(a,b)$ be the formula $\beta_B(a,b)$ of the proof for the diagonal $B = \{(u,u) : u \in \mathbb{R}\}$. Let $\gamma_0(a)$ be the formula

$$\sup_b\Big([b \ne 0] \wedge -\mathrm{nonatom}(b) \wedge \big[b \le \mathrm{at}(e_0) \vee \mathrm{at}(e_1)\big] \wedge \delta(a,b) \\ \wedge [\mathrm{Im}\, z_0(b) = 0] \wedge [\mathrm{Im}\, z_1(b) = 0] \wedge [\mathrm{Re}\, z_0(b) > 0] \wedge [\mathrm{Re}\, z_1(b) > 0]\Big).$$

Given $X \subseteq (0,1)$, take the model of the proof of Theorem A(2) on $\ell^2(\mathbb{N})$, with the standard unit vectors as the constants e_n and with ρ the projection onto $\mathbb{C}e_0$. In this model, let rep send the atom spanned by $\sqrt{t}\,e_0 + \sqrt{1-t}\,e_1$, for $0 < t < 1$, to the vector $\sqrt{t}\,e_0 + \sqrt{1-t}\,e_1$ itself if $t \in X$ and to its negative otherwise. For $0 < t < 1$, the atoms b below $\mathrm{at}(e_0) \vee \mathrm{at}(e_1)$ with $\mu(b) = t$ are the lines of the vectors $\sqrt{t}\,e_0 + \zeta\sqrt{1-t}\,e_1$ with $|\zeta| = 1$. A unit vector of

such a line has two real coordinates only when $\zeta = \pm 1$, and two positive real coordinates only when $\zeta = 1$ and the vector is $\sqrt{t}\, e_0 + \sqrt{1-t}\, e_1$ itself; for $t \in \{0, 1\}$, one coordinate vanishes. Hence the formula $\gamma_0(a)$ takes the value 1 in that model exactly when $\mu(a) \in X$, and -1 otherwise. Since the vocabulary is countable, there are at most $2^{\aleph_0}$ formulas, and hence at most $2^{\aleph_0}$ sets definable over T. The interval $(0, 1)$, by contrast, has $2^{2^{\aleph_0}}$ subsets, and some sets X that γ_0 defines in a single standard model are therefore not definable over T. In the model $\mathfrak{M}$ of the proof of Theorem 8.6, whose symbols are interpreted by Borel functions, a single model does suffice. Indeed, let $D \subseteq [0, 1]$ be a set such that some formula takes, at every element Q of the Solèr sort of $\mathfrak{M}$, the value 1 if $\mu(Q) \in D$ and -1 otherwise. The second half of the proof applies to this formula and shows that the set D is σ-projective, and hence definable over T.

In a model that is not standard, a formula that defines a set D over T need not take the value 1 exactly when $\mu(a) \in D$. By Remark 11.2, the theory T has a countable model $\mathfrak{N}$; the range of μ in $\mathfrak{N}$ is countable, and some $c \in [0, 1]$ is not a value of μ in $\mathfrak{N}$. By Theorem B, the formula $\sup_b \beta_{\mathbb{R}\times\{c\}}(a, b)$, with $\beta_{\mathbb{R}\times\{c\}}$ as in the proof of Theorem 8.6, defines $[0, 1]$ over T. In $\mathfrak{N}$, however, this formula takes the value -1 everywhere. The formulas $\beta_B(a)$ constructed in the proof of Lemma 5.2, by contrast, have no quantifiers, and in every structure that satisfies (Tr) they take the value 1 exactly when $\mu(a) \in B$. The conclusions of the remark on standard models at the end of Section 10, which rest on facts about all models of T, depend on these two properties of the formulas $\beta_B(a)$. For this reason that remark is stated for these formulas, whereas Theorem 8.6 is a theorem about the standard models, in which the measure takes every value in $[0, 1]$.

9. The proof of Theorem C

In this section we prove Theorem C: in no model of T, standard or not, does the measure μ take only the values 0 and 1. The theorem concerns every model of T, and its proof accordingly draws on none of the conclusions of Theorem A, which concern the standard models. By (H2), the constants e_n are orthonormal, and by (K4) the entries $\kappa_n\varepsilon$ of the constant ε satisfy $\kappa_n\varepsilon \doteq 2^{-n-1}e_n$, an identification up to the relation $\doteq$, since a model need not identify elements that no predicate distinguishes. Besides these constants, the proof uses only the eighteen vectors of Table 1, which are built from four of the vectors e_n. The Hilbert-space envelope of Lemma 6.3 supplies the geometry, while the rational terms built from the named basis vectors supply the vectors needed inside the model. The argument therefore requires neither the whole unit ball nor the whole lattice of closed subspaces.

The proof rests on four facts about the Hilbert-space envelope of Lemma 6.3, which hold in every model of T and which we gather in the following lemma. For a model of T, we let $\bar{E}$, the map $x \mapsto \hat{x}$ and the projections $\bar{P}_a$ be as in Lemma 6.3, applied to the reduct of the model to the vocabulary of T_0, and we let S_a be the range of $\bar{P}_a$, as in Lemma 6.4.

Lemma 9.1. *Let $\mathfrak{M}$ be a model of T, standard or not, and let $\bar{E}$, $\hat{x}$ and S_a be as above. Then the following hold:*

1. *The vectors $\hat{e}_n$, for $n \in \mathbb{N}$, form an orthonormal basis of $\bar{E}$.*
2. *For every x in the Hilbert sort with $\|x\| > 0$, the subspace $S_{\mathrm{at}(x)}$ is $\mathbb{C}\hat{x}$; consequently, $[\mathrm{at}(x) = \mathrm{at}(y)] = 1$ whenever $\|x\| > 0$ and $\hat{y} = \lambda\hat{x}$ for a nonzero $\lambda \in \mathbb{C}$.*
3. *The measure satisfies $\mu(0) = 0$, and $\mu(a \vee c) = \mu(a) + \mu(c)$ for all elements a, c of the Solèr sort with $S_c \subseteq S_a^{\perp}$.*
4. *We have $[\mathrm{span}(\varepsilon) = 1] = 1$, and hence $\mu(\mathrm{span}(\varepsilon)) = 1$.*

Proof. (1) By (H2), the vectors $\hat{e}_n$ are orthonormal. By Lemma 6.4, the projection $\bar{P}_{b_N}$ is the projection onto the span of $\hat{e}_0, \dots, \hat{e}_{N-1}$, and (H2) places every vector $\hat{x}$ in the closed span of the vectors $\hat{e}_n$; since the vectors $\hat{x}$ span a dense subspace, these vectors form an orthonormal basis.

(2) The first assertion is part (4) of Lemma 6.4. If $\hat{y} = \lambda\hat{x}$ with $\lambda \neq 0$, then $\|y\| > 0$ and $S_{\mathrm{at}(y)} = \mathbb{C}\hat{x} = S_{\mathrm{at}(x)}$, and part (3) of that lemma gives $[\mathrm{at}(x) = \mathrm{at}(y)] = 1$.

(3) By Lemma 6.4, the elements $c \wedge a^\perp$ and c satisfy $[c \wedge a^\perp = c] = 1$, and the elements $a \vee (c \wedge a^\perp)$ and $a \vee c$ satisfy the same equation. Axioms (M3) and (M4) give $\mu(a \vee c) = \mu(a) + \mu(c)$. For $a = c = 0$, which satisfies $S_0 = \{0\}$, this gives $\mu(0 \vee 0) = 2\mu(0)$; since $S_{0\vee 0} = S_0$, part (3) of Lemma 6.4 gives $[0 \vee 0 = 0] = 1$, and (M3) turns the equation into $\mu(0) = 2\mu(0)$, whence $\mu(0) = 0$.

(4) By (Sp1) and (K4), the subspace $S_{\mathrm{span}(\varepsilon)}$ contains every vector $2^{-n-1}\hat{e}_n$, and hence, by part (1), it is $\bar{E} = S_1$. Part (3) of Lemma 6.4 gives $[\mathrm{span}(\varepsilon) = 1] = 1$, and axioms (M3) and (M2) give $\mu(\mathrm{span}(\varepsilon)) = 1$. ☺

basis	vectors (coordinates with respect to an orthonormal basis g_0, g_1, g_2, g_3)			
1	$(0,0,0,1)$	$(0,0,1,0)$	$(1,1,0,0)$	$(1,-1,0,0)$
2	$(0,0,0,1)$	$(0,1,0,0)$	$(1,0,1,0)$	$(1,0,-1,0)$
3	$(1,-1,1,-1)$	$(1,-1,-1,1)$	$(1,1,0,0)$	$(0,0,1,1)$
4	$(1,-1,1,-1)$	$(1,1,1,1)$	$(1,0,-1,0)$	$(0,1,0,-1)$
5	$(0,0,1,0)$	$(0,1,0,0)$	$(1,0,0,1)$	$(1,0,0,-1)$
6	$(1,-1,-1,1)$	$(1,1,1,1)$	$(1,0,0,-1)$	$(0,1,-1,0)$
7	$(1,1,-1,1)$	$(1,1,1,-1)$	$(1,-1,0,0)$	$(0,0,1,1)$
8	$(1,1,-1,1)$	$(-1,1,1,1)$	$(1,0,1,0)$	$(0,1,0,-1)$
9	$(1,1,1,-1)$	$(-1,1,1,1)$	$(1,0,0,1)$	$(0,1,-1,0)$

TABLE 1. The configuration of Cabello, Estebaranz and García-Alcaine [CEGA96]: nine orthogonal bases of a four-dimensional space, formed by eighteen vectors, each of which occurs in exactly two bases. In the proof of Theorem C, the vectors $g_0, \dots, g_3$ are four consecutive vectors $e_{4j}, \dots, e_{4j+3}$, for some $j \in \mathbb{N}$, of the orthonormal basis $(e_n)_{n\in\mathbb{N}}$.

Proof of Theorem C. Suppose, towards a contradiction, that $\mathfrak{M}$ is a model of T in which μ takes only the values 0 and 1, and let $\bar{E}$, $\hat{x}$ and S_a be as in Lemma 9.1 for this model. We first show that exactly one of the constants e_n has an atom $\mathrm{at}(e_n)$ of measure 1. We then apply a parity argument to the bases of Table 1 within the block of four basis vectors that contains this vector. The entries $\kappa_n\varepsilon$ satisfy $\widehat{\kappa_n\varepsilon} = 2^{-n-1}\hat{e}_n$ by (K4), and part (2) of Lemma 9.1 gives $[\mathrm{at}(\kappa_n\varepsilon) = \mathrm{at}(e_n)] = 1$. Since the subspaces $S_{\mathrm{at}(e_n)} = \mathbb{C}\hat{e}_n$ are mutually orthogonal, repeated applications of part (3) of Lemma 9.1, combined with (M3), which replaces each $\mu(\mathrm{at}(\kappa_n\varepsilon))$ by $\mu(\mathrm{at}(e_n))$, give $\mu(c_N(\varepsilon)) = \sum_{n<N} \mu(\mathrm{at}(e_n))$ for every $N \in \mathbb{N}$. By (M5) and Lemma 9.1(4), we obtain

$$1 = \mu(\mathrm{span}(\varepsilon)) = \sum_n \mu(\mathrm{at}(e_n)).$$

As μ takes only the values 0 and 1, there is exactly one $n_0 \in \mathbb{N}$ with $\mu(\mathrm{at}(e_{n_0})) = 1$. Let $j \in \mathbb{N}$ satisfy $4j \le n_0 < 4j+4$, put $g_i = e_{4j+i}$ for $i = 0,1,2,3$, and let F be the span of $\hat{g}_0, \dots, \hat{g}_3$.

The basis $g_0, \dots, g_3$ of F contains exactly one vector, e_{n_0}, whose atom has measure 1. We show that every orthogonal basis of F whose vectors have coordinates in $\{0, 1, -1\}$ with respect to $\hat{g}_0, \dots, \hat{g}_3$ has the same property. Let $\mathcal{B}$ be such a basis, consisting of vectors $v_0, \dots, v_3$ of F. The coordinates of each vector $\frac{1}{4}v_i$ have absolute values with sum at most 1; by Lemma 6.3, some closed term t_i, built from the constants $g_0, \dots, g_3$ by the functions $\lambda x{+}\lambda' y$, therefore satisfies $\hat{t}_i = \frac{1}{4}v_i$. Hence the term $\frac{1}{2}\big(\frac{1}{2}\iota_0 t_0 + \frac{1}{2}\iota_1 t_1\big) + \frac{1}{2}\big(\frac{1}{2}\iota_2 t_2 + \frac{1}{2}\iota_3 t_3\big)$ denotes an element $s_{\mathcal{B}}$ of the sequence sort. Axiom (K2) gives $\langle \kappa_m \iota_n x \mid y\rangle = \langle \iota_n x \mid \iota_m y\rangle = \delta_{mn}\langle x \mid y\rangle$ for all x and y, whence $\widehat{\kappa_m \iota_n x} = \delta_{mn}\hat{x}$, because the vectors $\hat{y}$ span a dense subspace of $\bar{E}$; together with the linearity of κ_n asserted in (K2), this gives $\widehat{\kappa_n s_{\mathcal{B}}} = \frac{1}{16}v_n$ for $n < 4$. Let $c_{\mathcal{B}}$ be the element $\mathrm{at}(\kappa_0 s_{\mathcal{B}}) \vee \dots \vee \mathrm{at}(\kappa_3 s_{\mathcal{B}})$, that is, $c_4(s_{\mathcal{B}})$. For a nonzero vector v of F with coordinates in $\{0, 1, -1\}$, we fix a closed term t_v, built in the same way, with $\hat{t}_v = \frac{1}{16}v$, and we write $\mathrm{at}(v)$ for $\mathrm{at}(t_v)$; by Lemma 9.1(2), another choice of t_v changes $\mathrm{at}(v)$ only within the relation $[\cdot = \cdot] = 1$, and therefore, by (M3), leaves $\mu(\mathrm{at}(v))$ unchanged. By the same part of that lemma, we have $[\mathrm{at}(\kappa_i s_{\mathcal{B}}) = \mathrm{at}(v_i)] = 1$ for $i < 4$, and $[\mathrm{at}(v) = \mathrm{at}(-v)] = 1$ for every such vector v. By Lemma 6.4, the subspace $S_{c_{\mathcal{B}}}$ is the span of $v_0, \dots, v_3$, which is F. Hence $[c_{\mathcal{B}} = c_{\mathcal{B}'}] = 1$ for any two such bases $\mathcal{B}$ and $\mathcal{B}'$, and we let a_F be the element $c_{\mathcal{B}}$ for the basis $g_0, \dots, g_3$. By Lemma 9.1(3), applied three times, and by (M3),

$$\mu(a_F) = \mu(c_{\mathcal{B}}) = \sum_{v \in \mathcal{B}} \mu(\mathrm{at}(v)).$$

For the basis $\hat{g}_0, \dots, \hat{g}_3$, part (2) of Lemma 9.1 and (M3) give $\mu(\mathrm{at}(\hat{g}_i)) = \mu(\mathrm{at}(e_{4j+i}))$ for $i < 4$, and this sum therefore contains $\mu(\mathrm{at}(e_{n_0})) = 1$ and three zeros; hence $\mu(a_F) = 1$, and every such basis $\mathcal{B}$ contains exactly one vector v with $\mu(\mathrm{at}(v)) = 1$.

The nine bases of Table 1 are orthogonal bases of F of this kind, and their eighteen vectors span pairwise distinct rays. We conclude by counting in two ways the pairs $(\mathcal{B}, v)$ that consist of one of the nine bases $\mathcal{B}$ and a vector v of $\mathcal{B}$ with $\mu(\mathrm{at}(v)) = 1$. By the preceding paragraph, each basis contains exactly one such vector, and there are therefore exactly nine pairs. On the other hand, each of the eighteen vectors occurs in exactly two of the bases, and (M3) gives its atom the same measure in both. Each vector v with $\mu(\mathrm{at}(v)) = 1$ thus belongs to exactly two pairs, and the number of pairs is even. This contradiction proves the theorem. ☻

In physical terms, the Bell–Kochen–Specker theorem, Theorem C, excludes the noncontextual hidden-variable theories that Kochen and Specker considered. Such a theory assigns, for each value of the hidden variable, a definite truth value to every experimentally verifiable proposition about a quantum system, and this truth value, true or false, does not depend on the other propositions tested together with it. Contextual hidden-variable theories, such as that of Bohm [Boh52], in which the value revealed by a measurement depends on the whole experimental arrangement, lie outside the scope of the theorem. The experimentally verifiable propositions correspond to the closed subspaces [BvN36], whose role in a model of T is played by the elements of the Solèr sort, and a noncontextual assignment would therefore be recorded by a dispersion-free measure, one that takes only the values 0 and 1, for false and true respectively. By Theorem C, the axioms of T, σ-additivity among them, exclude such a measure as the interpretation of μ. Together with axiom (M1), the theorem yields the following corollary.

Corollary 9.2 (No dispersion-free measures). *In every model of T, standard or not, some element a of the Solèr sort satisfies $0 < \mu(a) < 1$.*

Proof. By (M1), the measure takes values in $[0, 1]$, and by Theorem C it does not take only the values 0 and 1. ☻

⧖ **Historical note.** Specker announced in 1960 that the closed subspaces of a three-dimensional Hilbert space cannot be mapped into a Boolean algebra in a way that respects the operations on orthogonal subspaces. Of the proof he said only that it rests on an elementary geometric argument [Spe60]. Bell derived from Gleason's theorem that no hidden-variable theory of the kind described above is possible, in a paper written in 1964 and published in 1966 [Bel66]. In 1967 Kochen and Specker gave a finite proof of this impossibility with 117 vectors in $\mathbb{R}^3$ [KS67]. The configurations then shrank: Peres found configurations of 33 vectors in $\mathbb{R}^3$ and of 24 in $\mathbb{R}^4$ in 1991 [Per91], and Kernaghan one of 20 vectors in $\mathbb{R}^4$ in 1994 [Ker94]. In 1996 Cabello, Estebaranz and García-Alcaine found the eighteen vectors of Table 1 [CEGA96]. Peres conjectured that no smaller configuration exists, and Xu, Chen and Gühne proved in 2020 that no Kochen–Specker set in any dimension has fewer than eighteen vectors [XCG20, BCG+22]. In dimension three the minimum remains unknown at the time of writing. Exhaustive computer searches raised the lower bound for this minimum to 22 vectors in 2016 [UW16] and to 24 in 2023. The bound of 24 was reached independently by Kirchweger, Peitl and Szeider and by Li, Bright and Ganesh [KPS23, LBG24]. The titles of the papers of Kernaghan and of Cabello, Estebaranz and García-Alcaine call the result the Bell–Kochen–Specker theorem, a name that records Bell's share and that Theorem C keeps.

10. The model existence game and finite obstructions

By Theorem C, the measure μ takes a value other than 0 and 1 in every model of the theory T of Theorem A, standard or not. In other words, the theory obtained from T by adjoining the axiom of Theorem C has no models. In first-order logic, compactness would turn this fact into a finite obstruction, a finite set of axioms that already has no models, much as finitely many vectors suffice in the Kochen–Specker theorem [KS67]. The countable suprema and infima of real-valued $L_{\omega_1\omega}$, through which σ-additivity is expressed, carry the logic beyond the reach of compactness. In place of compactness, the logic has a model existence theorem [ADI26], which concerns Henkin types, namely the assignments of values to sentences that satisfy the Henkin conditions recalled in Section 5.2. By that theorem, every Henkin type is the type of a canonical structure for the enlarged vocabulary, one in which every element is the value of a closed term. We recast this model existence argument as a game, in which finite obstructions arise at the terminal positions of a winning strategy. The same Henkin construction also underlies the omitting types theorem of Aubone, Dueñez and Iovino, proved by the Baire category theorem in the space of Henkin types. In Section A we compare these topological and game-theoretic treatments.

The game follows Väänänen's model existence game for two-valued logic [Vää11]. It is played by Olivia, who claims that a given countable theory has no models, and Josh, who claims that it has one. At each round Olivia makes a challenge, and Josh answers it by adding to a growing description of a structure (see Figure 6). Olivia wins as soon as Josh cannot answer without contradicting himself, and Josh wins if that never happens. The growing descriptions, which we call positions below, are the real-valued counterparts of the finite sets of sentences that serve as conditions in Keisler's forcing [Kei73, p. 100, Example 1.2], and the

rules for infima, suprema and countable joins follow the pattern of his ∀∃ formulas [Kei73, p. 107] (see Section A).

We show that the theory has no models exactly when Olivia has a winning strategy. Every winning strategy of Olivia, we show further, determines a well-founded tree of finite plays, at each leaf of which the commitments made along the play form a finite obstruction. A finite obstruction, in this sense, is a finite set of statements about the values of sentences that no structure satisfies. The tree branches infinitely only where Olivia challenges a countable supremum or infimum; once Josh's answers to those challenges are fixed, every play ends within a bounded number of rounds. The first three subsections depend only on Section 5. In the last, we return to the theory of Theorem C and construct a winning strategy of Olivia modeled on the proof of that theorem. Under this strategy, every play that lasts until Olivia has carried out the parity argument of that proof ends at a position whose commitments concern a configuration of the Kochen–Specker type; the other plays end earlier, at positions that are already satisfied in no structure.

10.1. **The setting.** Until the last subsection, we work with a countable many-sorted vocabulary $\mathcal{L}$ for the logic of Section 5 and with a nonempty countable theory T in $\mathcal{L}$. In the last subsection, the vocabulary and the theory are again those of Section 5.3 and of Theorem A. As usual, every sort of a structure is nonempty.

In the game, Josh commits the values of sentences to intervals, and the rules require for each sentence an interval fixed in advance that contains every possible value of that sentence. By Section 5, each formula φ has an a priori bound r_φ and takes its values in the interval $I_\varphi = [-\lceil r_\varphi \rceil, \lceil r_\varphi \rceil]$ in every structure; the endpoints of I_φ are integers, and they do not change when terms are substituted for the free variables of φ. In the game, the infima $\inf_x$ over a sort and the countable infima $\bigwedge_n$, which abbreviate $-\sup_x -\varphi$ and $-\bigvee_n -\varphi_n$, are treated as primitive, with rules of their own, and the interval of $\inf_x \varphi$ or $\bigwedge_n \varphi_n$ is that of $\sup_x \varphi$ or $\bigvee_n \varphi_n$. The formulas $s \cdot \varphi$, $\varphi + \psi$ and $\varphi \vee \psi$, by contrast, are handled by the rule (Cn) of Definition 10.2 through the functions $t \mapsto st$, $(t,u) \mapsto t+u$ and $(t,u) \mapsto \max(t,u)$ that interpret them. We call these functions the *connective functions*; each of them is continuous. Throughout this section we write $\bar{r}_\varphi$ for the integer $\lceil r_\varphi \rceil$, so that $I_\varphi = [-\bar{r}_\varphi, \bar{r}_\varphi]$. Finally, the logical unity constant $\mathbf{1}$, a nullary predicate symbol, takes only the value 1. For the definition of infeasible positions below, we write $\mathcal{V}_{\mathbf{1}} = \{1\}$ and $\mathcal{V}_R = [-r_R, r_R]$ for every other predicate symbol R.

The axioms enter the game through one of its rules, which is simplest to state when all axioms have the same form. As everywhere in the paper, an axiom of T is a closed condition required of all values of its free variables. If $\bar{x}$ lists the free variables of a formula φ, then, by the conventions of Section 5, an axiom $\varphi(\bar{x}) \le r$ is the closed condition $\sup_{\bar{x}} \varphi \le r$ and an axiom $\varphi(\bar{x}) \ge r$ is $\inf_{\bar{x}} \varphi \ge r$. An axiom $\varphi(\bar{x}) = r$ is the union of these two closed conditions. Every axiom of T may therefore be regarded as a closed condition $\psi \le r$ or $\psi \ge r$ for a sentence ψ and a rational number r, an axiom $\varphi(\bar{x}) = r$ counting as two. Theories that contain strict conditions are discussed in Remark 10.6.

To name the elements of a structure under construction, we add to $\mathcal{L}$ for each sort $\mathbf{S}$ a set $\mathcal{W}_{\mathbf{S}} = \{w_{\mathbf{S},i} : i \in \mathbb{N}\}$ of new constants of sort $\mathbf{S}$, the *witnesses*. We write $\mathcal{W}$ for the union of the sets $\mathcal{W}_{\mathbf{S}}$, and we denote by $\mathcal{L}_{\mathcal{W}}$ the vocabulary obtained by adding the witnesses to $\mathcal{L}$. We use the letters w and w' for witnesses. Constant symbols of $\mathcal{L}$ are treated as function symbols with no arguments, so that the rules below handle them together with the other function symbols. A sentence of $\mathcal{L}_{\mathcal{W}}$ is *basic* when it has the form $R(w_1, \dots, w_q)$, where R is

a predicate symbol and $w_1, \dots, w_q$ are witnesses. The number q may be 0; in particular, the sentence $\mathbf{1}$ is basic.

In the course of a play, Josh builds, in answer to Olivia's challenges, a partial description of a structure whose elements are named by witnesses. Such a description consists of two kinds of entries, commitments and records. A commitment says that the value of a sentence lies in a given interval, whereas a record says what a function symbol does to the elements named by given witnesses. Apart from the rules that bring in the axioms and make the function symbols total, the rules of the game (see Definition 10.2) follow the inductive definition of the value of a sentence. In accordance with that definition, a commitment on a sentence formed by a connective function or a countable supremum or infimum is challenged through the sentences from which that sentence is formed. A commitment on a supremum or an infimum over a sort is challenged through witnesses. A commitment on an atomic sentence, finally, is challenged through records, which replace the terms of that sentence by witnesses until a basic sentence remains.

Definition 10.1. A *commitment* is an expression $\varphi \in I$, where φ is a sentence of $\mathcal{L}_\mathcal{W}$ in which only finitely many witnesses occur and $I = [\alpha, \beta]$ is an interval with rational endpoints $\alpha \le \beta$; the interval may be a single point. A *record* is an expression $F(\bar{w}) \mapsto w'$, where F is a function symbol and $\bar{w}, w'$ are witnesses of the appropriate sorts. A *position* is a finite set $\mathfrak{p}$ of commitments and records that contains at most one record $F(\bar{w}) \mapsto w'$ for each function symbol F and each tuple $\bar{w}$. A witness *occurs* in $\mathfrak{p}$ when it occurs in a commitment or a record of $\mathfrak{p}$. In a structure for $\mathcal{L}$, a position is *satisfied* under an interpretation of the witnesses that occur in it when each of its commitments holds and, for each of its records $F(\bar{w}) \mapsto w'$, the term $F(\bar{w})$ and the witness w' have the same value. A position $\mathfrak{p}$ is *infeasible* when there is a basic sentence $R(\bar{w})$ such that the intervals I for which $\mathfrak{p}$ contains the commitment $R(\bar{w}) \in I$, together with $\mathcal{V}_R$, have empty intersection.

An infeasible position is contradictory in the most elementary way: it assigns to a basic sentence $R(\bar{w})$ finitely many rational intervals that have no common point in $\mathcal{V}_R$ (see Figure 8). Infeasibility is the elementary conflict detected by the rules; satisfiability asks whether all the commitments and records can hold together in a structure. A position may already be unsatisfiable without being infeasible, and Olivia's subsequent challenges may expose the contradiction at the level of basic sentences.

The rule for the connective functions asks Josh not for the exact values of the arguments of a connective but for small cells of a grid that contain those values. The grid is fine enough that the connective varies by at most a prescribed amount on each cell. The statement of the rule requires three notions: dyadic grids, a modulus of uniform continuity for each connective, and widened intervals. For $m \in \mathbb{N}$, the *dyadic grid* $\mathcal{G}_m$ is the set of intervals $[i2^{-m}, (i+1)2^{-m}]$, for $i \in \mathbb{Z}$. Let ℓ be 1 or 2, let $f\colon \mathbb{R}^\ell \to \mathbb{R}$ be a connective function, let $\varphi_1, \dots, \varphi_\ell$ be sentences, and let $k \in \mathbb{N}$ be given. The function f is uniformly continuous on the compact box $I_{\varphi_1} \times \cdots \times I_{\varphi_\ell}$; we write $m(f, \bar{\varphi}, k)$ for the least $m \in \mathbb{N}$ such that $|f(\xi) - f(\xi')| \le 2^{-k}$ for all points ξ, ξ' of the box with $|\xi_j - \xi'_j| \le 2^{-m}$ for $j = 1, \dots, \ell$. For an interval $I = [\alpha, \beta]$ and $k \in \mathbb{N}$, we write $I^{(k)} = [\alpha - 2^{-k}, \beta + 2^{-k}]$.

When Josh chooses a witness, the rules allow witnesses already in the position and one designated fresh witness. This restriction leaves finitely many choices at each such move, while permitting a previously unused witness whenever one is needed. The choices of an index for a countable supremum or infimum retain their countable branching; this distinction is the basis of the finite-obstruction analysis in the third subsection.

Definition 10.2 (The model existence game). The game $\mathrm{G}(T)$ is played by Olivia and Josh. It starts at the empty position; at each round, Olivia makes a *challenge*, and Josh *answers* it by adding to the current position $\mathfrak{p}$ commitments and records, if any, in the manner that the rule of the challenge prescribes. When Josh answers a challenge, the *first fresh witness* of a sort $\mathbf{S}$ is the witness $w_{\mathbf{S},i}$ with the least index i that occurs neither in $\mathfrak{p}$ nor in the challenge. In each rule other than (Ax) and (Tot), the commitment named by Olivia belongs to $\mathfrak{p}$, and its interval is written $I = [\alpha, \beta]$.

(Ax): Olivia names an axiom $\psi \le r$ or $\psi \ge r$ of T. Josh adds $\psi \in [\min(-\bar{r}_\psi, r), r]$ or $\psi \in [r, \max(r, \bar{r}_\psi)]$, respectively.

(Tot): Olivia names a function symbol F and witnesses $\bar{w}$ of the appropriate sorts. If $\mathfrak{p}$ contains no record $F(\bar{w}) \mapsto w'$, Josh adds such a record, choosing for w' a witness of the sort of the values of F that occurs in $\mathfrak{p}$ or is the first fresh witness of that sort; otherwise he adds nothing.

(Tm): Olivia names a commitment $R(\bar{t}) \in I$, where R is a predicate symbol and $\bar{t}$ is a tuple of terms, together with a term $F(\bar{w})$ that occurs in $\bar{t}$ and in which $\bar{w}$ is a tuple of witnesses (possibly empty). Josh first answers as in (Tot) for F and $\bar{w}$; then, with w' the witness such that the enlarged position contains the record $F(\bar{w}) \mapsto w'$, he adds $R(\bar{t}') \in I$, where $\bar{t}'$ is obtained from $\bar{t}$ by replacing every occurrence of $F(\bar{w})$ by w'.

(Cn): Olivia names a commitment $f(\varphi_1, \dots, \varphi_\ell) \in I$, where f is a connective function, and a number $k \in \mathbb{N}$. With $m = m(f, \bar{\varphi}, k)$, Josh chooses intervals $D_1, \dots, D_\ell$ of $\mathcal{G}_m$ such that each D_j meets I_{φ_j} and f maps $(D_1 \cap I_{\varphi_1}) \times \cdots \times (D_\ell \cap I_{\varphi_\ell})$ into $I^{(k)}$, and adds the commitments $\varphi_j \in D_j \cap I_{\varphi_j}$ (see Figure 7).

($\sup \le$): Olivia names a commitment $\sup_x \varphi(x) \in I$, where x is a variable of sort $\mathbf{S}$, and a witness $w \in \mathcal{W}_{\mathbf{S}}$. Josh adds $\varphi(w) \in [-\bar{r}_\varphi, \beta]$.

($\sup \ge$): Olivia names a commitment $\sup_x \varphi(x) \in I$, where x is a variable of sort $\mathbf{S}$, and a number $k \in \mathbb{N}$. Josh chooses a witness $w \in \mathcal{W}_{\mathbf{S}}$ that occurs in $\mathfrak{p}$ or is the first fresh witness of sort $\mathbf{S}$, and adds $\varphi(w) \in [\alpha - 2^{-k}, \bar{r}_\varphi]$.

($\inf \ge$): Olivia names a commitment $\inf_x \varphi(x) \in I$, where x is a variable of sort $\mathbf{S}$, and a witness $w \in \mathcal{W}_{\mathbf{S}}$. Josh adds $\varphi(w) \in [\alpha, \bar{r}_\varphi]$.

($\inf \le$): Olivia names a commitment $\inf_x \varphi(x) \in I$, where x is a variable of sort $\mathbf{S}$, and a number $k \in \mathbb{N}$. Josh chooses a witness w as in ($\sup \ge$), and adds $\varphi(w) \in [-\bar{r}_\varphi, \beta + 2^{-k}]$.

($\bigvee \le$): Olivia names a commitment $\bigvee_n \varphi_n \in I$, where $(\varphi_n)_{n \in \mathbb{N}}$ is a family of sentences with $r = \bar{r}_{\bigvee_n \varphi_n}$, and a number $n \in \mathbb{N}$. Josh adds $\varphi_n \in [-r, \beta]$.

($\bigvee \ge$): Olivia names a commitment $\bigvee_n \varphi_n \in I$ as in the preceding rule, and a number $k \in \mathbb{N}$. Josh chooses a number $n \in \mathbb{N}$ and adds $\varphi_n \in [\alpha - 2^{-k}, r]$.

($\bigwedge \ge$): Olivia names a commitment $\bigwedge_n \varphi_n \in I$, where $(\varphi_n)_{n \in \mathbb{N}}$ is a family of sentences with $r = \bar{r}_{\bigwedge_n \varphi_n}$, and a number $n \in \mathbb{N}$. Josh adds $\varphi_n \in [\alpha, r]$.

($\bigwedge \le$): Olivia names a commitment $\bigwedge_n \varphi_n \in I$ as in the preceding rule, and a number $k \in \mathbb{N}$. Josh chooses a number $n \in \mathbb{N}$ and adds $\varphi_n \in [-r, \beta + 2^{-k}]$.

An answer is *legal* when it obeys the rule and every interval that it adds is nonempty, that is, has its left endpoint at most its right endpoint. Josh *loses* at a round if he has no legal answer to Olivia's challenge, or if his answer makes the position infeasible. A *play* is a finite or infinite sequence of rounds played according to these rules; it ends at the round, if any, at which Josh loses, a round that consists of Olivia's challenge alone when he has no legal answer to it. Olivia wins a play if Josh loses at some round; Josh wins a play if he never loses, in which case the play is infinite. A *strategy* for a player chooses that player's next

move from the moves made so far; it is *winning* when that player wins every play in which the strategy is followed, whatever the opponent does.

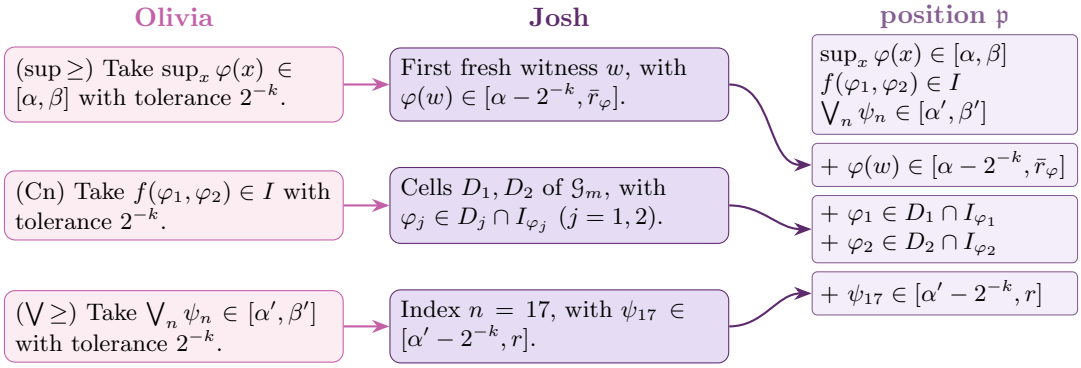


FIGURE 6. Three rounds of the game. Olivia's challenges are on the left and Josh's answers in the middle; the position, on the right, grows by one card at each answer. The first round concerns a supremum over a sort, the second a connective function and the third a countable supremum.

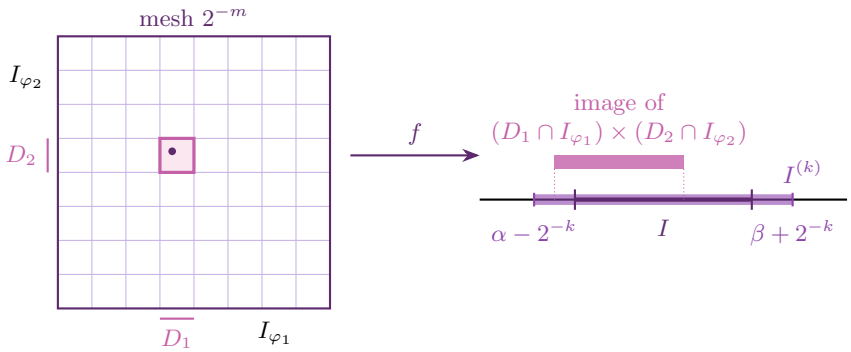


FIGURE 7. The rule (Cn) for a connective f of two variables. Josh must choose a cell $D_1 \times D_2$ of the dyadic grid of mesh 2^{-m} whose intersection with the box $I_{\varphi_1} \times I_{\varphi_2}$ is mapped by f into the widened interval $I^{(k)}$. The number k is the one named by Olivia, and the modulus of uniform continuity of f on that box, defined before Definition 10.2, is $m = m(f, \bar{\varphi}, k)$. By uniform continuity, a cell that contains the point (the dot) whose coordinates are the values of φ_1 and φ_2 in a structure satisfying the commitment $f(\varphi_1, \varphi_2) \in I$ always qualifies. The image of that cell, however, may leave I itself.

Josh's answers are of three kinds, according to the number of choices that the rules leave him. The rules (Ax), (sup $\leq$), (inf $\geq$), ($\bigvee \leq$) and ($\bigwedge \geq$) leave him no choice: the answer is determined by the challenge. Under (Tot), (Tm), (Cn), (sup $\geq$) and (inf $\leq$) he has finitely many choices because only finitely many witnesses occur in a position and only finitely many intervals of $\mathcal{G}_m$ meet a bounded interval. Only the rules ($\bigvee \geq$) and ($\bigwedge \leq$), which concern countable suprema and infima, leave him a choice among infinitely many answers. We call the challenges made under these two rules *countable challenges*, and Josh's answers to them *countable answers*.

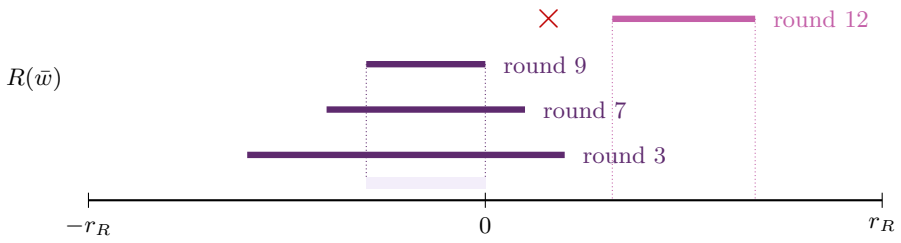


FIGURE 8. An infeasible position: at different rounds, Josh has committed the basic sentence $R(\bar{w})$ to intervals that have no common point.

10.2. **The model existence theorem as a game.** To extract a model from a play that Josh survives, we need Olivia to have made, sooner or later, every challenge open to her. A challenge is *available* at a round when the rules allow Olivia to make it there. Equivalently, a challenge is available at a round when the commitment that it names, if any, belongs to the position at the start of that round. Positions only grow, and a challenge that is available at a round therefore remains available at every later round. Since a position is finite and the vocabulary $\mathcal{L}_W$ and the theory T are countable, only countably many challenges are available at each round. Because T is nonempty, at least one challenge is available at each round. Moreover, every sentence occurring in a play is an atomic sentence of $\mathcal{L}_W$ or is obtained from a subformula of an axiom of T by a substitution. The substitution replaces the free variables of the subformula by closed terms of $\mathcal{L}_W$. The challenges that can occur in plays therefore form a countable set. A play is *fair* if every challenge that is available at some round is made by Olivia at that round or at a later one. Olivia has a *fair strategy*, one under which every infinite play is fair, whatever Josh does; we call the following one her *bookkeeping*. She fixes a bijection $i \mapsto (j(i), l(i))$ from $\mathbb{N}$ onto $\mathbb{N} \times \mathbb{N}$ with $j(i) \leq i$ for every $i \in \mathbb{N}$, and at each round she enumerates, with repetitions if necessary, the challenges available at that round. At round i she makes the $l(i)$-th challenge of her enumeration for round $j(i)$, which is still available because $j(i) \leq i$.

Theorem 10.3 (The model existence theorem as a game). *Let T be a nonempty countable theory of real-valued $L_{\omega_1\omega}$ in a countable vocabulary, whose axioms are closed conditions $\psi \leq r$ or $\psi \geq r$ for sentences ψ and rational numbers r, and let $\mathrm{G}(T)$ be the game of Definition 10.2. The following conditions are equivalent:*

(1) the theory T has a model;
(2) Josh has a winning strategy in $\mathrm{G}(T)$;
(3) Olivia has no winning strategy in $\mathrm{G}(T)$.

Proof. (1) implies (2). Let $\mathfrak{N}$ be a model of T; Josh answers every challenge truthfully in $\mathfrak{N}$. To do so, he keeps along the play a map h from the witnesses occurring in the current position to $\mathfrak{N}$. The map h makes every commitment of the position hold in $\mathfrak{N}$ when each witness w is interpreted as $h(w)$. Moreover, it satisfies the equation $F^{\mathfrak{N}}(h(\bar{w})) = h(w')$ for every record $F(\bar{w}) \mapsto w'$ of the position. When a challenge names a witness that does not yet occur, Josh first extends h to it arbitrarily, which is possible because the sorts of $\mathfrak{N}$ are nonempty. The commitment that he adds under (Ax) holds because $\mathfrak{N}$ is a model of T. Under (Tot) and (Tm), if the position contains no record $F(\bar{w}) \mapsto w'$, Josh adds one with w' the first fresh witness and extends h to w' by letting $h(w') = F^{\mathfrak{N}}(h(\bar{w}))$. Under (Tm), whether the

record is new or not, the value of $R(\bar{t}')$ in $\mathfrak{N}$ equals that of $R(\bar{t})$. To a challenge (Cn), Josh answers with cells $D_j \in \mathcal{G}_m$ that contain the values ξ_j of the sentences φ_j in $\mathfrak{N}$. Every point of $(D_1 \cap I_{\varphi_1}) \times \cdots \times (D_\ell \cap I_{\varphi_\ell})$ then has coordinates within 2^{-m} of those of $(\xi_1, \ldots, \xi_\ell)$. By the choice of m, the connective f maps every such point to within 2^{-k} of $f(\xi_1, \ldots, \xi_\ell)$, which lies in I. The commitment added under $(\sup \leq)$ holds because the value of $\varphi(w)$ is at most the supremum, hence at most β. Under $(\sup \geq)$, Josh takes the first fresh witness w. Since the supremum is at least α, he lets $h(w)$ be an element of $\mathfrak{N}$ at which φ takes a value at least $\alpha - 2^{-k}$. The remaining rules are handled in the same way; under $(\bigvee \geq)$, for instance, Josh names a number n for which the value of φ_n is at least $\alpha - 2^{-k}$. Every answer is legal, since the true values lie in the added intervals. Moreover, no position becomes infeasible, since the value in $\mathfrak{N}$ of a basic sentence $R(\bar{w})$ lies in every interval committed for that sentence and in $\mathcal{V}_R$. Since both of these facts hold at every round, Josh never loses.

(2) implies (1). Let Josh play a winning strategy against a fair strategy of Olivia. The resulting play is infinite and fair; let $\mathfrak{p}_0 \subseteq \mathfrak{p}_1 \subseteq \cdots$ be its positions, indexed by $i \in \mathbb{N}$, and let $\mathfrak{p}_\infty$ be their union. We define a structure $\mathfrak{M}$ for $\mathcal{L}_\mathcal{W}$ whose elements are the witnesses: the universe of sort $\mathbf{S}$ is $\mathcal{W}_\mathbf{S}$, and each witness is interpreted as itself. For a function symbol F and witnesses $\bar{w}$, Olivia makes the challenge (Tot) for F and $\bar{w}$ at some round, because the play is fair. The union $\mathfrak{p}_\infty$ therefore contains a record $F(\bar{w}) \mapsto w'$, which is unique because every $\mathfrak{p}_i$ is a position, and we define $F^\mathfrak{M}(\bar{w})$ to be w'. For a basic sentence $R(\bar{w})$, the intervals I for which $\mathfrak{p}_\infty$ contains $R(\bar{w}) \in I$ are compact, and so is $\mathcal{V}_R$. Every finite subfamily of the family formed by these intervals and $\mathcal{V}_R$ has a common point, because no position $\mathfrak{p}_i$ is infeasible. By the finite intersection property, the whole family has a common point, and we let $R^\mathfrak{M}(\bar{w})$ be one. As a point of $\mathcal{V}_R$, this value respects the a priori bound of R, and it equals 1 when R is $\mathbf{1}$. The logic requires of the interpretations $F^\mathfrak{M}$ and $R^\mathfrak{M}$ only that they have the appropriate sorts and, for $R^\mathfrak{M}$, that it respect the a priori bound of R (see Section 5). The interpretations chosen above therefore complete the definition of $\mathfrak{M}$.

We claim that every commitment $\varphi \in I$ of $\mathfrak{p}_\infty$ holds in $\mathfrak{M}$. The claim is proved by induction on the construction of φ and, for atomic sentences, on the number of occurrences of function symbols. For a basic sentence, the claim holds by the choice of $R^\mathfrak{M}$. A non-basic atomic sentence $R(\bar{t})$ contains a term $F(\bar{w})$ whose arguments $\bar{w}$ are witnesses (possibly none, when F is a constant symbol). By fairness, Olivia made the challenge (Tm) for the commitment $R(\bar{t}) \in I$ and this term, and Josh added $R(\bar{t}') \in I$, which holds by the induction hypothesis. The sentences $R(\bar{t})$ and $R(\bar{t}')$ have the same value in $\mathfrak{M}$, because the witness w' that replaces $F(\bar{w})$ is $F^\mathfrak{M}(\bar{w})$. For $f(\varphi_1, \ldots, \varphi_\ell) \in I$, fairness gives, for every $k \in \mathbb{N}$, intervals $D_1, \ldots, D_\ell$ such that $\mathfrak{p}_\infty$ contains the commitments $\varphi_j \in D_j \cap I_{\varphi_j}$ and the connective f maps $(D_1 \cap I_{\varphi_1}) \times \cdots \times (D_\ell \cap I_{\varphi_\ell})$ into $I^{(k)}$. By the induction hypothesis, the value of $f(\varphi_1, \ldots, \varphi_\ell)$ in $\mathfrak{M}$ lies in $I^{(k)}$ for every k, hence in I. For $\sup_x \varphi(x) \in [\alpha, \beta]$, fairness puts in $\mathfrak{p}_\infty$ a commitment $\varphi(w) \in [-\bar{r}_\varphi, \beta]$ for every witness w of the sort of x, that is, for every element of $\mathfrak{M}$ of that sort. Fairness also puts in $\mathfrak{p}_\infty$, for every $k \in \mathbb{N}$, a commitment $\varphi(w) \in [\alpha - 2^{-k}, \bar{r}_\varphi]$ for some witness w. By the induction hypothesis, the supremum lies in $[\alpha, \beta]$. The remaining cases of the induction are similar. Finally, every axiom $\psi \leq r$ of T was named in a challenge (Ax), to which Josh answered with the commitment $\psi \in [\min(-\bar{r}_\psi, r), r]$. By the claim, the structure $\mathfrak{M}$ satisfies $\psi \leq r$; the axioms $\psi \geq r$ are treated in the same way.

Hence the reduct of $\mathfrak{M}$ to $\mathcal{L}$ is a model of T.

(2) and (3) are equivalent. Regard $\mathrm{G}(T)$ as an infinite game of perfect information in which, after Josh has lost, the play continues with arbitrary moves, and in which a challenge that the rules do not allow loses for Olivia. Olivia then wins exactly when Josh loses at some

finite round before she has broken the rules. We give the space of infinite sequences of moves the topology whose basic open sets consist of the sequences that extend a given finite one; since each of Olivia's wins is decided at a finite round, the set of her wins is *open* in this topology. An infinite game of perfect information is *determined* when one of the two players has a winning strategy, and the Gale–Stewart theorem [GS53] asserts that such a game is determined whenever the set of wins of one of the players is open. Hence (3) implies (2), and since the two players cannot both have winning strategies, (2) implies (3).

Remark 10.4 (Baire category and the Banach–Mazur game). Olivia's bookkeeping can be read as a Baire category argument in the space of plays, and as a winning strategy in a Banach–Mazur game on that space. We set out both readings in Section A, beside the Baire category argument of Aubone, Dueñez and Iovino [ADI26] in the space of Henkin types.

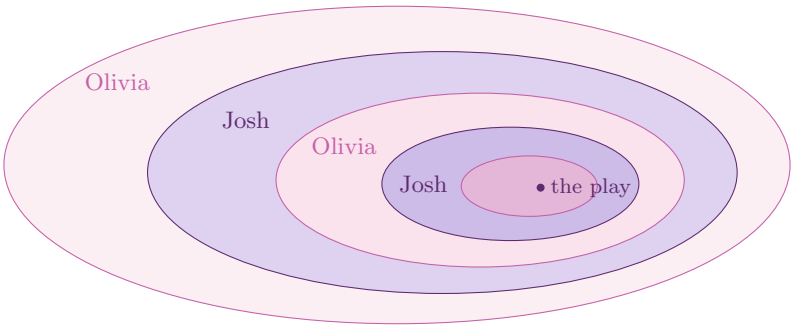


FIGURE 9. A play of the game as a sequence of nested basic open sets. Olivia (pink) and Josh (purple) move alternately, and each move, like a move of the Banach–Mazur game, shrinks the set of infinite plays that extend the current finite play. For an infinite play, the intersection of the nested sets consists of that play alone.

10.3. **Finite obstructions.** A winning strategy for Olivia is a certificate that a theory has no models, and its shape shows where finite obstructions suffice. Let σ be a strategy for Olivia. The *tree of* σ, denoted $\mathcal{P}_\sigma$, is the set of finite plays in which Olivia follows σ and Josh has not lost before the last round, ordered by extension. A finite play of $\mathcal{P}_\sigma$ in which Josh has not lost has as children its extensions by Olivia's next challenge and a legal answer of Josh. When Josh has no legal answer to that challenge, the extension of the play by the challenge alone is also a child of the play. A *leaf* of $\mathcal{P}_\sigma$ is a finite play with no children, that is, one at which Josh has lost. The strategy σ is winning exactly when $\mathcal{P}_\sigma$ has no infinite branch, that is, when $\mathcal{P}_\sigma$ is *well-founded*. The *rank* of a well-founded tree is defined by recursion: a leaf has rank 0, and every other node has as rank the least ordinal exceeding the ranks of all its children. The rank of the tree is the rank of its root. The tree $\mathcal{P}_\sigma$ is countably branching, hence countable; when it is well-founded, its rank is therefore a countable ordinal. That rank is finite exactly when the plays in which Olivia follows σ have bounded length.

In Corollary 10.5 below, we show that $\mathcal{P}_\sigma$ branches infinitely only at nodes where σ makes a countable challenge; the following notions keep track of Josh's answers to such challenges. The *countable record* of a node of $\mathcal{P}_\sigma$ is the finite sequence of the numbers n that Josh names in his countable answers along the branch from the root to the node. The numbers appear in that sequence in the order in which Josh names them. The *skeleton* of σ is the set of

countable records of nodes of $\mathcal{P}_\sigma$, ordered by extension; it is a tree of finite sequences of natural numbers.

Corollary 10.5 (Finite obstructions). *Let T be a nonempty countable theory of real-valued $L_{\omega_1\omega}$ in a countable vocabulary, whose axioms are closed conditions $\psi \le r$ or $\psi \ge r$ for sentences ψ and rational numbers r, and suppose that T has no models. Then Olivia has a winning strategy in $\mathrm{G}(T)$, and every winning strategy σ of Olivia has the following properties:*

(1) The tree $\mathcal{P}_\sigma$ is well-founded: every play in which Olivia follows σ ends after finitely many rounds, at a leaf.

(2) At every leaf, the last position $\mathfrak{p}$ is a finite set of commitments and records, involving finitely many axioms of T and finitely many witnesses, that is satisfied in no structure under any interpretation of the witnesses. Either $\mathfrak{p}$ is infeasible, in which case a basic sentence $R(\bar{w})$ receives finitely many rational intervals with no common point in $\mathcal{V}_R$, or Josh has no legal answer to Olivia's last challenge, which names a commitment of $\mathfrak{p}$ that holds in no structure.

(3) A node of $\mathcal{P}_\sigma$ has infinitely many children only if σ makes a countable challenge at it; every other node has finitely many children.

(4) If every node of $\mathcal{P}_\sigma$ has finitely many children, as is the case when σ makes no countable challenge, and in particular when no axiom of T contains a countable supremum or infimum, then the tree $\mathcal{P}_\sigma$ is finite. In that case there is a number N_σ such that every play in which Olivia follows σ ends within N_σ rounds, and some finite set of axioms of T has no models.

(5) The skeleton of σ is well-founded, and for every element $\vec{n}$ of the skeleton there is a number $N_\sigma(\vec{n})$ such that every play in which Olivia follows σ and whose countable record is $\vec{n}$ ends within $N_\sigma(\vec{n})$ rounds.

Proof. By Theorem 10.3, Olivia has a winning strategy; let σ be any winning strategy of Olivia. Part (1) holds because σ is winning. For part (2), let $\mathfrak{p}$ be the last position of a leaf, a finite set of commitments and records like every position. If Josh lost because his last answer made $\mathfrak{p}$ infeasible, then $\mathfrak{p}$ is satisfied in no structure, since the value of a basic sentence $R(\bar{w})$ in a structure lies in $\mathcal{V}_R$. Otherwise Josh had no legal answer to Olivia's last challenge, which therefore is not a challenge (Ax), (Tot) or (Tm), because under these rules a legal answer always exists. That challenge names a commitment of $\mathfrak{p}$, and the argument for "(1) implies (2)" in the proof of Theorem 10.3, which uses the axioms of T only under (Ax), produces a legal answer whenever this commitment holds in some structure under some interpretation of its witnesses. The commitment therefore holds in no structure, and neither does $\mathfrak{p}$. Part (3) follows from the count of Josh's possible answers in the paragraph after Definition 10.2.

Parts (4) and (5) rest on Kőnig's lemma: an infinite tree in which every node has finitely many children has an infinite branch. For part (4), the tree $\mathcal{P}_\sigma$ is well-founded and by hypothesis finitely branching; hence it is finite, and the length of its longest branch gives N_σ. By part (3), the hypothesis holds when σ makes no countable challenge. In particular, the hypothesis holds when no axiom of T contains a countable supremum or infimum. Indeed, every sentence that occurs in a play is atomic or is an instance of a subformula of an axiom. It remains to be shown that some finite set of axioms of T has no models. Let T_σ be the set of axioms named in challenges (Ax) at nodes of $\mathcal{P}_\sigma$, together with one axiom of T if there are none. In $\mathrm{G}(T_\sigma)$, every challenge that σ makes at a node of $\mathcal{P}_\sigma$ is allowed by the rules, and Josh has the same answers as in $\mathrm{G}(T)$. The strategy σ therefore wins $\mathrm{G}(T_\sigma)$, and the finite theory T_σ has no models by Theorem 10.3.

For part (5), let $\vec{n}$ be a finite sequence of natural numbers, and let $\mathcal{P}_\sigma^{\vec{n}}$ be the set of nodes of $\mathcal{P}_\sigma$ whose countable record is an initial segment of $\vec{n}$. The set $\mathcal{P}_\sigma^{\vec{n}}$ is a subtree of $\mathcal{P}_\sigma$, and every play in which Olivia follows σ and whose countable record is $\vec{n}$ runs inside that subtree. In $\mathcal{P}_\sigma^{\vec{n}}$, a node at which σ makes a countable challenge has at most one child. If Josh has no legal answer to the challenge, its only child is the extension by the challenge alone. Otherwise, if the countable record of the node is $\vec{n}$ itself, the node has no child in $\mathcal{P}_\sigma^{\vec{n}}$. In the remaining case, the only child of the node is the extension by the countable answer whose number is the next entry of $\vec{n}$. By part (3), every other node of $\mathcal{P}_\sigma^{\vec{n}}$ has finitely many children. Since $\mathcal{P}_\sigma^{\vec{n}}$ is well-founded, it is finite by Kőnig's lemma, and the length of its longest branch gives $N_\sigma(\vec{n})$. Finally, suppose, towards a contradiction, that the skeleton has an infinite branch $\vec{n}_1 \subsetneq \vec{n}_2 \subsetneq \cdots$. The set of nodes of $\mathcal{P}_\sigma$ whose countable record is an initial segment of some $\vec{n}_i$ then forms a finitely branching subtree of $\mathcal{P}_\sigma$. This subtree contains nodes of every countable record $\vec{n}_i$ and is therefore infinite. By Kőnig's lemma, it has an infinite branch, contradicting part (1); the skeleton is therefore well-founded.

For countable theories of the finitary part of the logic, part (4) of Corollary 10.5 yields the compactness theorem. That theorem asserts that if a countable theory of closed conditions without countable suprema and infima has no models, then some finite set of its axioms has no models. Part (3) locates the infinitary character of the obstruction: only at the countable challenges, where Josh chooses an index n among infinitely many, can the tree branch infinitely. By part (5), the skeleton (a well-founded countably branching tree) records the whole infinitary shape of the obstruction: once Josh's countable answers are fixed, the plays have bounded length (see Figures 10 and 11). For a theory with countable suprema and infima, the single finite obstruction of the compact case thus gives way to a well-founded tree of finite obstructions, one at each leaf.

Remark 10.6 (Strict conditions). The conditions of the logic are strict inequalities, and a closed condition is a countable theory of strict ones (see Section 5). For a theory that contains strict conditions, the game has one more rule.

(Str): Olivia names a condition $\varphi < r$ of T, for a sentence φ. Josh chooses a rational number $\varrho < r$ and adds $\varphi \in [-\bar{r}_\varphi, \varrho]$. For a condition $\varphi > r$ of T, Josh chooses a rational number $\varrho > r$ and adds $\varphi \in [\varrho, \bar{r}_\varphi]$.

With one more case for this rule, the proof of Theorem 10.3 applies to theories that consist of closed conditions as above and of strict conditions on sentences. In the additional case, a truthful Josh chooses ϱ between the value of φ and r, and in the structure built from a fair play the commitment $\varphi \in [-\bar{r}_\varphi, \varrho]$ gives $\varphi \leq \varrho < r$. Under (Str), however, Josh chooses among infinitely many answers, and the challenges (Str) are accordingly counted as countable challenges in Corollary 10.5. In the countable record, Josh's rational answers are coded by natural numbers; parts (1), (3) and (5) of the corollary then hold. Part (2) holds once the condition named in the last challenge is added to $\mathfrak{p}$. Indeed, Josh has no legal answer to (Str) for $\varphi < r$ exactly when $r \leq -\bar{r}_\varphi$, and none for $\varphi > r$ exactly when $r \geq \bar{r}_\varphi$. In either case, the condition named is itself unsatisfiable. With the same counting, part (4) holds, except for its clause on theories without countable suprema and infima. The exception arises because a challenge (Str) with a legal answer has infinitely many children and because such theories may have no models although each finite set of their axioms has one. For a nullary predicate symbol R with $r_R > 0$, the theory $\{R > 0\} \cup \{R < \frac{1}{n} : n \geq 1\}$ is such a theory. Olivia wins by naming first $R > 0$, to which Josh answers with some rational $\varrho > 0$ and the commitment $R \in [\varrho, \bar{r}_R]$, and then a condition $R < \frac{1}{n}$ with $\frac{1}{n} < \varrho$, to which every legal answer of Josh adds an interval disjoint from $[\varrho, \bar{r}_R]$ and thereby makes the position infeasible. The

tree of this strategy branches infinitely at its root. The contrast between the two kinds of conditions has a topological counterpart in the space of Henkin types of Aubone, Dueñez and Iovino [ADI26], which we describe in Section A.

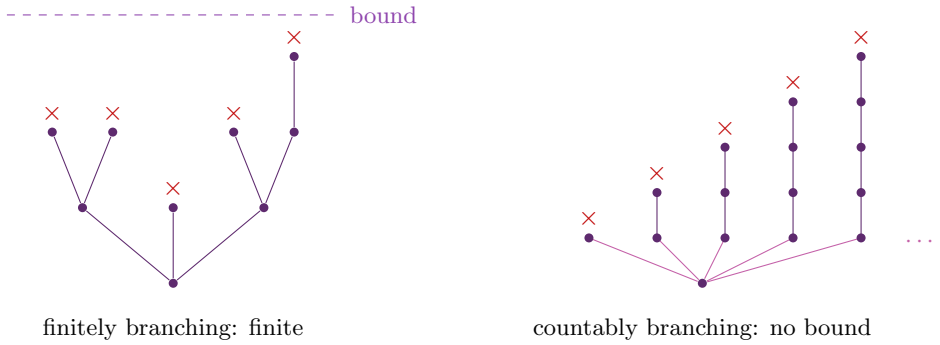


FIGURE 10. Trees grow upward from their roots. Left: a finitely branching tree without infinite branches is finite, by Kőnig's lemma, and its branches have bounded length. Right: a countably branching tree without infinite branches may have branches of every finite length, as in this fan with one branch of each length.

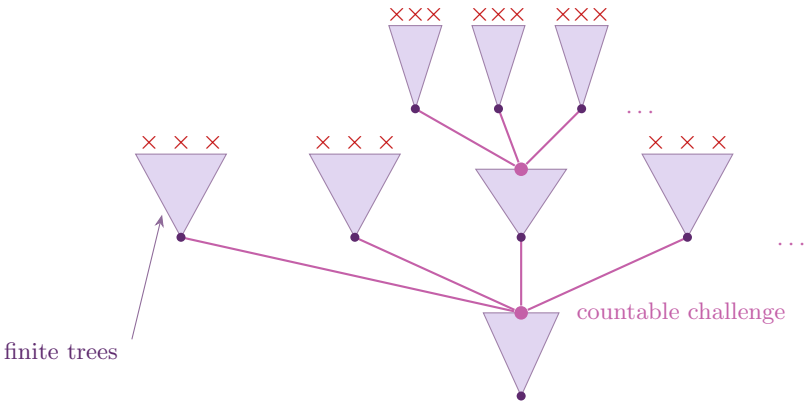


FIGURE 11. The tree of a winning strategy for Olivia, growing upward from its root. At each countable challenge (pink dot) to which Josh has a legal answer, the tree branches infinitely (pink fan), one branch for each number that he may name. The sequences of numbers that Josh names along the branches form the skeleton. The parts of the tree between countable challenges are finite (shaded), and at each of their leaves, marked $\times$, the commitments made along the play that ends there form a finite obstruction.

10.4. **The Bell–Kochen–Specker obstruction.** We now return to the theory T of Theorem A, with its measure μ, and to the axiom of Theorem C, which says that μ takes only the values 0 and 1. In the normal form described in the first subsection, that axiom becomes $\sup_a\big(\mu(a) \wedge (\mathbf{1} - \mu(a))\big) \leq 0$. By Theorem C, the theory obtained by adding the resulting axiom to T has no models, and the conclusions of Corollary 10.5 hold for it. We show that

the tree of every winning strategy of Olivia for this theory branches infinitely, and in Proposition 10.9 we construct a winning strategy modeled on the proof of Theorem C. Under that strategy, every play that lasts until Olivia has carried out the parity argument of Theorem C ends at a position whose commitments concern a configuration of the Kochen–Specker type.

Proposition 10.7. *Let T^{01} be the theory T of Theorem A together with the axiom*

$$\sup_a \big(\mu(a) \wedge (\mathbf{1} - \mu(a))\big) \leq 0.$$

(1) Olivia has a winning strategy in $\mathrm{G}(T^{01})$; every play in which she follows it ends after finitely many rounds, at a position that is satisfied in no structure.
(2) Every finite set of axioms of T^{01} has a model.
(3) The skeleton of every winning strategy of Olivia in $\mathrm{G}(T^{01})$ has more than one element: in some play in which Olivia follows such a strategy, Josh answers a countable challenge.

Proof. Part (1) follows from parts (1) and (2) of Corollary 10.5, since T^{01} has no models by Theorem C.

For part (2), recall that in the proof of Theorem C it is the instances of (K4), taken together, that make the entries of ε span the whole space, and that only then does the σ-additivity of μ along ε give some atom the measure 1. A finite set of axioms contains only finitely many of these instances, and we therefore look for its model among the structures whose sequences are all finitely supported, since in such a structure axiom (M5) involves only finite-dimensional subspaces, on which a measure that vanishes on every finite-dimensional subspace satisfies that axiom. Let a finite set of axioms of T^{01} be given, and let $Z \subseteq \mathbb{N}$ be the finite set of the numbers n for which this set contains the instance $\kappa_n\varepsilon \doteq 2^{-n-1}e_n$ of axiom (K4). A model of the given set is the following structure. Its Hilbert sort is the closed unit ball of the space ℓ^2 of square-summable complex sequences, and its Solèr sort is the lattice L_{fc} of Example 7.5. The symbols π, at, rep and the constants e_n are interpreted as in that example. The sequence sort consists of the finitely supported sequences in the closed unit ball of the space $\ell^2(\mathbb{N}, \ell^2)$ of square-summable sequences in ℓ^2. The interpretations of ι_n, κ_n and θ_N are those of the proof of Theorem 8.3, and the function span sends each element s of the sequence sort to the finite-dimensional span of its entries. The measure μ vanishes on the finite-dimensional subspaces and equals 1 on their orthogonal complements, and each truncation $R^{\pm}_{n,q}$ is interpreted by its intended value $\min\big(1, \max(0, \pm n(R-q))\big)$, so that the axioms (Tr) hold. The constant ε is the finitely supported sequence whose entry of index n is $2^{-n-1}e_n$ for $n \in Z$ and 0 otherwise; its norm is less than 1. The axioms of T_0 hold as in Example 7.5, and the axioms (K1)–(K3) and (Sp) hold as in the proof of Theorem 8.3, since the span of finitely many vectors is closed; in this structure, $[S = S'] = 1$ exactly when $S = S'$. Of the instances of (K4), the only axiom in which ε occurs, exactly those with $n \in Z$ hold. Axiom (M3) holds because $[S = S'] = 1$ only when $S = S'$; axioms (M1), (M2) and the axiom $\sup_a\big(\mu(a) \wedge (\mathbf{1} - \mu(a))\big) \leq 0$ hold because μ takes only the values 0 and 1, with $\mu(1) = 1$. For (M4), the subspaces a and $b \wedge a^{\perp}$ are orthogonal, and at most one of them is infinite-dimensional; the measure of their join is 1 exactly when one of them has measure 1. For (M5), let s be an element of the sequence sort; the element $\mathrm{span}(s)$ and the elements $c_N(s)$ are finite-dimensional subspaces, and both sides of (M5) vanish. The structure is therefore a model of the given finite set of axioms.

For part (3), suppose that the skeleton of a winning strategy σ of Olivia had only one element, the empty sequence. Then Josh would give no countable answer at any node of $\mathcal{P}_\sigma$: a node at which σ makes a countable challenge would have only one child, the extension by

the challenge alone. Every other node would have finitely many children by Corollary 10.5(3). By Corollary 10.5(4), the theory T^{01} would then have a finite set of axioms without models, contradicting part (2). ☻

Proof of Theorem D. In the model existence game, Olivia is the player who claims that the theory has no models, and the tree of plays of a strategy σ of hers is the tree $\mathcal{P}_\sigma$. Let T' be a countable theory that consists of closed conditions and has no models, with its axioms put in the normal form of the first subsection. Since T' is countable, its axioms involve only countably many sorts and symbols, and every structure for the vocabulary formed by these expands to a structure for the whole vocabulary, by arbitrary nonempty sets for the remaining sorts and arbitrary interpretations of the remaining symbols; we may therefore assume, as in Section 10, that the vocabulary is countable. The theory T' is then nonempty, since the empty theory has models, and the assertions of Theorem D about T' are those of Theorem 10.3 and Corollary 10.5, with T' in place of T. The theory of Theorem C, written in this normal form, is the theory T^{01} of Proposition 10.7. By part (2) of that proposition, every finite set of axioms of T^{01} has a model; the model built in the proof of that part even has a measure that takes only the values 0 and 1. It follows from part (3) that Josh gives a countable answer at some node of the tree of every winning strategy of Olivia. The interval that Josh adds under the rules $(\bigvee \geq)$ and $(\bigwedge \leq)$ does not depend on the number he names. Every number is therefore a legal answer at that node, and the tree branches infinitely there. ☻

The proof of Theorem C has two stages: countable additivity locates an atom of measure 1, and a finite parity argument gives the contradiction in the block containing that atom. To follow these stages in the game, Olivia must obtain finite commitments from consequences of whole axiom schemes. The following continuation argument supplies this passage and specifies when each auxiliary sequence of challenges ends. It uses fair bookkeeping, rather than a uniform bound on the number of challenges.

Observing values. Olivia can obtain intervals of arbitrarily small width for the value of a base predicate at named elements. Indeed, she brings into play the instance of (Tr) with $n = 1$ and $q = 0$ for that predicate, and follows the finite chain of connective functions to the occurrence of the predicate by challenges (Cn). At the last step she challenges the addition $R(\bar{w}) - 0$; a tolerance 2^{-k} yields an interval of width at most 2^{-k} for $R(\bar{w})$. All the functions involved have the compact domains fixed in Definition 10.2, and the mesh in (Cn) is at least this fine for addition on a nondegenerate coordinate interval. A degenerate coordinate interval already gives an exact value. The challenges may, of course, expose a contradiction before the observation is completed. Closed terms can first be named, from their innermost subterms outward, by finitely many challenges (Tot); the resulting records make these names denote the terms in every structure satisfying the position. An observation of $R(\bar{t})$ means these records together with an interval commitment on the resulting basic sentence $R(\bar{w})$; no rule copying that commitment back to the original syntactic sentence is being added. Bounds on the original value follow from the records.

Name ε by a witness w_ε, retaining the record for that constant. One further value will be observed below: the sentence

$$\chi = \bigvee_{N \in \mathbb{N}} \mu(c_N(w_\varepsilon)).$$

The instance of (M5) at w_ε, obtained by (Ax) and the universal rules, contains χ as an argument of the negation in $\mu(\mathrm{span}(w_\varepsilon)) - \chi$. Challenges (Cn) along this finite chain give

intervals of arbitrarily small width for χ as well. Here the observation concerns the value of the join, not the choice of one of its members; that choice will be made separately by $(\bigvee \geq)$.

Lemma 10.8 (Finite continuations). *Let S be a nonempty countable subtheory of T^{01}, regarded as a theory in the full vocabulary, and let $\mathfrak{p}$ be a position reached in the game. Let $\psi_1, \dots, \psi_m$, for a positive integer m, be sentences whose values can be observed as above using axioms of S. Let $U \subseteq \mathbb{R}^m$ be open, and suppose that every model of S satisfying $\mathfrak{p}$, under an interpretation of its witnesses, has*

$$(\psi_1, \dots, \psi_m) \in U.$$

Olivia has a continuation, using only axioms of S, under which every play either ends in her favour or reaches, after finitely many rounds, a position whose commitments confine this tuple of values to a nonempty closed box contained in U.

Proof. Fix an enumeration of all potential challenges generated by S, the initial position and the observing procedures, restricting (Ax) to axioms of S but including challenges to every commitment subsequently introduced, as well as (Tot). At stage $k \geq 1$, Olivia makes every challenge among the first k entries that is available at the start of the stage, and then observes all the listed values at tolerance 2^{-k}, in a fixed order. Each stage has finitely many rounds unless Josh loses, and every challenge that becomes available is eventually made, since availability persists. For each observed sentence she intersects its recorded intervals with its fixed interval of possible values; these intersections form nested closed intervals, and the observations make their widths tend to zero. She ends the continuation when their product is nonempty and contained in U.

Suppose that a play continued indefinitely without either outcome. It would be fair for S, and the construction in the proof of Theorem 10.3, applied from the initial position $\mathfrak{p}$, would give a model of S satisfying every commitment and record made in the play. This construction does not require the initial position to be empty: its commitments are included in the same induction on formulas, and its records in the definition of the functions. The tuple of observed values in this model belongs to U by hypothesis and lies in every recorded box. Since the boxes have widths tending to zero and U is open, some box is contained in U, a contradiction. An empty intersection cannot occur in such an infinite surviving play, since the constructed model supplies a point of every interval. Thus every play has one of the asserted finite outcomes. ☺

The open set in this lemma expresses a numerical margin, such as a value exceeding a prescribed rational threshold or a finite sum differing from another value by less than a prescribed tolerance. The stopping test concerns the intervals already recorded, not an assumption that the current position satisfies S. Only a hypothetical infinite surviving play is used to construct a model of S. This distinction is what permits the semantic lemmas of Section 9 to guide finite play without treating their conclusions as additional axioms.

Proposition 10.9 (The localization and parity strategy). *Olivia has a winning strategy in $\mathrm{G}(T^{01})$ with the following two phases. Unless Josh loses earlier, the first phase obtains a countable answer $N \geq 1$ and selects an index $n_0 < N$ whose atom has measure greater than $15/16$ throughout an interval recorded for it. The second phase establishes finite geometric commitments for the nine contextual bases of Table 1 in the block containing e_{n_0}, and concludes by a finitely branching continuation using finitely many instances of (M1), (M3), (M4) and the dispersion-free axiom. Every play ends after finitely many rounds. The geometric commitments in the second phase are obtained using no axiom that mentions μ.*

Proof. Localization. In every model of T satisfying the record for w_ε, the sentence χ has value 1, by (M5) and Lemma 9.1(4). Apply Lemma 10.8 with $S = T$ and the open interval $(15/16, 17/16)$. Unless Josh loses first, the intersection of the recorded intervals for χ has lower endpoint greater than $15/16$. That endpoint is the maximum of finitely many recorded lower endpoints and the fixed lower bound -2; hence one actual commitment for χ has lower endpoint $\alpha > 15/16$. A challenge $(\bigvee \geq)$ with tolerance $1/16$ then makes him name an index $N \in \mathbb{N}$ and commit

$$\mu(c_N(w_\varepsilon)) \geq \alpha - 1/16 > 7/8.$$

If $N = 0$, the selected term is $c_0(w_\varepsilon) = 0$; fair bookkeeping for T exposes a contradiction after finitely many rounds, since an infinite surviving play would give a model of T with $\mu(0) > 7/8$, contrary to Lemma 9.1(3). We may therefore suppose that $N \geq 1$.

Write $h_n = \mathrm{at}(\kappa_n w_\varepsilon)$ for $n < N$. In every model of T satisfying the retained record for w_ε, the finite-additivity calculation in the proof of Theorem C gives

$$\sum_{n<N} \mu(h_n) = \mu(c_N(w_\varepsilon)).$$

A further application of Lemma 10.8, with $S = T$, confines the values $\mu(h_n)$ to a closed box on which their sum exceeds $3/4$. Put $\delta_N = \min(1/16, 1/(4N))$. Now use only (M1), the dispersion-free axiom and (Tr) for μ. In every model of these axioms each $\mu(h_n)$ is 0 or 1. The lemma therefore gives a refinement of the recorded box in which each coordinate interval lies either within $(-\delta_N, \delta_N)$ or within $(1 - \delta_N, 1 + \delta_N)$. If all intervals were of the first kind, their sum would be less than $N\delta_N \leq 1/4$, contrary to the preceding box condition. Choose the least index $n_0 < N$ whose interval is of the second kind. Its lower endpoint is greater than $1 - \delta_N \geq 15/16$. To pass to the named basis vector, first apply the lemma with $S = T_1$ to the atomic norms defining $[h_{n_0} = \mathrm{at}(e_{n_0})]$, obtaining a positive lower bound for this equality formula. The geometric identity holds in every model of T_1 satisfying the records, by (K4) and Lemma 9.1(2), whose geometric assertion uses no measure axiom. With this bound recorded, (M3) forces $\mu(\mathrm{at}(e_{n_0})) = \mu(h_{n_0})$ in every model of (M3) satisfying the position. An application of the lemma using (M3) and (Tr) for μ therefore records an interval for $\mu(\mathrm{at}(e_{n_0}))$ with lower endpoint greater than $15/16$. This completes the first phase; no fixed number of its auxiliary challenges has been asserted.

Geometric identities. Choose $j \in \mathbb{N}$ with $4j \leq n_0 < 4j + 4$ and use the closed terms $s_{\mathcal{B}}$ and $\mathrm{at}(v)$ of the proof of Theorem C for the nine bases of Table 1 in that block. The four contextual atom terms of a basis $\mathcal{B}$ are $a_{\mathcal{B},i} = \mathrm{at}(\kappa_i s_{\mathcal{B}})$, for $i < 4$. Also consider the basis of the block formed by its four named vectors, ordered with e_{n_0} first. For this basis use the direct atom terms $\mathrm{at}(e_{n_0})$ and the atoms of the other three named vectors, and let a_F be their join in that order. For any of these ten ordered bases, write its atom terms as $a_0, \ldots, a_3$, and form the terms

$$d_1 = a_0, \qquad d_{i+1} = d_i \vee a_i, \qquad q_i = a_i \wedge d_i^{\perp}, \qquad r_i = d_i \vee q_i \quad (1 \leq i \leq 3).$$

Let $\mathcal{D}$ be the following finite list of pairs of closed terms of the Solèr sort: the two contextual terms of each shared ray; the pairs (q_i, a_i) and (r_i, d_{i+1}) for each ordered basis; the pairs (d_4, a_F) for these bases; and the pair $(h_{n_0}, \mathrm{at}(e_{n_0}))$. By Lemmas 6.3, 6.4 and 9.1(2), each pair (u, v) in this list satisfies $[u = v] = 1$ in every model of T_1 satisfying the records. These conclusions use no measure axiom.

For such a pair, the formula $[u = v]$ is the minimum of $1 - 2\|\mathrm{rep}(u \wedge (u \wedge v)^{\perp})\|$ and the corresponding expression with u and v interchanged. The norms in these expressions vanish in every model of T_1 satisfying the records. Apply Lemma 10.8 with $S = T_1$ to these finitely

many atomic norms and the open box in which each has absolute value less than $1/8$. Unless Josh loses earlier, the resulting basic commitments ensure

$$[u = v] > 3/4 \qquad ((u, v) \in \mathcal{D})$$

in every structure satisfying the position. Name the finitely many closed subterms involved by (Tot), retaining their records. Thus both the contextual ray identities and the identities needed for finite additivity have been recorded to a definite margin before any probability is carried across them.

The finite parity continuation. Retain the finite set $\mathfrak{p}_0$ consisting of these function records and norm commitments, together with the interval for $\mu(h_{n_0})$ whose lower endpoint exceeds $15/16$. Bring into play the instances of (M3) at the pairs in $\mathcal{D}$, the instances of (M4) at (d_i, a_i) for the ten bases, and the instances of (M1) and the dispersion-free axiom at the finitely many terms occurring in these measure conditions. Each instance is obtained by (Ax) followed by the universal rules, using the witnesses that name its terms; an equation contributes its two closed inequalities. Denote the resulting finite family of ground commitments by $\mathfrak{q}$. All its formulas are finitary in the primitive connectives of Section 5.

There is no assignment of values to its finitely many basic sentences, within their prescribed bounds and the intervals of $\mathfrak{p}_0$, that satisfies these ground conditions when terms are evaluated through their records. Indeed, (M3) and the positive antecedents give $\mu(u) = \mu(v)$ for each pair in $\mathcal{D}$. For each ordered basis, (M4) gives $\mu(r_i) = \mu(d_i)+\mu(q_i)$, and the recorded identities consequently give

$$\mu(d_{i+1}) = \mu(d_i) + \mu(a_i), \qquad \mu(a_F) = \sum_{i<4} \mu(a_i).$$

In the named basis of the block, the first summand is $\mu(\mathrm{at}(e_{n_0})) = \mu(h_{n_0}) > 15/16$ and hence is 1 by the dispersion-free condition and (M1). All other summands are nonnegative and $\mu(a_F) \leq 1$ by (M1), so $\mu(a_F) = 1$. Each of the nine contextual bases therefore has exactly one atom of measure 1. The paired occurrences have equal measures, giving the same odd–even contradiction as in the proof of Theorem C.

Olivia now fairly challenges only the finitary commitments of $\mathfrak{p}_0 \cup \mathfrak{q}$ and their descendants under (Cn) and (Tm). All their closed terms already have records; (Tm) merely replaces them by their recorded witnesses. If this continuation survived indefinitely, the compact intervals for its finitely many basic sentences would have common values, because no position was infeasible. The induction on finitary formulas in the proof of Theorem 10.3, using all tolerances in the fair schedule, would make those values satisfy $\mathfrak{p}_0 \cup \mathfrak{q}$, contrary to the preceding paragraph. Thus the continuation wins. It makes no countable challenge and is finitely branching, so its tree is finite by Kőnig's lemma. Other commitments made earlier in the play may expose a contradiction sooner; they are not needed for this final argument.

Each of the preceding continuations has a finite outcome, there are only finitely many such continuations after N is named, and the final parity continuation is finite. Fixing the stated enumerations and the least choices therefore defines a winning strategy at every live history, with the two asserted phases. ☺

The construction supplies finite geometric commitments rather than literal equalities between contextual terms. Their positive margins activate (M3), and its instances carry the probabilities across the corresponding identities in the finite parity argument. The last continuation uses only the finite incidence pattern and finitary measure conditions; the earlier

continuations are where the countable challenges needed to establish the geometric and localization margins may occur. Their lengths need not have a common bound, so the construction does not answer Question 10.10. The two phases are pictured in Figures 12 and 13.

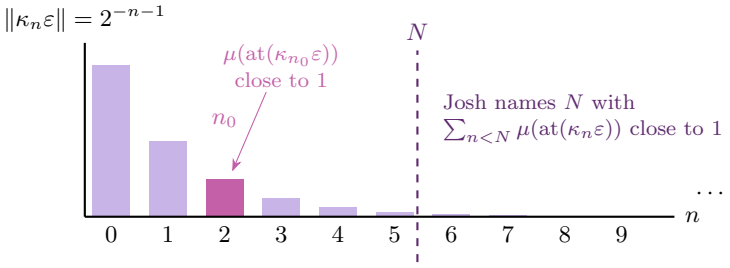


FIGURE 12. The localization phase of Proposition 10.9, shown schematically. The entries of ε have norms 2^{-n-1}. Josh names N with $\mu(c_N(\varepsilon))$ close to 1; the subsequent finite continuations obtain a large sum $\sum_{n<N} \mu(\mathrm{at}(\kappa_n\varepsilon))$ and select some $n_0 < N$ whose atom has value close to 1. The auxiliary challenges are not drawn.

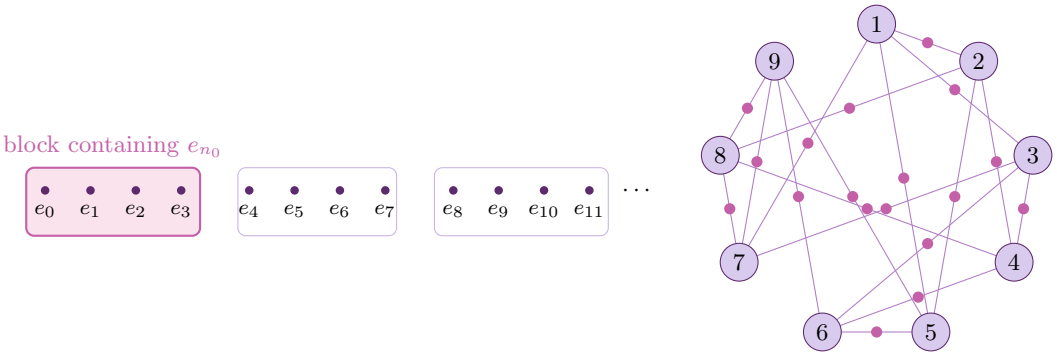


FIGURE 13. The parity configuration used in the final continuation of Proposition 10.9, shown schematically. Left: the basis vectors e_n grouped in blocks of four, with the block that contains e_{n_0} highlighted (here $n_0 = 2$, as in Figure 12). Right: the nine bases of Table 1 (numbered circles) and their eighteen rays (dots), each ray drawn on the segment that joins the two bases containing it. The element a_F of the proof of Theorem C, which stands for the span of the block, has measure 1. A $\{0,1\}$-valued measure would therefore give the value 1 to exactly one ray in each basis. The pairs formed by a basis and its ray of value 1 would then number nine when counted basis by basis. Counted ray by ray, their number would be even, since each ray lies in exactly two bases. The drawing omits the ground commitments and auxiliary challenges specified in the proposition.

Josh gives countable answers in some plays of every winning strategy of Olivia in $\mathrm{G}(T^{01})$, by Proposition 10.7(3). By Corollary 10.5(5), the plays of such a strategy have bounded length once those answers are fixed. Whether the length can be bounded independently of them remains open.

Question 10.10. Does Olivia have a winning strategy in $\mathrm{G}(T^{01})$ whose tree has finite rank, that is, such that every play in which she follows it ends within a fixed number of rounds?

Remark 10.11 (Standard models). The game $\mathrm{G}(T)$ concerns all models of T, standard or not, whereas Theorem B concerns the standard models, which are singled out by a hypothesis on the model rather than by axioms. For a Borel set B that meets $[0,1]$, let T_B be the theory T together with the axiom $\sup_a \beta_B(a) \leq -1$, the normal form of the axiom of Theorem B. Here the formula $\beta_B(a)$, given by Lemma 5.2, takes the value 1 at a when $\mu(a) \in B$ and -1 otherwise, in every structure that satisfies (Tr). By Theorem 10.3, Olivia has a winning strategy in $\mathrm{G}(T_B)$ exactly when T_B has no models at all, standard or not. This is the case when B contains 0 or 1, by Lemma 9.1(3) and (M2), and when B contains $(0,1)$, by Corollary 9.2. Some other Borel sets B give theories with models. Indeed, the range of μ in a countable model $\mathfrak{N}$ of T (see Remark 11.2) is countable and contains 0 and 1 and, by Corollary 9.2, a point of $(0,1)$. For the complement B of that range in $[0,1]$, the structure $\mathfrak{N}$ is a model of T_B. We do not know which of the remaining Borel sets B give theories T_B with models.

Historical note. The Banach–Mazur game began as Problem 43 of the Scottish Book, posed by Mazur. On 4 August 1935 Banach wrote in the book that Mazur's conjecture is true, but Banach's proof was never published, and Oxtoby published one in 1957 [Mau81, Oxt57]. Gale and Stewart proved in 1953 that open and closed games are determined [GS53]. The analysis of finite obstructions rests, in addition, on the lemma on finitely branching trees that Dénes Kőnig had published in 1927 [Kőn27]. The witnesses of the game descend from the new constants of Henkin's completeness proof of 1949 [Hen49]. For Väänänen there are essentially three kinds of games in logic, the semantic game, the model existence game and the Ehrenfeucht–Fraïssé game; his book of 2011 is built on them [Vää11].

11. Remarks

The remarks of this section concern the choices made in setting up the theory and the limits of what its axioms can say; none of them is used in the proofs of the main theorems. We first explain why the vocabulary links vectors to elements of the Solèr sort through projections rather than through a membership relation, and why the hypothesis of standardness cannot be removed from part (1) of Theorem A or from Theorem B. Axiom (K4) is indispensable to Theorem C in a similar way, and after showing this we describe the changes that real and quaternionic scalars require, closing with the logical form of the axioms and the redundancies among them.

Remark 11.1 (Projections rather than membership). The vocabulary links a vector and an element of the Solèr sort through the projection π rather than through a membership relation. In terms of π, a vector x belongs to an element a of the Solèr sort exactly when $\|\frac{1}{2}x - \frac{1}{2}\pi(a,x)\| = 0$. The choice matters for joins: with a membership relation, the members of $a \vee b$ would naturally be described existentially. These members would be described as the vectors x with $x = y + z$ for some member y of a and some member z of b. In the logic, an existential statement of this kind is expressed by the vanishing of an infimum, here the infimum of a scaled norm of $x - y - z$ over such y and z. Consider a standard model whose Hilbert sort is the closed unit ball of a Hilbert space H. In that model, the infimum of the scaled norm of $x-y-z$ vanishes exactly at the sums $y+z$ of such members, since these sums form a closed set. Indeed, bounded sequences in the closed subspaces S_a and S_b spanned by the members of a and of b have weakly convergent subsequences. When $S_a + S_b$ is not closed, these sums therefore miss the vectors of the unit ball that lie in the join $\overline{S_a + S_b}$ of S_a and S_b in $L(H)$ but not in $S_a + S_b$. Axiom (Sub2) reaches the join through the projections instead, by von Neumann's alternating projection theorem. The atoms are reached in the

same manner, since axiom (At1) describes $\mathrm{at}(x)$ through the projection onto it: in a standard model of T_0, the functions at and rep provide the bijection between the rays of H, which correspond to the unit vectors up to a phase, and the atoms of the quotient of the Solèr sort by its equality (see Proposition 7.3). Completeness of that quotient, by contrast, is imposed by no axiom: in the standard models of T_1 it is a theorem, which the sequence sort yields through the function span (see Theorem 8.3).

Remark 11.2 (Nonstandard models). Standardness is essential to Theorem B and to part (1) of Theorem A, because by the downward Löwenheim–Skolem theorem [ADI26, Lemma 3] the theory T has models whose sorts are countable, and no such model is standard. In each of these models the range of μ is countable, and hence different from $[0,1]$; likewise, the quotient of the Solèr sort by the relation $[a = b] = 1$, being countable, is not isomorphic to the lattice of closed subspaces of an infinite-dimensional Hilbert space, which is uncountable.

In a nonstandard model of T_0, even the conclusion of Gleason's theorem for T_0 (see Theorem 7.6) can fail, as the pre-Solèr lattice of Example 4.3 shows. Recall from that example that E_0 is the space of finitely supported complex sequences with the inner product of ℓ^2, and that Λ_0 is the lattice formed by the finite-dimensional subspaces of E_0 and their orthogonal complements in E_0. Take the closed unit ball of E_0 as Hilbert sort, the standard unit vectors as the constants e_n for $n \in \mathbb{N}$, and the lattice Λ_0 as Solèr sort. The function π sends (a, x) to the orthogonal projection of x onto a, and the function at sends x to $\mathbb{C}x$. Finally, the function rep assigns to each $a \neq \{0\}$ a unit vector $\mathrm{rep}(a)$ chosen in a, and to $\{0\}$ the vector 0. With the truncations interpreted by their intended values, the resulting structure is a model of T_0 that is not standard. Every maximal orthonormal family of E_0 is countable, and its rays have join 1 in Λ_0. A σ-additive state ν on Λ_0 would therefore give a function f on the unit vectors of E_0, defined by $f(x) = \nu(\mathbb{C}x)$, with values in $[0,1]$. Its sum over every maximal orthonormal family would be 1, and its sum over an orthonormal basis of any finite-dimensional subspace M would be $\nu(M)$, independently of that basis. Thus f would be a nonnegative weak frame function in the sense used by Dvurečenskij [Dvu95, Theorem 3.1], with maximal-family sums uniformly bounded below by 1 (compare Hamhalter and Pták [HP87]). His theorem would force E_0 to be complete, a contradiction. Hence Λ_0 has no σ-additive state, although every density operator ρ on ℓ^2 defines a map $a \mapsto \mathrm{Tr}(\rho P_a)$ on Λ_0, with P_a the projection of ℓ^2 onto the closure of a. The identification of σ-additive states with all such trace maps therefore fails in this nonstandard model.

In standard models, the axioms of Theorem B exclude every range other than $[0,1]$, one Borel set at a time, or more generally one σ-projective set at a time (see Theorem 8.6). No sentence, however, and indeed no countable theory, says that the range of μ is disconnected. Indeed, suppose that a countable theory T' of real-valued $L_{\omega_1\omega}$ in the vocabulary of T were satisfied, among the models of T, exactly by those in which the range of μ is disconnected; a sentence ψ that took the value 1 exactly in these models would give such a theory, namely the closed condition $\psi = 1$. Let $\mathfrak{M}$ be a standard model of T, which exists by Theorem A(2). By Theorem B, the range of μ in $\mathfrak{M}$ is the connected set $[0,1]$, and $\mathfrak{M}$ is therefore not a model of T'. By the downward Löwenheim–Skolem theorem [ADI26, Lemma 3 and its proof], applied to a countable fragment that contains the sentences of the axioms of T and of the conditions of T', the model $\mathfrak{M}$ has a substructure $\mathfrak{N}$ with countable sorts in which every sentence of that fragment keeps its value. Hence $\mathfrak{N}$ is a model of T but not of T', yet the range of μ in $\mathfrak{N}$ is countable and contains $\mu(0) = 0$ and $\mu(1) = 1$, whence it is disconnected, a contradiction.

We leave open whether the measure μ of a nonstandard model of T can differ from every map $a \mapsto \mathrm{Tr}(\rho \bar{P}_a)$ with ρ a density operator on a complex Hilbert space $\bar{E}$ as in Lemma 6.3. Here $\bar{P}_a$ is the orthogonal projection of $\bar{E}$ that extends $\pi(a, \cdot)$.

Remark 11.3 (Axiom (K4)). Axiom (K4) is essential to Theorem C, which concerns every model of T. Indeed, the structure described in the proof of Proposition 10.7(2), taken with Z empty, satisfies every axiom of T except (K4), and its measure takes only the values 0 and 1. In the standard models the axiom is not needed, since the proof of Theorem A does not use it and Gleason's theorem then excludes such a measure; it is the sequence sort of a nonstandard model, which may lack the sequences whose spans exhaust the space, that makes (K4) indispensable.

Remark 11.4 (Scalars). The vocabulary fixes the complex numbers as scalars. For real Hilbert spaces, replace $\mathbb{Q}(i)$ by $\mathbb{Q}$ and omit the predicate $\mathrm{Im}\langle x \mid y\rangle$. For quaternionic spaces, use the quaternions with rational coordinates, acting on the left, and four predicates for the real components of the form. In Proposition 7.3, "complex number of modulus one" then becomes "real number of modulus one" or "quaternion of modulus one". Each choice yields the analogue of Theorem 8.3 for the corresponding case of Theorem 3.6, and the proof of Theorem C applies in all three cases, since the vectors of Table 1 have real coordinates.

Remark 11.5 (The logical form of the axioms). Every axiom of T is a closed condition in the logic of Aubone, Dueñez and Iovino, built with the connectives $s \cdot \varphi$, $\varphi + \psi$, $\varphi \vee \psi$, the suprema over the sorts and countable joins of families with bounded a priori bounds. The axioms (H1), (OL), (P), (Sub0), (Sub3), (At1), (At2), (K1), (K2), (K4), (Sp1), (Tr) and (M1)–(M4) involve no countable join, except through the squares of Lemma 5.1 in (H1), (K1) and (At1). Axioms (H2), (Sub1), (Sub2), (K3), (Sp2) and (M5) use countable suprema or infima, and (At3) uses the countable join $\lceil \|x\| > 0 \rceil$. The hypothesis of the implication in (M3) is the formula $[a = b]$ of Section 5.3, and that of (Sp2) is the countable conjunction of the formulas $[\mathrm{at}(\kappa_n s) \leq b]$; these formulas are built from the norms of the vectors $\mathrm{rep}(c)$, and by Lemma 6.4 they, and hence the conjunction, are two-valued in every model of T_1, whose reduct to the vocabulary of T_0 is a model of T_0, and there they express the order and the equality of the Solèr sort. Each axiom (Tr), for given R, n and q, consists of closed conditions on formulas without countable joins, whose a priori bounds grow with n; the formulas $\lceil R(\bar{t}) > q \rceil$ and $\lceil R(\bar{t}) < q \rceil$, which gather the truncations for all n, are countable joins of members with a priori bound 1. The functions at and span take values in the Solèr sort, on which the formula $d(a, b) = \frac{1}{2}(1 - [a = b])$ defines, in every model of T, a pseudometric with the values 0 and 1. Its topology induces the discrete topology on the quotient by the equality. In a standard model, neither function is continuous from the norm topology of its vector sort to this topology, since at separates any two nonzero vectors that are not multiples of each other, however close they may be, and span does the same to the sequences $\iota_0 x$ and $\iota_0 y$, because by Lemma 8.2 the element $\mathrm{span}(\iota_0 x)$ has the class of $\mathrm{at}(x)$. The logic accommodates this discontinuity, since it places no condition on the interpretation of a function symbol beyond its sorts. This observation concerns the norm topology and the discrete quotient topology just specified. It does not preclude a pre-metric expansion in Keisler's sense: for a countable vocabulary such an expansion exists with a distance predicate that is a uniform limit of formulas [Kei23, Theorem 3.3.4], and its topology need not be the norm topology.

Remark 11.6 (Redundancy). The axioms (OL) hold in every structure that satisfies the other axioms of T_0, since the proof of Lemma 6.4, which embeds the quotient of the Solèr sort into the lattice of closed subspaces of a Hilbert space, does not use them. In standard models, axiom (Sub2) likewise follows from (Sub1), (Sub3) and (OL). The two redundancies cannot be exploited together, because the proof of Lemma 6.4 uses (Sub2) to compute joins, whereas the derivation of (Sub2) uses (OL); both groups of axioms are kept nonetheless, so that each property has an axiom of its own.

Historical note. Löwenheim proved in 1915 that a first-order sentence with a model has a countable model [Lö15]. Skolem extended the result to countable sets of sentences in 1920, and in 1922 he observed that axiomatic set theory, in which uncountable sets exist, therefore has countable models [Sko70]. The countable models of Remark 11.2 are of the same kind: they satisfy every axiom of a theory whose standard models are uncountable. That an incomplete inner product space can be recognized by the absence of states goes back to Hamhalter and Pták. In 1987 they proved that a separable inner product space is complete exactly when the lattice of its orthogonally closed subspaces admits a σ-additive state [HP87]. The theorem of Dvurečenskij of 1995 used in Remark 11.2, which detects completeness through frame functions rather than through states on the whole lattice of orthogonally closed subspaces, belongs to the same line of work [Dvu95].

12. Finite obstructions and Kochen–Specker configurations

Kochen and Specker certified, by a finite configuration of rays in a finite-dimensional space, that no quantum probability measure on the lattice of closed subspaces of that space takes only the values 0 and 1. Such a configuration is a finite set of rays in a space of finite dimension, considered together with its bases, the sets of mutually orthogonal rays of the configuration whose number equals the dimension. An assignment of the values 0 and 1 to its rays is *admissible* if it gives the value 1 to exactly one ray of each of these bases and to no two orthogonal rays. The configurations of Kochen and Specker have no admissible assignment.[4] The model existence game certifies that the theory of Theorem C, whose constants e_n name an infinite orthonormal family, has no models. The certificates of the game are finite sets of commitments that stand at the leaves of a well-founded tree of plays. A finite obstruction of the game is the last position of a play won by Olivia (see Corollary 10.5). In this section we compare the two kinds of certificate: their material, their demands on the measure, their location, and the topological property of the logic behind their finiteness.

12.1. **Events, probabilities and hidden-variable responses.** The comparison rests on three notions that the foundations of quantum mechanics keep apart: an outcome event, its representation by a projection, and the response that a hidden-variable model assigns to it. The model existence game keeps them apart as well, since the geometric identities between projections and the probabilistic requirements on the measure enter a play through different axioms.

An outcome event is a specified outcome of a specified measurement procedure. For a projective measurement on a complex Hilbert space H, quantum theory represents the measurement by a family of mutually orthogonal projections resolving the identity, called here a *context*, and each outcome event by a member of that family. For a density operator ρ on H, a positive operator of trace 1, the Born probability of an event represented by a projection P is $\mathrm{Tr}(\rho P)$. Two outcome events are *operationally equivalent* when their probabilities agree for every preparation procedure. With all quantum preparations available, two projective outcome events are operationally equivalent exactly when their projections are equal.

An *ontological model* introduces an underlying state λ and assigns to an outcome event E a response $\xi_E(\lambda) \in [0,1]$, interpreted as its conditional probability at that state. For each underlying state, the responses to the outcomes of a measurement sum to 1. The preparation procedure determines a distribution of underlying states, and averaging the responses against

[4] An interactive presentation of the smallest configuration found so far in our ongoing computational search is maintained on the UTSA Quantum Mathematics website, where the current example and search status are recorded.

that distribution reproduces the operational probabilities. *Measurement noncontextuality* requires operationally equivalent events to have equal responses at every underlying state; *outcome determinism* requires the responses to take only the values 0 and 1 [Spe05]. These requirements concern the underlying responses, whereas operational equivalence concerns their preparation-dependent averages over underlying states.

Consider an ontological model that reproduces the quantum probabilities for all preparations and all finite projective measurements on H. At each fixed underlying state, measurement noncontextuality and outcome determinism then give a normalized, finitely additive $\{0,1\}$-valued map on the projections of H, since noncontextuality makes the response depend on the projection alone, and the measurements $\{P, Q, 1-P-Q\}$ and $\{P+Q, 1-P-Q\}$, for orthogonal projections P and Q, then give finite additivity. The theory T^{01} of Proposition 10.7, obtained from T by adjoining the axiom of Theorem C, represents such a map by the single predicate μ; through axiom (M5), however, it requires in addition countable additivity along the spans of sequences, a hypothesis that the ray-localization argument below uses. Quantum probabilities and hidden-variable responses thus play different roles: a density operator gives a probability measure on projections, while the dispersion-free hypothesis concerns the responses at a fixed underlying state. The geometric axioms derive identities between the presentations of a projection in different contexts, and axiom (M3) carries the values of the measure along those identities.

12.2. **Contextual presentations and geometric identities.** Throughout the remainder of this section, T^{01} is the theory of Proposition 10.7, obtained by adjoining the dispersion-free axiom to the theory T of Theorem A. We consider the strategy of Olivia described in Section 10.4 and the plays that last until Olivia has carried out its parity argument. In the comparison that follows, the vertices are contextual: each of them indexes an occurrence of a term that represents a ray in a specified basis. In every model of T, the two terms that represent a ray shared by two bases denote elements a and b of the Solèr sort with $[a = b] = 1$, that is, elements with the same projection $\bar{P}_a = \bar{P}_b$ (see Lemma 6.3 and part (3) of Lemma 6.4); these elements need not coincide, since the vocabulary has no equality symbol and a nonstandard model may contain distinct elements with the same projection. In Proposition 10.9, the finite position records positive lower bounds on the formulas expressing these geometric identities. The ground instances of (M3) then force exact equality of the corresponding measure values; this implication needs only a positive antecedent, not the value 1.

Representing contextual occurrences separately has precedents in quantum foundations. For the eighteen-ray configuration, Kunjwal and Spekkens [KS15, Fig. 2] represent measurement events by distinct nodes and display the operational equivalences between them. Our analysis places the relevant projection identities within a theory of vectors, sequences and subspaces, and follows their consequences through the probability predicate and the model existence game.

Other frameworks distinguish contextual occurrences through different mathematical requirements. In contextuality-by-default, variables with a common content in different contexts are represented separately. A coupling is a jointly distributed family reproducing the joint distribution in each context; the noncontextuality criterion imposes specified maximal-agreement requirements on occurrences with common content [KDL15, DKL15]. In the sheaf-theoretic framework of Abramsky and Brandenburger, compatible probability distributions on contexts agree on overlaps. Noncontextuality requires a probability distribution on assignments of outcomes to all measurements whose marginal on each context reproduces the

distribution assigned to that context [AB11]. The present comparison concerns the representation of cross-context relations: our certificates track the derivation of projection identities and their use by a dispersion-free valuation.

The comparison with the usual ray configuration is summarized in Tables 2 and 3, which separate what holds for every winning strategy of Olivia from what belongs to the strategy of Section 10.4. For every winning strategy of Olivia in $\mathrm{G}(T^{01})$, the results of Section 10 provide a well-founded tree of plays and, at each of its leaves, a finite position that no structure satisfies, whose interval commitments record rational tolerances on the values of the hypothetical measure. The contextual term occurrences, the four-dimensional block and the eighteen-ray pattern, by contrast, belong to the strategy of Section 10.4 and to the plays that last until Olivia has carried out its parity argument; in those plays, countable additivity along the named basis selects the block in which the parity argument is carried out.

	Kochen–Specker configurations	finite obstructions of the model existence game for T^{01}
framework	partial Boolean algebras: in dimension at least three, the projections admit no homomorphism into the two-element Boolean algebra	real-valued $L_{\omega_1\omega}$: the theory T^{01} has no models, although each of its finite sets of axioms has one
topological property of the logic	compactness, the topological form of finiteness: each condition on a basis involves finitely many rays and defines a closed set, whence the theorem of J. S. Bell yields a finite configuration (J. L. Bell, Pitowsky); Kochen and Specker exhibited one	Čech-completeness of the space of Henkin types (see Section A), in which the Baire category theorem holds: σ-additivity along the named basis, which locates the unit-weight block under the strategy of Section 10.4, involves infinitely many rays at once, and the inconsistency of T^{01} is genuinely infinitary; the finite certificates come from the game, whose analysis rests on determinacy, and Olivia's bookkeeping is a Baire category argument on the space of plays, the analogue of the one on the space of Henkin types
		for every winning strategy of Olivia, who claims that T^{01} has no models
tree of plays	none: one configuration serves against every assignment	well-founded, and branching infinitely at some node
certificate of impossibility	no assignment of 0 and 1 to the rays gives 1 to exactly one ray of each basis and to no two orthogonal rays	at each leaf, a finite position of commitments and function records satisfied in no structure

TABLE 2. The certificates of the model existence game for T^{01}, with its axioms as stated in Section 6 and Proposition 10.7, beside the finite coloring certificate of Kochen and Specker. The assertions about the tree of plays and its leaves hold for every winning strategy of Olivia.

In a configuration of Kochen and Specker, a vertex is a ray, a single vertex in all the bases that contain it. That a ray receives one value in all its bases is thus part of the data. Under the strategy of Section 10.4, a vertex indexes the presentation of a ray in a particular basis. As in the proof of Theorem C, each of the nine bases $\mathcal{B}$ of Table 1, with vectors $v_0, \dots, v_3$, gives a closed term $s_{\mathcal{B}}$ of the sequence sort whose entries $\kappa_i s_{\mathcal{B}}$ satisfy $\widehat{\kappa_i s_{\mathcal{B}}} = \frac{1}{16} v_i$ for $i < 4$. Since at sends a nonzero vector x to the element whose members are the multiples of x, the vertices of

	Kochen–Specker configurations	finite obstructions of the model existence game for T^{01}
		under the strategy of Section 10.4, in a play that lasts until Olivia has carried out its parity argument
bases	orthonormal bases of the space	the nine bases of Table 1 in the span F of a block, each completed by $F^{\perp}$
vertex	a ray	a contextual occurrence of a term representing a ray in one of the nine bases
identifications	none needed: a ray is one vertex in all the bases that contain it (the hypothesis of noncontextuality)	eighteen identifications of rays and nine of the joins of the bases with F, all derived from axioms of T that do not mention μ, along which axiom (M3) carries probabilities; the parity argument needs all eighteen identifications of rays
values	0 or 1 at each ray	an interval with rational endpoints, within δ of 0 or of 1, for the probability of each ray
constraint on a basis	exactly one ray of value 1	an interval with rational endpoints, within η of 1, for the sum of the probabilities of its rays; any tolerances with $4\delta + \eta < 1$ suffice
location of the rays	fixed in advance	the eighteen rays of Table 1, in the block determined by Josh's answers
space	finite dimension, spanned by the rays	infinite dimension; the rays lie in the span of a block $e_{4j}, \dots, e_{4j+3}$ of the orthonormal basis $(e_n)_{n\in\mathbb{N}}$ named by constants of T
role of σ-additivity	none: every basis is finite; in infinite dimension, the eighteen projections $P_v \otimes 1$ on $\mathbb{C}^4 \otimes \ell^2$ need only finite additivity	locates the unit-weight block under the strategy of Section 10.4: Josh names a number N such that the probabilities of the rays of $e_0, \dots, e_{N-1}$ add up nearly to 1, and the rays lie in the block that contains some e_{n_0} with $n_0 < N$

TABLE 3. The eighteen-ray comparison under the strategy of Section 10.4, for plays that last until Olivia has carried out its parity argument. The contextual vertices index occurrences of terms; their geometric identities are recorded by commitments.

$\mathcal{B}$ are indexed by the terms $\mathrm{at}(\kappa_i s_{\mathcal{B}})$, for $i < 4$. Every ray of the table lies in two bases and is therefore represented by two contextual occurrences of terms. The instances of (K2), together with those of the axioms behind Lemmas 6.3 and 6.4, give $[\mathrm{at}(\kappa_i s_{\mathcal{B}}) = \mathrm{at}(v_i)] = 1$, where $\mathrm{at}(v_i)$ is as in the proof of Theorem C; hence the two contextual terms of a ray denote elements that are equal in the sense of the formula $[a = b]$. The same instances give $[c_{\mathcal{B}} = a_F] = 1$ for the nine elements $c_{\mathcal{B}} = \mathrm{at}(\kappa_0 s_{\mathcal{B}}) \vee \dots \vee \mathrm{at}(\kappa_3 s_{\mathcal{B}})$, where the element a_F stands for the span F of the block. All these identifications follow from axioms that do not mention μ. Axiom (M3), $[a = b] \wedge |\mu(a) - \mu(b)| \leq 0$, which every quantum probability measure satisfies, says that μ respects the equality of the Solèr sort. This axiom requires the measure to take the same value at the elements denoted by the two contextual terms of a ray.

For the following minimality assertion, retain only the Boolean variables attached to the thirty-six contextual occurrences, the nine exactly-one basis constraints and the eighteen equalities between paired occurrences. Each equality is indispensable to the inconsistency of

this constraint system. Indeed, the nine bases can be arranged in the cycle $1, 2, 4, 3, 6, 5, 9, 8, 7$ of their numbers in Table 1, in which any two consecutive bases (the last and the first included) share a ray. If the identification attached to a ray of a basis $\mathcal{B}$ is omitted, give the value 1 to the vertex of that ray in $\mathcal{B}$. Next, pair off the other eight bases, which follow one another along the cycle, by four of the rays that consecutive bases share. Finally, give the value 1 to both vertices of each of these four rays and the value 0 to all the other vertices. Each basis then has exactly one vertex of value 1. Without identifications, a vertex of value 1 could be chosen in each basis independently. An independent choice is possible in the dense families of orthonormal bases that Clifton and Kent constructed in each finite dimension. In each of these families, any two bases generate Boolean algebras of projections with only 0 and the identity in common [CK00].

12.3. **Quantitative stability of the parity obstruction.** The interval commitments admit a precise stability statement for the parity argument. The tolerances concern numerical values of the hypothetical valuation, while the rays and their geometric identifications remain exact.

Theorem 12.1 (Quantitative parity obstruction). *Let $\mathcal{R}$ be the set of the eighteen rays determined by Table 1, and let $\mathfrak{B}$ be the family of the nine four-ray bases specified there. Let δ and η be nonnegative real numbers with $4\delta + \eta < 1$. There is no family of numbers $(p_{B,r})_{B\in\mathfrak{B},\, r\in B}$ in $[0,1]$ satisfying all three conditions:*

(i) $p_{B,r} = p_{B',r}$ whenever $B, B' \in \mathfrak{B}$ and $r \in B \cap B'$;
(ii) $p_{B,r} \in [0,\delta] \cup [1-\delta, 1]$ for every $B \in \mathfrak{B}$ and every $r \in B$;
(iii) $\left|\sum_{r\in B} p_{B,r} - 1\right| \le \eta$ for every $B \in \mathfrak{B}$.

Proof. Since $4\delta + \eta < 1$, the intervals $[0,\delta]$ and $[1-\delta, 1]$ are disjoint. Call an occurrence, that is, a pair (B, r) with $B \in \mathfrak{B}$ and $r \in B$, high when $p_{B,r}$ belongs to $[1-\delta, 1]$. A basis with no high occurrence has sum at most $4\delta < 1 - \eta$, contrary to (iii). A basis with at least two high occurrences has sum at least $2 - 2\delta > 1 + \eta$, again contrary to (iii). Thus each of the nine bases has exactly one high occurrence. Condition (i) gives both occurrences of each ray the same value, and each ray belongs to exactly two bases. Counting high occurrences basis by basis therefore gives nine, whereas counting them ray by ray gives an even number, a contradiction. ☺

The finite parity continuation of Proposition 10.9 uses the ground instances of the measure axioms and the recorded positive geometric antecedents. These antecedents activate (M3), giving exact transport between paired occurrences. Localization has recorded an atom value greater than 15/16. Under the dispersion-free ground conditions that value must be 1, and finite additivity, nonnegativity and the upper bound on the block measure then force the block measure, and each of the nine basis sums, to be 1. The final fair refinement exposes the incompatibility of these finitary conditions at a basic sentence. The quantitative theorem isolates the same parity obstruction when endpoint and basis-sum conditions are allowed numerical tolerances: for bases of d rays, with an integer $d \ge 2$, the bound $d\delta + \eta < 1$ forces exactly one high value per basis. Any configuration whose exactly-one constraints have no Boolean solution then gives the same contradiction.

The parameters δ and η bound errors in commitments about a hypothetical dispersion-free valuation. Born probabilities may take intermediate values, so condition (ii) is a restriction on that valuation rather than an estimate of experimental frequency errors. An experimental

noncontextuality inequality would additionally require specified preparation and measurement procedures, operational equivalences and a statistical bound for the relevant class of ontological models. Theorem 12.1 thus establishes the quantitative stability of the obstruction to a dispersion-free valuation when the rays are exact; the distinct problem of imperfect measurements has its own analyses [SBZ01, Lar02].

A configuration of Kochen and Specker is fixed in advance, and one configuration serves against every assignment. Such a configuration can be realized in the laboratory, each of its bases as a measurement; D'Ambrosio and his coauthors implemented the nine bases of Table 1 with single photons in 2013 [DHA+13]. Under the strategy of Section 10.4, the location of the ray configuration is determined by Josh's answers. Consider the first phase of the strategy, where the constants e_n of T name an orthonormal basis $(e_n)_{n\in\mathbb{N}}$ and the constant ε names the sequence $(2^{-n-1}e_n)_{n\in\mathbb{N}}$. In this phase, a countable challenge to (M5) at the constant ε makes Josh name a number N for which the probabilities of the rays of $e_0, \dots, e_{N-1}$ add up nearly to 1. His later answers single out some $n_0 < N$ for which the probability of the ray of e_{n_0} is close to 1. The eighteen rays then lie in the span F of the block $e_{4j}, \dots, e_{4j+3}$ with $4j \leq n_0 < 4j + 4$. Since Olivia's challenges are dictated by her strategy, a play is determined by Josh's answers. Because every finite set of axioms of T^{01} has a model, Theorem D shows that the tree of every winning strategy of Olivia for T^{01} branches infinitely at some node, at which every natural number is a legal countable answer for Josh (see Proposition 10.7). The localization in a four-dimensional block, by contrast, is a feature of the strategy of Section 10.4 alone. Whether the length of the plays can be bounded independently of Josh's answers is the subject of Question 10.10.

12.4. **Countable localization and infinite-rank projections.** The ray-localization argument uses the countable-additivity scheme along the named orthonormal basis. In the models of the finite sets of axioms of T^{01} constructed in the proof of Proposition 10.7(2), the measure vanishes on every finite-dimensional subspace. These structures satisfy (M5) on the sequences present in their restricted sequence sorts. Their measures are therefore not σ-additive on the lattice L_{fc} of Example 7.5: the rays of the vectors e_n are mutually orthogonal and have join 1 in that lattice, yet each of them has measure 0. It is σ-additivity over all the vectors e_n, through infinitely many instances of (K4), that excludes a measure vanishing on every finite-dimensional subspace. In finite dimension this step is automatic, since a configuration of Kochen and Specker spans the space. Under the strategy of Section 10.4, this is the step that selects a unit-weight finite-dimensional sector. A different finite obstruction is available on the full projection lattice. Write a separable infinite-dimensional Hilbert space H as $\mathbb{C}^4 \otimes \ell^2$, and, for each vector v of Table 1, let P_v be the projection of $\mathbb{C}^4$ onto the ray of v. For each basis $\mathcal{B}$ of the table, the four projections $P_v \otimes 1$ with v in $\mathcal{B}$ are pairwise orthogonal with sum 1. The eighteen projections $P_v \otimes 1$, which have infinite rank, therefore form a configuration that is fixed in advance and refutes every finitely additive $\{0, 1\}$-valued probability measure on $L(H)$ by the same parity argument. More generally, Döring proved the Kochen–Specker theorem for von Neumann algebras without summands of types I_1 and I_2 [Dör05]. His argument passes through the finitely additive form of Gleason's theorem due to Christensen, Yeadon, and Bunce and Wright [Chr82, Yea83, BW92] (see Hamhalter [Ham03]). The Solèr sort of the models of finite sets of axioms of T^{01} constructed in the proof of Proposition 10.7(2) is the lattice L_{fc} of Example 7.5. This lattice omits the amplified projections of infinite rank and infinite corank. Its measures therefore supply finite-fragment models without contradicting the finite-additivity obstruction on the full lattice $L(H)$. The

strategy of Section 10.4 instead uses rays and records how the hypothetical measure selects their four-dimensional sector.

A configuration of Kochen and Specker certifies the impossibility by the absence of an admissible assignment, whereas a finite obstruction certifies it by a finite position of commitments and function records that no structure satisfies. Behind each kind of certificate stands a topological property of the logic. The assignments of 0 and 1 to the rays form a compact space. In that space, the condition that a finite basis contain exactly one ray of value 1 involves finitely many rays and defines a closed set. It is through these closed sets that compactness, the topological form of finiteness, turns the theorem of J. S. Bell into a finite configuration [Bel96, Pit98]. The condition that the probabilities of the rays of the vectors e_n add up to 1 involves infinitely many rays at once and defines no closed set. Indeed, as n grows, the assignments that give the value 1 to the ray of e_n alone converge to the assignment that vanishes everywhere, which violates that condition. Such conditions are expressed by the countable suprema and infima of real-valued $L_{\omega_1\omega}$, whose space of Henkin types is Čech-complete, whence the Baire category theorem holds in it (see Section A). The finite certificates come from the game, whose analysis rests on the determinacy theorem of Gale and Stewart. Olivia's bookkeeping is a Baire category argument on the space of plays, the analogue of the one in the space of Henkin types. The passage from compactness to Čech-completeness is thus the passage from what finite configurations of rays can witness to what σ-additivity demands. The game returns what σ-additivity demands as finite certificates, one at each leaf of a well-founded tree.

Historical note. In 1990 Peres gave a proof of the Kochen–Specker theorem by observables on two qubits, valid for a particular state. Building on that proof, Mermin found a finite obstruction of another kind, formed by nine such observables arranged in three rows and three columns of commuting observables. No assignment of the values ± 1 to these nine observables respects the multiplicative relations along the rows and the columns [Per90, Mer90]. The array thus gives a state-independent proof of the theorem. In 1999 Meyer observed that a coloring of Godsil and Zaks [GZ88] assigns 0 and 1 to the unit vectors of $\mathbb{R}^3$ with rational coordinates, which are dense in the unit sphere. In this assignment, each orthogonal triple contains exactly one vector of value 1, and no two orthogonal vectors have the value 1 [Mey99]. Kent extended the result in the same year to dense sets of resolutions of the identity in every finite dimension [Ken99], and the families of Clifton and Kent date from 2000 [CK00].

Appendix A. The space of Henkin types and the model existence game

The model existence theorem of Section 10 and the omitting types theorem of Aubone, Dueñez and Iovino [ADI26] are two treatments of the Henkin construction. In the first, the description of a structure whose elements are named by witnesses is built move by move in the plays of a game. In the second, such descriptions are points of the space of Henkin types recalled in Section 5, and structures are obtained from the Baire category theorem. In this appendix we recall the facts about the space of Henkin types which bear on the game, referring to the paper of Aubone, Dueñez and Iovino for the proofs. We also compare the positions and the rules of the game with the conditions and the $\forall\exists$ formulas of Keisler's model-theoretic forcing [Kei73]. Throughout, we distinguish the correspondences that rest on proof from those that we present as analogies.

The space of Henkin types. The vocabulary $\mathcal{L}$ and the theory T are as in the setting of Section 10. The vocabulary is countable, and T has countably many axioms, whence the fragment $\mathcal{F}$ generated by the sentences of the axioms of T is countable (see [ADI26,

Section 6]). For the set C of new constants of Section 5 we take the set $\mathcal{W}$ of witnesses of Section 10. The fragment $\mathcal{F}_C$ that $\mathcal{F}$ generates in $\mathcal{L}_\mathcal{W}$ then has a countable set $S_C^\mathcal{F}$ of sentences. The set $S_C^\mathcal{F}$ contains every sentence that occurs in a play of the game $\mathrm{G}(T)$ of Definition 10.2. Indeed, a sentence that occurs in a play is an atomic sentence of $\mathcal{L}_\mathcal{W}$ or is obtained from a subformula of an axiom by substituting closed terms. We write $\prod S_C^\mathcal{F}$ for the product of the closed intervals $[-r_\varphi, r_\varphi]$, for $\varphi \in S_C^\mathcal{F}$, and $\varphi^\mathfrak{m}$ for the coordinate at φ of a point $\mathfrak{m}$ of the product. By Tychonov's theorem, the product is a compact Hausdorff space; it contains the space $\mathcal{H}_C^\mathcal{F}$ of Henkin types as a subspace. For a set T' of closed conditions on sentences of $S_C^\mathcal{F}$, we write $\langle\!\langle T' \rangle\!\rangle$ for the set of points that satisfy every condition of T'. The set $\langle\!\langle \emptyset \rangle\!\rangle$ is thus the whole product.

The correspondence rests on three results of [ADI26]:

(i) A structure for $\mathcal{L}_\mathcal{W}$ is *canonical* when every element is the value of a closed term. The family of the values of the sentences of $S_C^\mathcal{F}$ in a canonical structure, its $S_C^\mathcal{F}$*-type*, belongs to $\mathcal{H}_C^\mathcal{F}$. Conversely, every member of $\mathcal{H}_C^\mathcal{F}$ is the $S_C^\mathcal{F}$-type of a canonical structure, the term structure of its restriction to the atomic sentences [ADI26, Lemma 2].

(ii) For a countable vocabulary and a countable fragment, every structure has a substructure whose sorts are countable and in which every sentence of the fragment keeps its value [ADI26, Lemma 3 and its proof]. A structure for $\mathcal{L}$ whose sorts are countable has an expansion to $\mathcal{L}_\mathcal{W}$ that names every element by a witness, and this expansion is canonical [ADI26, proof of Proposition 4].

(iii) The space $\mathcal{H}_C^\mathcal{F}$ is a G_δ subset of $\prod S_C^\mathcal{F}$ [ADI26, Proposition 5]. Every Henkin condition defines a closed set except the upper bounds on suprema, each of which defines a countable intersection of open sets. For a supremum $\sup_x \varphi(x)$, the upper bound says that its value does not exceed the supremum of the values of the sentences $\varphi(t)$. For a countable join, the upper bound says that the value of the join does not exceed the supremum of the values of its members.

These facts are proved in the paper of Aubone, Dueñez and Iovino for the logic of this paper, which is theirs. The same paper proves Henkin Realizability [ADI26, Proposition 4] and the Čech-completeness of the space of Henkin types [ADI26, Corollary 6].

By (iii), the space $\mathcal{H}_C^\mathcal{F}$ is a G_δ subset of the product. The set $\langle\!\langle T \rangle\!\rangle \cap \mathcal{H}_C^\mathcal{F}$ is likewise a G_δ subset of the product, since each axiom $\psi \le r$ or $\psi \ge r$ of T defines the intersection of the open sets $\{\mathfrak{m} : \psi^\mathfrak{m} < r + \frac{1}{n}\}$ or $\{\mathfrak{m} : \psi^\mathfrak{m} > r - \frac{1}{n}\}$, for $n \ge 1$. Let K be either of these two sets. The closure of K in $\prod S_C^\mathcal{F}$ is compact (see Engelking [Eng89, Theorem 3.1.2]), hence a compactification of K. The remainder of this compactification, the trace of the complement of a G_δ set, is a countable union of closed sets. By Theorem 3.9.1 of Engelking [Eng89] and the definition that follows it, the space K is Čech-complete. Aubone, Dueñez and Iovino define Čech-completeness by the equivalent internal characterization of Theorem 3.9.2 in the same book. In K, the union of countably many nowhere dense subsets has dense complement, by the Baire category theorem [Eng89, Theorem 3.9.3].

Positions and plays. Let $\mathfrak{p}$ be a position of $\mathrm{G}(T)$ whose sentences belong to $S_C^\mathcal{F}$, as do those of every position reached in a play. The position $\mathfrak{p}$ defines the set $\langle\!\langle \mathfrak{p} \rangle\!\rangle$ of the points $\mathfrak{m} \in \prod S_C^\mathcal{F}$ that satisfy two requirements. The first is that $\varphi^\mathfrak{m} \in I$ for every commitment $\varphi \in I$ of $\mathfrak{p}$. The second is that $\theta(F(\bar{w}))^\mathfrak{m} = \theta(w')^\mathfrak{m}$ for every record $F(\bar{w}) \mapsto w'$ of $\mathfrak{p}$ and every atomic formula $\theta(x)$ of $\mathcal{L}_\mathcal{W}$ whose only free variable is x. These equations say that $F(\bar{w})$ and w' are indiscernible by atomic formulas with parameters, as equal elements are. Since the projections are continuous, the sets $\langle\!\langle \mathfrak{p} \rangle\!\rangle$ and $\langle\!\langle T' \rangle\!\rangle$ are closed. A position, which

commits sentences to closed intervals, thus defines a closed set and not, as finitely many strict conditions would, a basic open set of the product. Recall from Definition 10.1 that a position is satisfied in a structure under an interpretation of the witnesses that occur in it when its commitments hold and, for each of its records $F(\bar{w}) \mapsto w'$, the term $F(\bar{w})$ and the witness w' have the same value.

Proposition A.1. *Let $\mathfrak{p}$ be a position whose sentences belong to $S_C^{\mathcal{F}}$, and let T' be a subset of T. The set $\langle\!\langle \mathfrak{p} \rangle\!\rangle \cap \langle\!\langle T' \rangle\!\rangle \cap \mathcal{H}_C^{\mathcal{F}}$ is nonempty if and only if $\mathfrak{p}$ is satisfied in some model of T' under some interpretation of the witnesses. In particular, the theory T has a model if and only if $\langle\!\langle T \rangle\!\rangle \cap \mathcal{H}_C^{\mathcal{F}}$ is nonempty.*

Proof. Let $\mathfrak{m}$ be a point of the set; by (i), it is the $S_C^{\mathcal{F}}$-type of a canonical structure $\mathfrak{N}$. The structure $\mathfrak{N}$ satisfies the commitments of $\mathfrak{p}$, but a record $F(\bar{w}) \mapsto w'$ asks for equal values, whereas the defining equations guarantee that the term $F(\bar{w})$ and the witness w' agree in every atomic formula with parameters, without requiring their values to coincide; we pass to a quotient to make the records hold as literal equalities. Let two elements of the same sort of $\mathfrak{N}$ be equivalent when they agree in every atomic formula with parameters. This relation is a congruence for every function symbol, since substituting a term for a variable of an atomic formula yields an atomic formula; the quotient map therefore preserves the value of every atomic formula and, by induction on formulas, of every formula. The reduct of this quotient to $\mathcal{L}$ is therefore a model of T', which satisfies $\mathfrak{p}$ when each witness is interpreted as the class of its value. Indeed, the equations defining $\langle\!\langle \mathfrak{p} \rangle\!\rangle$ place the values of $F(\bar{w})$ and w' in one class. Conversely, let a model $\mathfrak{M}$ of T' satisfy $\mathfrak{p}$ under an interpretation h. We apply (ii) to the expansion of $\mathfrak{M}$ by constants for the elements $h(w)$ and to the fragment generated by $\mathcal{F}$ and the sentences of $\mathfrak{p}$. The structure $\mathfrak{M}$ thus has a substructure $\mathfrak{M}_0$ with countable sorts that contains these elements and satisfies T' and $\mathfrak{p}$ under h, the records included. The infinitely many witnesses of each sort that do not occur in $\mathfrak{p}$ can name all the elements of $\mathfrak{M}_0$. The resulting expansion is canonical, and by (i) its $S_C^{\mathcal{F}}$-type belongs to $\langle\!\langle \mathfrak{p} \rangle\!\rangle \cap \langle\!\langle T' \rangle\!\rangle \cap \mathcal{H}_C^{\mathcal{F}}$. ☺

In these terms, Josh's truthful answers in a model of T, in the proof that (1) implies (2) in Theorem 10.3, lead to positions that the same model satisfies. These answers therefore keep the set $\langle\!\langle \mathfrak{p} \rangle\!\rangle \cap \langle\!\langle T \rangle\!\rangle \cap \mathcal{H}_C^{\mathcal{F}}$ nonempty. At a leaf of the tree of a winning strategy of Olivia, on the other hand, the last position $\mathfrak{p}$ is satisfied in no structure (see Corollary 10.5(2)). By the proposition, applied with T' empty, the closed set $\langle\!\langle \mathfrak{p} \rangle\!\rangle$ does not meet $\mathcal{H}_C^{\mathcal{F}}$.

Through the sets $\langle\!\langle \mathfrak{p} \rangle\!\rangle$, the game can also be compared with Keisler's model-theoretic forcing [Kei73]. Keisler adjoins a countable set C of new constants and takes as forcing conditions the finite sets of sentences with constants from C that are satisfiable in a given class of models, ordered by inclusion [Kei73, pp. 98 and 100, Example 1.2]. The witnesses play the role of his constants, and a position, which grows along a play, is the real-valued counterpart of such a finite set. By Proposition A.1, a position $\mathfrak{p}$ is satisfied in some model of T under some interpretation of the witnesses exactly when $\langle\!\langle \mathfrak{p} \rangle\!\rangle \cap \langle\!\langle T \rangle\!\rangle \cap \mathcal{H}_C^{\mathcal{F}}$ is nonempty; such positions are the counterparts of his conditions for the class of models of T. A record $F(\bar{w}) \mapsto w'$ is the counterpart, in a logic without equality, of an atomic sentence $c = \tau$. For every closed term τ, Keisler's definition of a forcing property requires each condition to have an extension that contains such a sentence for some $c \in C$ [Kei73, p. 99, clause (iv)], and the rule (Tot) meets the corresponding requirement in the game. A fair play that Josh survives corresponds to the increasing sequence of conditions by which Keisler obtains a generic set [Kei73, p. 101, Lemma 1.4], and the structure built from it is canonical, as his generic models are [Kei73, p. 101].

The alternation of challenges and answers follows the pattern of Keisler's $\forall\exists$ formulas [Kei73, p. 107]. His formulas have the form $\forall\bar{x}\bigvee_n \exists\bar{y}(\varphi_{n1}\wedge\cdots\wedge\varphi_{nj_n})$. Consider an axiom $\inf_{\bar{x}}\bigvee_n \sup_{\bar{y}}\theta_n \geq r$ of the same shape, in which each formula θ_n is a minimum of finitely many formulas and the a priori bounds of the formulas θ_n have a finite supremum, as the formation of the countable join requires. Once Josh has committed its value under (Ax), the rule $(\inf\geq)$ lets Olivia choose the witnesses for $\bar{x}$, whereas $(\bigvee\geq)$ and $(\sup\geq)$ require Josh to choose the index n and the witnesses for $\bar{y}$, up to the tolerances that Olivia names. The universal choices are thus Olivia's and the existential choices Josh's. Keisler's general omitting types theorem asks that every condition remain satisfiable together with every instance, at constants from C, of the given $\forall\exists$ formulas [Kei73, p. 108]. In the game, the corresponding requirement is that Josh answer each challenge at a position satisfied in a model of T by a position with the same property, and the proof that (1) implies (2) in Theorem 10.3 shows that he can do so whenever T has a model. When T has no model, no position is satisfied in a model of T, and a winning strategy of Olivia makes this fact finite. At each leaf of its tree, the last position is a finite set of commitments and records satisfied in no structure, and hence contained in no position that some structure satisfies, that is, in no counterpart of a forcing condition. The contextual presentations of Section 12 (see Table 3) picture such positions, since their vertices, identifications and intervals are given by commitments and records of the last position of a play. The correspondence is not literal in three respects: commitments confine the values of sentences to closed intervals instead of asserting sentences, the axioms of T need not have the $\forall\exists$ shape, and Keisler's logic has equality [Kei73, p. 97], whose role in the game is taken by the records.

A fair play that Josh survives determines a nonempty subset of $\langle\!\langle T\rangle\!\rangle\cap\mathcal{H}_C^{\mathcal{F}}$ that is closed in $\mathcal{H}_C^{\mathcal{F}}$. Let $\mathfrak{p}_0\subseteq\mathfrak{p}_1\subseteq\cdots$ be its positions, and let $\mathfrak{p}_\infty$ be their union. The set $\bigcap_i\langle\!\langle\mathfrak{p}_i\rangle\!\rangle\cap\mathcal{H}_C^{\mathcal{F}}$, which is closed in $\mathcal{H}_C^{\mathcal{F}}$, is contained in $\langle\!\langle T\rangle\!\rangle$, since every axiom of T is named in a challenge (Ax). The structure $\mathfrak{M}$ built in the proof that (2) implies (1) in Theorem 10.3 is canonical, since its elements are the witnesses, each named by itself. It satisfies every commitment and every record of $\mathfrak{p}_\infty$; by (i), its $S_C^{\mathcal{F}}$-type lies in the set, which is therefore nonempty. The set need not consist of a single point, since a play leaves freedom to the structure that it describes. Suppose, for instance, that a predicate symbol R other than $\mathbf{1}$, with $r_R>0$, occurs in no axiom of T; then no sentence that occurs in the play contains R. The construction leaves the value of $R^{\mathfrak{M}}$ at each tuple of witnesses free in $[-r_R,r_R]$, and different choices yield canonical structures whose $S_C^{\mathcal{F}}$-types differ at the sentences $R(\bar{w})$.

Closed conditions, strict conditions and compactness. The contrast between closed and strict conditions drawn in Remark 10.6 has a topological counterpart. The Henkin types that satisfy a closed condition form a closed subset of $\mathcal{H}_C^{\mathcal{F}}$, and those that satisfy a strict condition an open subset. In the compact space $\prod S_C^{\mathcal{F}}$, a family of closed sets with empty intersection has a finite subfamily with empty intersection, whereas a family of open sets need not. For the theory $\{R>0\}\cup\{R<\frac{1}{n}:n\geq1\}$ of Remark 10.6, each finite part has a model, whence the open set that it defines meets the space of Henkin types by (i) and (ii). The whole theory, on the other hand, defines the empty set. The tree of Olivia's strategy for that theory branches infinitely at its root. If $\mathcal{H}_C^{\mathcal{F}}$ were closed, the compactness of the product would yield, whenever the closed-axiom theory T has no models, a finite subset T' of T with $\langle\!\langle T'\rangle\!\rangle\cap\mathcal{H}_C^{\mathcal{F}}=\emptyset$. By Proposition A.1, the set T' would be a finite set of axioms without models. The space $\mathcal{H}_C^{\mathcal{F}}$ need not be closed in the product or compact; as a G_δ subset of the compact product, it is always Čech-complete. On Čech-completeness rest the Baire category theorem in that space and, with it, the omitting types theorem recalled below.

By (iii), the Henkin requirements that need not define closed sets are the upper bounds on suprema, which in the game Josh certifies by naming an instance. Under $(\sup \geq)$ and $(\inf \leq)$ the instance is a witness, chosen among the finitely many that the freshness of witnesses leaves him. Under $(\bigvee \geq)$ and $(\bigwedge \leq)$ it is an index, chosen among infinitely many. Parts (3) and (4) of Corollary 10.5 record these two cases. This correspondence concerns the rules of the game, from which Corollary 10.5 is proved directly, without recourse to the topology of $\mathcal{H}^{\mathcal{F}}_C$.

Baire category. Olivia's bookkeeping can also be read as a Baire category argument. Let σ' be a winning strategy for Josh, and let $X_{\sigma'}$ be the set of infinite plays in which Josh follows σ'. We equip $X_{\sigma'}$ with the topology whose basic open sets consist of the plays that extend a given finite play. Since the moves range over countable sets, the space $X_{\sigma'}$ is homeomorphic to a closed subspace of the Baire space $\mathbb{N}^{\mathbb{N}}$. It is therefore a Polish space, in which the Baire category theorem holds. For each challenge and each round, the set of plays in which the challenge, if available at that round, is made at that round or later is open and dense in $X_{\sigma'}$. The fair plays therefore form a countable intersection of dense open subsets of $X_{\sigma'}$, which by the Baire category theorem is dense, hence nonempty. The bookkeeping can be read, too, through the *Banach–Mazur game*. In that game, two players alternately choose nonempty open sets of a space, each contained in the one before. The second player wins if the intersection of the chosen sets is contained in a given set (see Figure 9). Consider the Banach–Mazur game on $X_{\sigma'}$ in which the players choose basic open sets and the second player wins if the resulting play is fair. In this game the second player has a winning strategy modeled on Olivia's bookkeeping (compare Oxtoby [Oxt57]). At the move of index k of the second player, for each $k \in \mathbb{N}$, this strategy extends the current finite play by one round. In that round, Olivia makes the $l(k)$-th challenge of her enumeration for round $j(k)$, and Josh answers by σ'. The challenge is available because at least k rounds precede the new one and because $j(k) \leq k$. In the proof of Theorem 10.3, the category argument reduces to the bookkeeping, because only countably many challenges are available at each round.

The Baire category argument on plays is the analogue of the one by which Eagle [Eag14] proved the omitting types theorem for $[0,1]$-valued $L_{\omega_1\omega}$, applying Čech's form of the Baire category theorem to a space of countable metric structures. Aubone, Dueñez and Iovino prove the omitting types theorem for their logic [ADI26, Theorem 8] by Morley's Baire category method, applied in the Čech-complete space of Henkin types. The points of that space are the types of structures built from closed terms, as Keisler and Miller build models in logic without equality [KM01, Section 4]. In the terms of this appendix, for a sequence $(\Xi_n(\bar{x}_n))_{n\in\mathbb{N}}$ of types, sets of closed conditions $\varphi(\bar{x}_n) \geq r$ with $\varphi \in \mathcal{F}$ and r rational, their theorem can be stated as follows. Suppose that T has a model and that, for every $n \in \mathbb{N}$ and every tuple $\bar{t}$ of closed terms of $\mathcal{L}_{\mathcal{W}}$ of the appropriate sorts, the set of points of $\langle\!\langle T \rangle\!\rangle \cap \mathcal{H}^{\mathcal{F}}_C$ that satisfy the instance $\Xi_n(\bar{t})$ has empty interior in that space. Then T has a model whose sorts are countable and in which no tuple satisfies any $\Xi_n(\bar{x}_n)$. Indeed, the complements of these sets are countably many dense open subsets of the nonempty space $\langle\!\langle T \rangle\!\rangle \cap \mathcal{H}^{\mathcal{F}}_C$, in which the Baire category theorem holds. By (i), a point of the intersection of these complements is the $S^{\mathcal{F}}_C$-type of a canonical structure, whose tuples consist of values of closed terms. In the reduct of this structure to $\mathcal{L}$, no tuple satisfies any $\Xi_n(\bar{x}_n)$. The hypothesis of empty interior at tuples of closed terms explicates, as a statement about interiors in a Čech-complete space, the hypothesis of Keisler's general omitting types theorem [Kei73, p. 108] in the case in which his $\forall\exists$ formulas are taken over the whole fragment (see [ADI26, Introduction]). The two applications of the Baire category theorem take place in different spaces, the Polish space

of plays and the Čech-complete space of Henkin types. They correspond as analogues rather than through a map. Indeed, as observed above, a fair play determines a closed set of Henkin types, not a single point. Nor do we know of any continuous map from the one space to the other that carries one argument onto the other.

Historical note. The omitting types theorem goes back to Henkin, in 1954 [Hen54], and to Orey, in 1956 [Ore56]. Robinson introduced forcing into model theory in an article published in 1971 [Rob71], adapting the method that Cohen had created for set theory in 1963 [Coh63]. Barwise extended Robinson's forcing to infinitary logic and used it to give a new proof of the omitting types theorem (see Keisler [Kei73, p. 96]). In 1973 Keisler built countable models by forcing with finite sets of sentences satisfiable in a given class of models [Kei73, p. 100, Example 1.2]. He described his treatment of forcing as a reformulation, in terms of ordinary models, of the Boolean-valued models of Rasiowa and Sikorski [RS63], and he regarded its basic result, the Generic Model Theorem, as essentially due to them [Kei73, p. 97]. From generic models he derived a general omitting types theorem, in which $\forall\exists$ formulas take the place of principal types [Kei73, pp. 107–108]. In 2009 Ben Yaacov and Iovino developed model-theoretic forcing for an infinitary continuous logic [BYI09].

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